

The Fabius Function and Rvachev's Up-Function

Exact arithmetic, probability, Fourier analysis, and corrected sharp asymptotics

Human-readable synthesis of the formal development

Vladimir Reshetnikov's ProveIt repository

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Abstract

The Fabius function is the unique smooth distribution function that is constant outside $[0, 1]$, symmetric about $1/2$, and satisfies the self-similar differential equation $F'(x) = 2F(2x)$ on the left half of the unit interval. Folding it about the midpoint gives Rvachev's compactly supported even function up, one of the canonical examples of a C^∞ function with compact support that is given by an exact refinement equation. The same object admits several sharply different descriptions: a contraction fixed point, a probability law of an infinite weighted sum of independent uniform random variables, an infinite convolution, an infinite product of sinc functions, a Thue–Morse translate sum, and a family of exact rational recurrences.

This article develops these descriptions and proves their equivalence. Particular emphasis is placed on two subjects for which the formal development goes substantially beyond the usual short accounts. First, every dyadic value is evaluated by an explicit finite Thue–Morse/moment formula, and also by a fast terminating recursion that removes one set bit of the numerator at each step. Second, the small-argument asymptotic is derived by an exact negative-Laplace decomposition and a quantitative saddle-point argument. The correct main term contains a genuine nonconstant periodic fluctuation in the lower Lambert coordinate. Its Fourier coefficients are explicit Gamma-zeta values. Omitting that fluctuation, as in a frequently reproduced elementary formula, destroys the claimed inverse-logarithmic error bound. The first saddle correction and the full all-orders expansion are also described.

The presentation follows the theorem structure of the Lean formalization under `Analysis / FabiusFunction`, while translating the machine-oriented decomposition into conventional mathematical arguments. Source-level endpoint and normalization defects are corrected rather than repeated. In particular, a centered finite-spline approximation is given with an explicit uniform error, and this approximation is turned into a primitive-recursive dyadic algorithm proving computability of the bounded Fabius function in the sense of Grzegorzczuk. Binary reduction also yields an absolutely convergent signed-global series; finite half-base q -binomial identities give arbitrary-complex translated presentations of its summands, and a bounded-variation Toeplitz argument proves the corresponding discrete-limit formula.

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1 Historical note and scope

The probability distribution now attached to the name of Fabius is older than Fabius's two-page note. It occurs in the 1935 work of Jessen and Wintner on infinite convolutions of independent random variables [1], was rediscovered several times, and entered a different mathematical culture through the compactly supported "finite function" of V. A. Rvachev. Fabius used it in 1966 as a particularly economical probabilistic construction of a nowhere analytic C^∞ function [2]. Independently, and for entirely different reasons, the Rvachevs introduced the same object in 1971 as the canonical compactly supported solution of a functional–differential equation [3], and V. A. Rvachev later developed it into the theory of *atomic functions* used in approximation theory and numerical analysis [4, 5, 6]. Rvachev's normalization emphasizes the even compactly supported density, whereas the Fabius normalization emphasizes its cumulative distribution function. Arias de Reyna's 1982 paper gave a systematic treatment of the compactly supported function, including its Fourier product, probability interpretation, partitions of unity, derivatives, and dyadic values; an English translation appeared in 2017 [7]. His later arithmetic paper organized the exact rational sequences and denominator questions [8]. Haugland's note [9] gives a practical algorithm for evaluating F at dyadic rationals, closely related to the bit-recursion of Section 9.

The present article is not merely a paraphrase of those papers. It follows the proof dependency graph of the Lean development in the `ProveIt` repository [21]. That development first constructs the bounded distribution function, then derives the Rvachev and signed-global forms, and finally builds the arithmetic and asymptotic layers on top. This order has two advantages. It prevents identities valid for the signed extension from being silently applied to the bounded distribution, and it makes the exact dyadic algorithms independent of any numerical approximation.

The formalization also settles several questions that are only implicit, conjectural, or incorrectly normalized in the source material. Most importantly, the sharp endpoint asymptotic contains a nonconstant periodic function of an exact lower-Lambert phase. The periodic term is not a cosmetic improvement: every nonzero Fourier mode is explicit and nonzero, and an elementary formula without it fails at the error scale usually claimed. The development further proves a full inverse-Lambert expansion, establishes an exact bit-recursive evaluator for every dyadic rational, and supplies an absolutely and uniformly convergent Fourier–Legendre expansion of up .

Our convention is fixed throughout:

- F is the bounded cumulative distribution function, equal to 0 on $(-\infty, 0]$ and to 1 on $[1, \infty)$;
- up is the even compactly supported density on $[-1, 1]$;
- $\mathcal{F}$ is the signed global extension satisfying $\mathcal{F}'(x) = 2\mathcal{F}(2x)$ on all of $\mathbb{R}$.

All logarithms are natural logarithms. We write $L = \log 2$ in the asymptotic sections, $\text{wt}_2(n)$ for the sum of the binary digits of n , and $\text{sinc } z = \sin z/z$ with the removable value 1 at the origin.

2 Three functions that must not be conflated

The literature often uses the same letter F for three related, but globally different, functions. That economy of notation is harmless in a local calculation and dangerous in a long proof. We therefore begin by separating them.

2.1 The bounded Fabius distribution function

Definition 2.1 (Bounded Fabius function). A function $F : \mathbb{R} \rightarrow [0, 1]$ is called a *bounded Fabius function* if

$$F(x) = 0 \quad (x \leq 0), \quad (1)$$

$$F(x) = 1 \quad (x \geq 1), \quad (2)$$

$$F(1-x) = 1 - F(x) \quad (0 \leq x \leq 1), \quad (3)$$

$$F'(x) = 2F(2x) \quad (0 \leq x \leq \tfrac{1}{2}), \quad (4)$$

and F is infinitely differentiable on $\mathbb{R}$.

The derivative equation on the right half follows from symmetry:

$$F'(x) = 2F(2-2x), \quad \tfrac{1}{2} \leq x \leq 1. \quad (5)$$

In particular $F'(x) > 0$ for $0 < x < 1$, once positivity in the interior is known. Thus F is a genuine cumulative distribution function, not merely a smooth interpolation between 0 and 1.

Because F is constant on both exterior half-lines and is C^∞ , every derivative of positive order vanishes at both endpoints:

$$F^{(m)}(0) = F^{(m)}(1) = 0 \quad (m \geq 1). \quad (6)$$

This flatness is the source of both its extraordinary endpoint decay and its failure to be analytic there.

2.2 Rvachev's compactly supported bump

Definition 2.2 (Rvachev up-function). Given F , define

$$\text{up}(x) = \begin{cases} F(x+1), & x \leq 0, \\ F(1-x), & x > 0. \end{cases} \quad (7)$$

Equivalently,

$$\text{up}(x) = F(1 - |x|). \quad (8)$$

Here the right side is interpreted using the globally bounded function F .

It follows immediately that up is even, nonnegative, supported on $[-1, 1]$, and $\text{up}(0) = 1$. On the unit interval the two functions are simply translates:

$$F(x) = \text{up}(x-1), \quad 0 \leq x \leq 1. \quad (9)$$

The graph of F is an infinitely flat sigmoid; the graph of up is its symmetric fold.

2.3 The signed global extension

Let $\text{wt}_2(n)$ denote the sum of the binary digits of n , and put

$$\varepsilon_n = (-1)^{\text{wt}_2(n)}. \quad (10)$$

The sequence (ε_n) is the $\{\pm 1\}$ form of the Thue–Morse sequence [14].

Definition 2.3 (Signed extension). Define

$$\mathcal{F}(x) = \sum_{n \geq 0} \varepsilon_n \text{up}(x - 2n - 1). \quad (11)$$

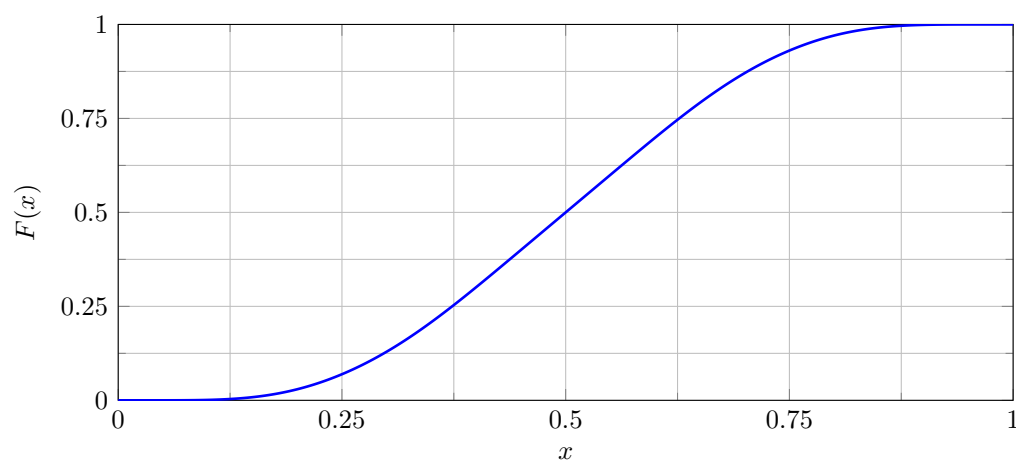


Figure 1: The bounded Fabius function on $[0, 1]$, drawn through exact dyadic values on a 2^{-8} grid. At this resolution the straight segments are visually indistinguishable from the smooth curve.

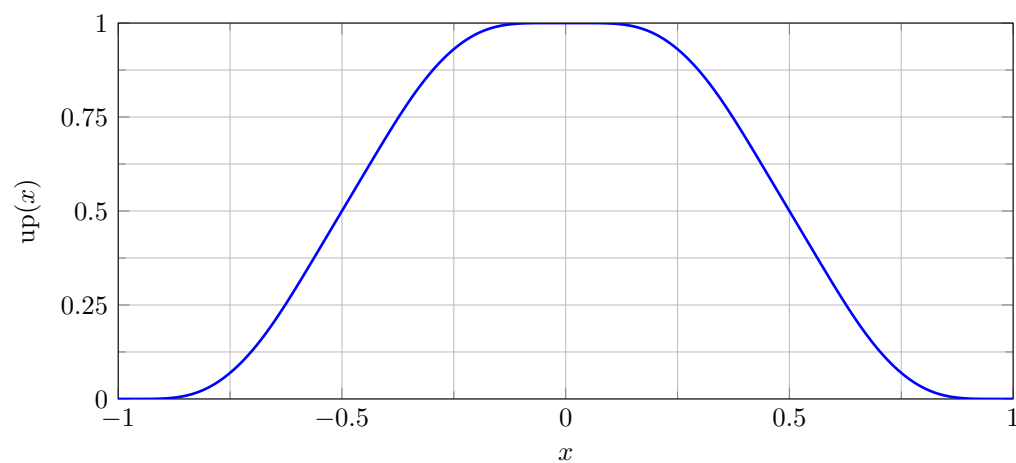


Figure 2: Rvachev's up-function on its support $[-1, 1]$, obtained by folding the same exact dyadic data.

At a fixed x only finitely many summands can be nonzero, so this is a locally finite sum. On every block $[2b, 2b + 2]$ there is exactly one nonzero translate:

$$\mathcal{F}(x) = \varepsilon_b \text{up}(x - 2b - 1), \quad 2b \leq x \leq 2b + 2. \quad (12)$$

It vanishes for $x \leq 0$, and it agrees with the bounded F on $[0, 1]$. It does *not* remain in $[0, 1]$ to the right of that interval. This is the global object that satisfies the cleanest differential equation:

$$\mathcal{F}'(x) = 2\mathcal{F}(2x) \quad (x \in \mathbb{R}). \quad (13)$$

Arias de Reyna's arithmetic paper uses this extension. Statements about its values outside $[0, 1]$ must therefore not be transferred blindly to the bounded cumulative distribution.

Notation used in this article. F always means the bounded distribution function; up always means the compactly supported even bump; and $\mathcal{F}$ always means the signed global extension. On $[0, 1]$ we have $F = \mathcal{F}$, but globally they are different functions.

2.4 Exact shape: positivity, bijectivity, and unimodality

The three defining equations (1)–(4) already determine the qualitative picture, but the statements that are actually used later are the sharp ones: not $F \geq 0$ but $F(x) > 0$ exactly for $x > 0$; not $\text{up} \leq 1$ but $\text{up}(x) = 1$ exactly at the origin. We record them here, in the form of equivalences, because the strict versions are what make the convexity and Lipschitz theorems below sharp.

Everything rests on a single formula that merges (4) and (5) into one identity valid on the whole line.

Lemma 2.4 (Unified derivative formula). *For every real x ,*

$$F'(x) = 2 \text{up}(2x - 1). \quad (14)$$

Proof. By (8), $\text{up}(2x - 1) = F(1 - |2x - 1|)$. For $0 \leq x \leq \frac{1}{2}$ one has $|2x - 1| = 1 - 2x$, so $\text{up}(2x - 1) = F(2x)$ and (14) is (4). For $\frac{1}{2} \leq x \leq 1$ one has $|2x - 1| = 2x - 1$, so $\text{up}(2x - 1) = F(2 - 2x)$ and (14) is (5). For $x < 0$ or $x > 1$ one has $|2x - 1| > 1$, hence $1 - |2x - 1| < 0$ and $\text{up}(2x - 1) = 0$ by (1), while $F'(x) = 0$ because F is constant on each exterior half-line by (1) and (2). $\square$

Since $\text{up} \geq 0$, formula (14) shows at once that F is nondecreasing on all of $\mathbb{R}$. It is also convenient to note that the reflection symmetry (3), stated on the unit interval, in fact holds globally:

$$F(1 - x) = 1 - F(x) \quad (x \in \mathbb{R}). \quad (15)$$

Indeed, for $x < 0$ both sides equal 1 by (1) and (2), and for $x > 1$ both sides equal 0.

Theorem 2.5 (Positivity, strict monotonicity, bijectivity). *The bounded Fabius function satisfies the following.*

1. $F(x) > 0$ if and only if $x > 0$.
2. $F(x) < 1$ if and only if $x < 1$.
3. $F'(x) > 0$ for $0 < x < 1$, and F is strictly increasing on $[0, 1]$.
4. F maps $[0, 1]$ bijectively onto $[0, 1]$.

Proof. The heart of the matter is the following doubling step: if $0 \leq x \leq \frac{1}{2}$ and $F(x) = 0$, then $F(2x) = 0$. To see it, observe that $F \geq 0$ everywhere, so a zero of F is a global minimum of a function differentiable on all of $\mathbb{R}$; Fermat's rule gives $F'(x) = 0$, and (4) turns this into $2F(2x) = 0$.

Suppose now that $F(x) = 0$ for some $x > 0$. Since $2^n x \rightarrow \infty$, there is a smallest integer $n \geq 0$ with $2^n x \geq \frac{1}{2}$. For $0 \leq m < n$ we have $2^m x < \frac{1}{2}$, so the doubling step applies at $2^m x$ and propagates the vanishing: from $F(x) = 0$ we obtain $F(2^m x) = 0$ for every $m \leq n$, and in particular $F(2^n x) = 0$. But F is nondecreasing and $2^n x \geq \frac{1}{2}$, so $F(2^n x) \geq F(\frac{1}{2}) = \frac{1}{2}$, where the midpoint value comes from (3) at $x = \frac{1}{2}$. This contradiction proves that F has no zero on the positive half-line; combined with (1) it proves (i).

Assertion (ii) follows from (i) and (15): $F(x) < 1$ if and only if $F(1 - x) > 0$, that is, if and only if $1 - x > 0$.

For (iii), let $0 < x < 1$. Then $2x - 1 \in (-1, 1)$, so $1 - |2x - 1| \in (0, 1]$, and (i) gives $\text{up}(2x - 1) = F(1 - |2x - 1|) > 0$. Now (14) yields $F'(x) > 0$. A continuous function on $[0, 1]$ whose derivative is positive throughout the interior is strictly increasing on $[0, 1]$.

For (iv), F is continuous with $F(0) = 0$ and $F(1) = 1$, so the intermediate value theorem makes $F([0, 1])$ contain $[0, 1]$; since F takes values in $[0, 1]$, the image is exactly $[0, 1]$. Strict monotonicity from (iii) gives injectivity. $\square$

The fold (8) now transfers the whole picture to Rvachev's bump, and it does so in strict form.

Theorem 2.6 (Strict unimodality of up). *Rvachev's function satisfies the following.*

1. up is strictly increasing on $[-1, 0]$ and strictly decreasing on $[0, 1]$.
2. $\text{up}(x) = 1$ if and only if $x = 0$; the maximum value 1 is attained at the origin and nowhere else.
3. $\text{up}(x) = 0$ if and only if $x \notin (-1, 1)$. Thus up is nonzero exactly on $(-1, 1)$, and $\text{supp } \text{up} = [-1, 1]$.
4. $\text{up}'(x) > 0$ for $-1 < x < 0$ and $\text{up}'(x) < 0$ for $0 < x < 1$.

Proof. By (8), $\text{up}(x) = F(1 - x)$ for $x \geq 0$. As x increases through $[0, 1]$ the argument $1 - x$ decreases through $[0, 1]$, on which F is strictly increasing by theorem 2.5(iii); hence up is strictly decreasing on $[0, 1]$, and evenness gives (i) on $[-1, 0]$.

For (ii), $\text{up}(x) = F(1 - |x|)$ and theorem 2.5(ii) say that $\text{up}(x) < 1$ unless $1 - |x| \geq 1$, that is, unless $x = 0$; and $\text{up}(0) = F(1) = 1$. For (iii), $\text{up}(x) > 0$ if and only if $1 - |x| > 0$ by theorem 2.5(i), and $\text{up} \geq 0$, so the zero set is exactly the complement of $(-1, 1)$; the closed support is the closure $[-1, 1]$.

For (iv), the identity $\text{up}(x) = F(1 - x)$ holds on the whole open half-line $x > 0$, hence may be differentiated there: $\text{up}'(x) = -F'(1 - x)$, which is negative for $0 < x < 1$ by theorem 2.5(iii). Evenness gives $\text{up}'(x) = -\text{up}'(-x)$, so $\text{up}' > 0$ on $(-1, 0)$. $\square$

Parts (i) and (iii) are the classical description of the bump [6, 7]; the strict forms, and the equivalences in theorem 2.5, are exactly what the formal development records.

2.5 Convexity and the unique inflection point

Formula (14) converts the first-order shape of up into the second-order shape of F . The affine substitution $x \mapsto 2x - 1$ carries $(-\infty, \frac{1}{2}]$ onto $(-\infty, 0]$ and $[\frac{1}{2}, \infty)$ onto $[0, \infty)$, which are precisely the two halves on which up is monotone.

Theorem 2.7 (Convexity and the inflection point). *The derivative F' is nondecreasing on $(-\infty, \frac{1}{2}]$ and nonincreasing on $[\frac{1}{2}, \infty)$, and it is strictly increasing on $[0, \frac{1}{2}]$ and strictly decreasing on $[\frac{1}{2}, 1]$.*

Consequently F is convex on $(-\infty, \frac{1}{2}]$ and concave on $[\frac{1}{2}, \infty)$, strictly convex on $[0, \frac{1}{2}]$ and strictly concave on $[\frac{1}{2}, 1]$. The midpoint $x = \frac{1}{2}$ is the unique inflection point, and

$$F'(\frac{1}{2}) = 2 = \max_{x \in \mathbb{R}} F'(x), \quad (16)$$

the maximum being attained at $x = \frac{1}{2}$ and nowhere else.

Proof. By theorem 2.6, up is nondecreasing on $(-\infty, 0]$: it vanishes on $(-\infty, -1]$ and increases strictly on $[-1, 0]$. Symmetrically it is nonincreasing on $[0, \infty)$. Composing with $x \mapsto 2x - 1$ and using (14) shows that F' is nondecreasing on $(-\infty, \frac{1}{2}]$ and nonincreasing on $[\frac{1}{2}, \infty)$. If $0 \leq x < y \leq \frac{1}{2}$ then $-1 \leq 2x - 1 < 2y - 1 \leq 0$ and the increase of up on $[-1, 0]$ is strict, so $F'(x) < F'(y)$; the second half is symmetric.

A differentiable function on an interval is convex precisely when its derivative is nondecreasing there, and strictly convex when the derivative is strictly increasing; the concave statements are the same assertions for $-F$. The intervals of strictness cannot be enlarged: F vanishes identically on $(-\infty, 0]$ and equals 1 identically on $[1, \infty)$, so it is affine, hence not strictly convex or concave, on either exterior half-line.

Since F is convex on the whole of $(-\infty, \frac{1}{2}]$ and concave on the whole of $[\frac{1}{2}, \infty)$, the sense of convexity can change only at the midpoint, and it does change there, because F is strictly convex immediately to the left of $\frac{1}{2}$ and strictly concave immediately to the right. So $\frac{1}{2}$ is the unique inflection point. Finally $F'(x) = 2 \text{up}(2x - 1) \leq 2 \text{up}(0) = 2$ by theorem 2.6(ii), with equality if and only if $2x - 1 = 0$. $\square$

Differentiating (14) once more gives $F''(x) = 4 \text{up}'(2x - 1)$, so $F''(\frac{1}{2}) = 4 \text{up}'(0) = 0$: the origin is an interior maximum of the differentiable function up. Together with $F(\frac{1}{2}) = \frac{1}{2}$ this locates the steepest point of the sigmoid exactly at its centre of symmetry, where the graph crosses its own midpoint with slope 2.

2.6 The optimal Lipschitz constant

The bound $F' \leq 2$ of (16) is a Lipschitz estimate, and because the bound is attained the estimate cannot be improved.

Theorem 2.8 (Two is the least Lipschitz constant). *Both F and up are 2-Lipschitz on $\mathbb{R}$:*

$$|F(x) - F(y)| \leq 2|x - y|, \quad |\text{up}(x) - \text{up}(y)| \leq 2|x - y| \quad (x, y \in \mathbb{R}). \quad (17)$$

Moreover 2 is the least admissible constant for each: if $K \geq 0$ is such that $|F(x) - F(y)| \leq K|x - y|$ for all x, y , or such that $|\text{up}(x) - \text{up}(y)| \leq K|x - y|$ for all x, y , then $K \geq 2$.

Proof. By (14) and $0 \leq \text{up} \leq 1$ we have $0 \leq F' \leq 2$ on all of $\mathbb{R}$, so the mean value theorem gives the first estimate in (17). The map $x \mapsto 1 - |x|$ is 1-Lipschitz, so the second estimate follows from the first through the fold (8).

For optimality, let K be admissible for F . Dividing $|F(x) - F(\frac{1}{2})| \leq K|x - \frac{1}{2}|$ by $|x - \frac{1}{2}|$ and letting $x \rightarrow \frac{1}{2}$ gives $K \geq |F'(\frac{1}{2})| = 2$ by (16). For up the same limit at the same point works: the proof of theorem 2.6(iv) gives $\text{up}'(\frac{1}{2}) = -F'(\frac{1}{2}) = -2$, so $K \geq 2$. $\square$

The two extremal points are the same point in the two pictures: $\frac{1}{2}$ is the centre of the sigmoid and the midpoint of the decreasing half of the bump, and the fold sends one to the other.

Corollary 2.9 (Linear majorants at the endpoints). *For all admissible arguments,*

$$F(x) \leq 2x \quad (x \geq 0), \quad (18)$$

$$1 - F(x) \leq 2(1 - x) \quad (x \leq 1), \quad (19)$$

$$\text{up}(x) \leq 2(1 - |x|) \quad (|x| \leq 1). \quad (20)$$

Proof. Applying (17) with $y = 0$ and using $F(0) = 0$ gives (18). Replacing x by $1 - x$ there and using (15) gives (19). Substituting $1 - |x| \geq 0$ into (18) and using (8) gives (20). $\square$

The content of (18) lies on $[0, \frac{1}{2}]$; for $x \geq \frac{1}{2}$ it is weaker than $F \leq 1$. Its value is that it holds with no hypothesis at all on the size of x , so it can be used as a linear majorant in any estimate near the left endpoint, and (20) says that the bump vanishes at least linearly at both ends of its support. These majorants and the optimality of the constant 2 are contributions of the formal development; the plain 2-Lipschitz bound is used implicitly throughout [7, 6].

3 Existence and uniqueness by a contraction

The most economical construction of F is a Banach fixed-point argument. It has an additional advantage: the contraction estimate later gives a clean uniqueness theorem, without presupposing Fourier analysis or probability.

3.1 The admissible function space

Let $I = [0, 1]$ and let $C(I)$ carry the supremum norm. Define $\mathcal{A} \subset C(I)$ by

$$\mathcal{A} = \{f \in C(I) : 0 \leq f \leq 1, f(1 - x) = 1 - f(x)\}. \quad (21)$$

The set $\mathcal{A}$ is closed in the Banach space $C(I)$, hence complete. It is nonempty; for example $f(x) = x$ belongs to it.

Extend $f \in \mathcal{A}$ to a continuous function $\tilde{f}$ on $\mathbb{R}$ by clamping the argument to I :

$$\tilde{f}(t) = \begin{cases} f(0), & t \leq 0, \\ f(t), & 0 \leq t \leq 1, \\ f(1), & t \geq 1. \end{cases} \quad (22)$$

Symmetry gives

$$\int_0^1 f(t) dt = \frac{1}{2}. \quad (23)$$

Indeed, replacing t by $1 - t$ and adding the two integrals gives $2 \int_0^1 f = 1$.

3.2 The integral transform

For $f \in \mathcal{A}$, define Tf by

$$(Tf)(x) = \begin{cases} \int_0^{2x} \tilde{f}(t) dt, & 0 \leq x \leq \frac{1}{2}, \\ 1 - \int_0^{2-2x} \tilde{f}(t) dt, & \frac{1}{2} < x \leq 1. \end{cases} \quad (24)$$

The two definitions agree at $x = 1/2$ by (23). Positivity and the upper bound $f \leq 1$ show that $0 \leq Tf \leq 1$, and a direct substitution shows $(Tf)(1 - x) = 1 - (Tf)(x)$. Hence $T : \mathcal{A} \rightarrow \mathcal{A}$.

The central estimate is stronger than the naive integral bound.

Lemma 3.1 (Contraction estimate). *For $f, g \in \mathcal{A}$,*

$$\|Tf - Tg\|_\infty \leq \frac{1}{2} \|f - g\|_\infty. \quad (25)$$

Proof. Write $h = f - g$. Since both f and g have integral $1/2$, $\int_0^1 h = 0$. For $0 \leq y \leq 1$ we may therefore estimate the primitive in two ways:

$$\begin{aligned} \left| \int_0^y h(t) dt \right| &\leq y \|h\|_\infty, \\ \left| \int_0^y h(t) dt \right| &= \left| \int_y^1 h(t) dt \right| \leq (1-y) \|h\|_\infty. \end{aligned}$$

Consequently

$$\left| \int_0^y h(t) dt \right| \leq \min(y, 1-y) \|h\|_\infty \leq \frac{1}{2} \|h\|_\infty.$$

Every value of $Tf - Tg$ is one of these primitives, possibly with a minus sign. Taking the supremum proves (25). $\square$

Theorem 3.2 (Existence of the bounded Fabius function). *The transform T has a unique fixed point $f_* \in \mathcal{A}$. Extending f_* by 0 on $(-\infty, 0]$ and by 1 on $[1, \infty)$ produces the bounded Fabius function F .*

Proof. The space $\mathcal{A}$ is complete and T is a strict contraction with constant $1/2$, so Banach's fixed-point theorem gives a unique fixed point. On $[0, 1/2]$ its fixed-point equation is

$$F(x) = \int_0^{2x} F(t) dt. \quad (26)$$

Differentiating gives $F'(x) = 2F(2x)$. The second half of (24) supplies symmetry and the matching right-hand formula. The endpoint values are immediate. $\square$

Starting from any admissible f_0 , the iterates $f_{n+1} = Tf_n$ converge uniformly to F , with the explicit estimate

$$\|f_n - F\|_\infty \leq 2^{-n} \|f_0 - F\|_\infty. \quad (27)$$

This gives a conceptually simple approximation scheme, although exact dyadic evaluation will be much faster.

3.3 The differential equation for up and smoothness

Folding (4) and (5) yields the refinement-type differential equation

$$\text{up}'(x) = 2(\text{up}(2x+1) - \text{up}(2x-1)) \quad (x \in \mathbb{R}). \quad (28)$$

At $x = 0$ the two one-sided derivatives glue because $\text{up}(1) = \text{up}(-1) = 0$. Once up is continuously differentiable, (28) bootstraps regularity: if $\text{up} \in C^r$, then the right side is C^r , so $\text{up} \in C^{r+1}$. Induction gives $\text{up} \in C^\infty$ and then $F \in C^\infty$.

Proposition 3.3 (Basic shape). *The functions F and up have the following properties.*

1. F is strictly increasing on $(0, 1)$ and satisfies $0 < F(x) < 1$ there.
2. up is even, strictly positive on $(-1, 1)$, and zero outside $[-1, 1]$.
3. $\int_{-1}^1 \text{up}(x) dx = 1$.
4. Every derivative of up vanishes at $x = \pm 1$.

Proof. For $0 < x \leq 1/2$, (4) gives $F'(x) = 2F(2x)$. Positivity follows first near $1/2$ from $F(1/2) = 1/2$, and then propagates toward zero by repeatedly doubling the argument; symmetry handles the right half. This proves strict monotonicity and, by (8), positivity of up in the interior of its support. Finally,

$$\int_{-1}^1 \text{up}(x) dx = 2 \int_0^1 F(t) dt = 1,$$

where the middle identity uses the fold and the last uses symmetry. Flatness at the support endpoints follows from smoothness and the fact that up is identically zero just outside the support. $\square$

3.4 The original Rvachev characterization

The preceding construction is equivalent to the characterization historically associated with Rvachev.

Theorem 3.4 (Original characterization). *There is a unique nonzero C^∞ function $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ such that*

$$\text{supp } \varphi = [-1, 1], \quad \int_{\mathbb{R}} \varphi(x) dx = 1, \quad (29)$$

and

$$\varphi'(x) = k(\varphi(2x+1) - \varphi(2x-1)) \quad (30)$$

for some real scale k . Necessarily $k = 2$, and $\varphi = \text{up}$.

Proof architecture. Integrating (30) against suitable test functions, or taking its Fourier transform, first forces $k = 2$ under the normalization (29). With the Fourier convention

$$\widehat{\varphi}(z) = \int_{\mathbb{R}} \varphi(x) e^{-2\pi i x z} dx, \quad (31)$$

one obtains

$$\widehat{\varphi}(z) = \text{sinc}(\pi z) \widehat{\varphi}(z/2), \quad \text{sinc } z = \begin{cases} \sin z/z, & z \neq 0, \\ 1, & z = 0. \end{cases} \quad (32)$$

Iterating N times gives a finite sinc product multiplied by $\widehat{\varphi}(z/2^N)$. Continuity and $\widehat{\varphi}(0) = 1$ determine the Fourier transform uniquely in the limit. Fourier inversion then determines φ . The function constructed above satisfies all the hypotheses, so it is that unique solution. $\square$

4 The probability law behind the functions

The fixed-point construction is analytic. A probabilistic construction explains the same self-similarity almost without calculation and makes monotonicity and positivity transparent.

4.1 An infinite weighted sum of uniforms

Let $U_0, U_1, \dots$ be independent random variables, each uniformly distributed on $[0, 1]$, and define

$$X = \sum_{n=0}^{\infty} \frac{U_n}{2^{n+1}}. \quad (33)$$

The series converges absolutely for every outcome and satisfies $0 \leq X \leq 1$. Removing the first coordinate gives an independent copy X' of X , and

$$X = \frac{U_0 + X'}{2}. \quad (34)$$

Reflecting every coordinate, $U_n \mapsto 1 - U_n$, preserves the product measure and sends X to $1 - X$.

Let $H(x) = \mathbb{P}(X \leq x)$ be the cumulative distribution function. Conditioning on U_0 in (34) gives

$$H(y) = \int_0^1 H(2y - u) \, du. \quad (35)$$

For $0 \leq y \leq 1/2$, the integrand vanishes when $u > 2y$, so

$$H(y) = \int_0^{2y} H(t) \, dt. \quad (36)$$

Reflection gives $H(1 - y) = 1 - H(y)$. Thus the restriction of H to $[0, 1]$ is a fixed point of the contraction T from section 3.

Theorem 4.1 (Probabilistic representation). *For every $x \in \mathbb{R}$,*

$$F(x) = \mathbb{P}\left(\sum_{n=0}^{\infty} \frac{U_n}{2^{n+1}} \leq x\right). \quad (37)$$

For $-1 \leq y \leq 0$,

$$\text{up}(y) = \mathbb{P}(X \leq y + 1). \quad (38)$$

Proof. The preceding calculation shows that H is an admissible fixed point of T . Uniqueness of the fixed point gives $H = F$. Equation (38) is then just the left branch of (7). $\square$

The more symmetric random variable is

$$Y = 2X - 1 = \sum_{n=0}^{\infty} \frac{V_n}{2^{n+1}}, \quad V_n = 2U_n - 1 \sim \text{Unif}[-1, 1]. \quad (39)$$

The density of Y is precisely up . Indeed, for $-1 < y < 1$,

$$f_Y(y) = \frac{1}{2} F'\left(\frac{y+1}{2}\right) = \text{up}(y). \quad (40)$$

This identity is a compact way to remember why up integrates to one.

4.2 Finite convolutions and step approximants

For each n , the summand $V_n/2^{n+1}$ is uniform on $[-2^{-n-1}, 2^{-n-1}]$ and has density

$$b_n(x) = 2^n \mathbf{1}_{[-2^{-n-1}, 2^{-n-1}]}(x). \quad (41)$$

Hence the partial sum

$$Y_N = \sum_{n=0}^N \frac{V_n}{2^{n+1}} \quad (42)$$

has density

$$u_N = b_0 * b_1 * \cdots * b_N. \quad (43)$$

These compactly supported piecewise-polynomial densities converge to up . Weak convergence is immediate from almost-sure convergence of Y_N ; the refinement equation and uniform bounds upgrade the convergence to the strong forms used in the formal development. This constructs up as an infinite convolution of shrinking boxes, a perspective useful in approximation theory and wavelet analysis.

4.3 Centered finite splines and an explicit error bound

The preceding finite random sums also give a completely explicit polynomial approximation to F . Put

$$T_p = \sum_{i=0}^{p-1} \frac{U_i}{2^{i+1}}, \quad C_p(x) = \mathbb{P}(T_p \leq x - 2^{-p-1}), \quad (44)$$

where $p \geq 1$. The shift by half of the omitted tail is important: it centers the one-sided approximation and improves its error by a factor of two.

Let

$$N_p(x) = \max\left(0, \left\lfloor 2^p x + \frac{1}{2} \right\rfloor\right), \quad \varepsilon_r = (-1)^{\text{wt}_2(r)}, \quad (45)$$

and interpret a sum with $N_p(x) = 0$ as empty. Define

$$S_p(x) = \frac{(-1)^p}{2^{\binom{p}{2}} p!} \sum_{r=0}^{N_p(x)-1} \varepsilon_r \left(r - 2^p x + \frac{1}{2}\right)_+^p. \quad (46)$$

Equivalently,

$$S_p(x) = \frac{1}{2^{\binom{p}{2}} p!} \sum_{r \geq 0} \varepsilon_r \left(2^p x - \frac{1}{2} - r\right)_+^p, \quad (47)$$

where only the terms $r < N_p(x)$ can be nonzero and $t_+ = \max(t, 0)$. The cutoff in (45) is a safe natural-number form of the inclusive upper limit

$$0 \leq r \leq \left\lfloor 2^p x - \frac{1}{2} \right\rfloor.$$

It handles the empty case without assigning a negative upper endpoint to a finite sum.

Theorem 4.2 (Centered spline as a finite distribution function). *For $p \geq 1$ and $0 \leq x \leq 1$,*

$$S_p(x) = C_p(x). \quad (48)$$

Consequently $0 \leq S_p(x) \leq 1$ on the unit interval.

Proof. For $p = 1$, $T_1 = U_0/2$, and direct evaluation gives

$$C_1(x) = \max\left(0, \min\left(1, 2x - \frac{1}{2}\right)\right).$$

Formula (47) gives exactly the same broken-linear function: the term $r = 0$ starts at $x = 1/4$, and the term $r = 1$ subtracts the excess after $x = 3/4$.

For the induction step, split off the first uniform coordinate. The finite sums satisfy

$$T_{p+1} \stackrel{d}{=} \frac{U_0 + T'_p}{2},$$

where T'_p is an independent copy of T_p . Conditioning on U_0 and accounting for the midpoint shifts gives the exact smoothing recurrence

$$C_{p+1}(x) = \int_0^1 C_p(2x - u) du. \quad (49)$$

On the spline side, integrate each truncated power in (47). The two binary children $2r$ and $2r + 1$ have opposite Thue–Morse signs, because $\varepsilon_{2r} = \varepsilon_r$ and $\varepsilon_{2r+1} = -\varepsilon_r$. After the change of scale this yields

$$S_{p+1}(x) = \int_0^1 S_p(2x - u) du. \quad (50)$$

The normalization is exactly the one required by

$$\sum_{r=0}^{2^p-1} \varepsilon_r r^p = (-1)^p 2^{\binom{p}{2}} p!,$$

while every lower power sum vanishes. Thus the two recurrences and their common base case prove (48). $\square$

Theorem 4.3 (Quantitative centered approximation). *For $p \geq 1$ and $0 \leq x \leq 1$,*

$$F(x - 2^{-p-1}) \leq S_p(x) \leq F(x + 2^{-p-1}), \quad (51)$$

and hence

$$|S_p(x) - F(x)| \leq 2^{-p}. \quad (52)$$

Proof. Write $X = T_p + R_p$, where the omitted tail satisfies

$$0 \leq R_p = \sum_{i=p}^{\infty} \frac{U_i}{2^{i+1}} \leq 2^{-p}.$$

If $X \leq x - 2^{-p-1}$, then certainly $T_p \leq x - 2^{-p-1}$. Conversely, if $T_p \leq x - 2^{-p-1}$, then $X \leq x + 2^{-p-1}$. Taking probabilities and using (37) and (48) proves (51).

The bounded Fabius function is globally 2-Lipschitz. Indeed, on the open unit interval its derivative is a rescaled value of up (or of the signed extension), whose absolute value is at most 1, and outside the unit interval F is constant. Thus

$$|F(x \pm 2^{-p-1}) - F(x)| \leq 2 \cdot 2^{-p-1} = 2^{-p}.$$

Combining these two inequalities with the sandwich gives (52). $\square$

The bound is uniform on the closed interval and includes both endpoints. It is stronger than mere weak convergence of the finite distributions, and it is the quantitative input for the computability proof in [section 20.6](#).

The same finite formula also has the correct signed global behavior. For $x \leq 0$ its safe cutoff is empty, so $S_p(x) = \mathcal{F}(x) = 0$. On the nonnegative half-line, binary block translation and midpoint reflection extend the local identification. Consequently, for every $x \in \mathbb{R}$,

$$|S_p(x)| \leq 1, \quad S_p(x) \longrightarrow \mathcal{F}(x). \quad (53)$$

This global form is the analytic input for the complex-shift and discrete-limit results below; it should not be confused with uniform convergence on the whole half-line.

5 Fourier transform, entire products, and Poisson summation

5.1 The sinc product

We use the Fourier convention

$$\widehat{f}(z) = \int_{\mathbb{R}} f(x) e^{-2\pi i x z} dx. \quad (54)$$

The characteristic function of a uniform variable on $[-a, a]$ is $\text{sinc}(2\pi a z)$. Independence in (39) therefore yields the product formula.

Theorem 5.1 (Fourier product). *For every $z \in \mathbb{C}$,*

$$\widehat{\text{up}}(z) = \prod_{n=0}^{\infty} \text{sinc}\left(\frac{\pi z}{2^n}\right). \quad (55)$$

The product converges locally uniformly and hence defines an entire function.

Proof. For real z , the characteristic function of the partial sum Y_N is the finite product through $n = N$. Since $Y_N \rightarrow Y$ almost surely and the exponential has modulus one, dominated convergence gives the real identity. On compact subsets of $\mathbb{C}$,

$$\text{sinc } w = 1 - \frac{w^2}{6} + \mathcal{O}(w^4),$$

so the sum of the deviations of the factors from 1 is dominated by $C_K \sum 4^{-n}$. The Weierstrass product theorem gives local uniform convergence to an entire function. Equality on the real axis and the identity theorem extend the formula to all complex z . $\square$

Separating the first factor gives the refinement equation already seen in (32):

$$\widehat{\text{up}}(z) = \text{sinc}(\pi z) \widehat{\text{up}}(z/2). \quad (56)$$

Conversely, continuity at zero, normalization $\widehat{\text{up}}(0) = 1$, and (56) force (55) by iteration. This is the core of the Fourier uniqueness proof.

5.2 Rapid decay and inversion

Since $\text{up} \in C_c^\infty(\mathbb{R})$, repeated integration by parts gives, for every integer $N \geq 0$,

$$|\widehat{\text{up}}(\xi)| \leq C_N (1 + |\xi|)^{-N}. \quad (57)$$

Thus the Fourier product is rapidly decreasing on the real axis despite being entire of finite exponential type. Fourier inversion is absolutely convergent:

$$\text{up}(x) = \int_{\mathbb{R}} \widehat{\text{up}}(\xi) e^{2\pi i x \xi} d\xi. \quad (58)$$

5.3 Poisson summation and partitions of unity

Rapid decay on both sides makes Poisson summation available without distributional qualifications.

Theorem 5.2 (Poisson summation). *For $u > 0$ and $t \in \mathbb{R}$,*

$$\sum_{k \in \mathbb{Z}} \text{up}(t + uk) = \frac{1}{u} \sum_{k \in \mathbb{Z}} \widehat{\text{up}}(k/u) e^{2\pi i kt/u}. \quad (59)$$

Both series converge absolutely and locally uniformly.

For a positive integer m , every nonzero term $\widehat{\text{up}}(mk)$ vanishes because the first sinc factor in (55) vanishes at nonzero integers. Taking $u = 1/m$ in (59) gives the exact partition of unity

$$\sum_{k \in \mathbb{Z}} \text{up}\left(t + \frac{k}{m}\right) = m. \quad (60)$$

In particular,

$$\sum_{k \in \mathbb{Z}} \text{up}(t + k) = 1. \quad (61)$$

This identity is one reason up is useful as a smooth compactly supported averaging kernel.

5.4 Moment expansion of the Fourier transform

Define the even moments

$$c_n = \int_{-1}^1 x^{2n} \text{up}(x) \, dx. \quad (62)$$

Evenness eliminates the odd moments, and expansion of the exponential in (54) gives the entire series

$$\widehat{\text{up}}(z) = \sum_{n=0}^{\infty} \frac{(-1)^n c_n}{(2n)!} (2\pi z)^{2n}. \quad (63)$$

Comparing this with the sinc refinement will produce the exact rational recurrence for c_n in section 7.

6 The global extension and exact derivative identities

The signed extension $\mathcal{F}$ is not merely a device for shortening notation. It turns the piecewise differential structure of F and up into a global scaling law, and from that law one can read off all higher derivatives.

6.1 Thue–Morse splitting

The binary weight satisfies

$$\text{wt}_2(2n) = \text{wt}_2(n), \quad \text{wt}_2(2n+1) = \text{wt}_2(n) + 1. \quad (64)$$

Therefore

$$\varepsilon_{2n} = \varepsilon_n, \quad \varepsilon_{2n+1} = -\varepsilon_n. \quad (65)$$

Differentiate the locally finite sum (11) term by term and use (28). The result is a sum over even and odd indices. Reindexing those two subseries with (65) makes them recombine into $2\mathcal{F}(2x)$.

Theorem 6.1 (Global differential equation). *The signed extension is infinitely differentiable and*

$$\mathcal{F}'(x) = 2\mathcal{F}(2x) \quad (66)$$

for every real x .

Proof. Local finiteness permits differentiation of (11). By (28),

$$\begin{aligned} \mathcal{F}'(x) &= 2 \sum_{n \geq 0} \varepsilon_n [\text{up}(2x - 4n - 1) - \text{up}(2x - 4n - 3)] \\ &= 2 \sum_{n \geq 0} \varepsilon_{2n} \text{up}(2x - 2(2n) - 1) + 2 \sum_{n \geq 0} \varepsilon_{2n+1} \text{up}(2x - 2(2n+1) - 1) \\ &= 2 \sum_{m \geq 0} \varepsilon_m \text{up}(2x - 2m - 1) = 2\mathcal{F}(2x). \end{aligned}$$

The same local-finiteness argument shows that the sum is C^∞ because each translate of up is. $\square$

6.2 All iterated derivatives

Repeated differentiation of (66) gives a triangular power of two.

Theorem 6.2 (Derivative scaling). *For every integer $m \geq 0$,*

$$\mathcal{F}^{(m)}(x) = 2^{\binom{m+1}{2}} \mathcal{F}(2^m x). \quad (67)$$

For $-1 \leq x \leq 1$,

$$\text{up}^{(m)}(x) = 2^{\binom{m+1}{2}} \mathcal{F}(2^m x + 2^m). \quad (68)$$

Proof. The first assertion is immediate by induction. If it holds for m , then

$$\begin{aligned}\mathcal{F}^{(m+1)}(x) &= 2^{\binom{m+1}{2}} 2^m \mathcal{F}'(2^m x) \\ &= 2^{\binom{m+1}{2}} 2^m \cdot 2 \mathcal{F}(2^{m+1} x) \\ &= 2^{\binom{m+2}{2}} \mathcal{F}(2^{m+1} x).\end{aligned}$$

On $[-1, 1]$, $\text{up}(x) = \mathcal{F}(x+1)$, so $\text{up}^{(m)}(x) = \mathcal{F}^{(m)}(x+1)$ and the second formula follows. $\square$

The formula is striking: the m th derivative is not an approximation or an asymptotic rescaling of the original function; it is an exact rescaling of the signed extension. The large factor $2^{m(m+1)/2}$ quantifies the rapid growth of high derivatives, while the argument $2^m x$ accounts for the increasingly fine oscillatory structure.

6.3 Sharp uniform bounds for all derivatives

Equation (67) converts any bound on $\mathcal{F}$ itself into a bound on all of its derivatives. The one ingredient it needs is that the signed extension is bounded by one in absolute value. That is not visible in (11), whose terms carry no uniform sign, but it is immediate from the block form (12).

Lemma 6.3 (The signed extension is bounded by one). *For every real x one has $-1 \leq \mathcal{F}(x) \leq 1$.*

Proof. For $x \leq 0$ the extension vanishes. For $x > 0$ let $b = \lfloor x/2 \rfloor$, a nonnegative integer with $2b \leq x \leq 2b+2$. Then (12) gives $\mathcal{F}(x) = \varepsilon_b \text{up}(x-2b-1)$ with $\varepsilon_b = \pm 1$ by (10), and $0 \leq \text{up} \leq 1$. $\square$

Theorem 6.4 (Sharp uniform derivative bounds). *For every integer $m \geq 0$ and every real x ,*

$$|\mathcal{F}^{(m)}(x)| \leq 2^{\binom{m+1}{2}}, \quad |\text{up}^{(m)}(x)| \leq 2^{\binom{m+1}{2}}, \quad |F^{(m)}(x)| \leq 2^{\binom{m+1}{2}}. \quad (69)$$

Each of the three constants is attained:

$$\mathcal{F}^{(m)}(2^{-m}) = F^{(m)}(2^{-m}) = \text{up}^{(m)}(2^{-m} - 1) = 2^{\binom{m+1}{2}}. \quad (70)$$

Hence $2^{\binom{m+1}{2}}$ is not merely an upper bound but the exact supremum of $|\mathcal{F}^{(m)}|$, of $|\text{up}^{(m)}|$, and of $|F^{(m)}|$ over $\mathbb{R}$, and each supremum is a maximum.

Proof. The bound for $\mathcal{F}$ is (67) combined with theorem 6.3. For up and $|x| \leq 1$ the bound is (68) combined with the same lemma; for $|x| > 1$ the function up vanishes identically on a neighbourhood of x by theorem 2.6(iii), so every derivative of up vanishes at x .

For the bounded function, note that F and $\mathcal{F}$ agree on $(-\infty, 1]$: both vanish on $(-\infty, 0]$ and they agree on $[0, 1]$. Hence they have the same germ at every point $x < 1$, so $F^{(m)}(x) = \mathcal{F}^{(m)}(x)$ there and the bound holds for $x < 1$. For $x > 1$ the function F is constant near x , so $F^{(m)}(x) = 0$ when $m \geq 1$, while for $m = 0$ the bound reads $F \leq 1$. The single remaining point $x = 1$ follows by letting $x \uparrow 1$, since $F^{(m)}$ is continuous and the bound is a closed condition.

For (70), take $x = 2^{-m}$ in (67): then $2^m x = 1$ and $\mathcal{F}(1) = 1$, so $\mathcal{F}^{(m)}(2^{-m}) = 2^{\binom{m+1}{2}}$. When $m \geq 1$ the point 2^{-m} lies in $(0, 1)$, where F and $\mathcal{F}$ have the same germ, so $F^{(m)}$ takes the same value there; when $m = 0$ the assertion is $F(1) = 1 = 2^0$. Finally take $x = 2^{-m} - 1 \in [-1, 0]$ in (68): then $2^m x + 2^m = 1$ and the same computation applies. $\square$

For $m = 1$ the theorem returns (16), with the extremal point $2^{-1} = \frac{1}{2}$; for $m = 2$ the sharp constant is $2^3 = 8$, for $m = 3$ it is $2^6 = 64$. Two features deserve emphasis. First, no estimate of the form $|F^{(m)}| \leq C_m$ can have $C_m < 2^{\binom{m+1}{2}}$, so the triangular power of two in (67) is not an artefact of the induction. Second, the extremal points 2^{-m} march towards the origin as m grows: the derivatives become large exactly in the region where F itself is flattest. The next subsection quantifies that flatness, and it will do so at the very same points 2^{-m} .

6.4 Effective flatness at the origin

Equation (6) says that F vanishes to infinite order at the origin, so that $F(x) = o(x^n)$ as $x \rightarrow 0^+$ for every n . That statement carries no constant. Two elementary arguments supply one, and the second is strictly better than the first.

Lemma 6.5 (Self-improving estimate). *For $0 \leq x \leq \frac{1}{2}$ one has $F(x) \leq 2x F(2x)$.*

Proof. At $x = 0$ both sides vanish. For $0 < x \leq \frac{1}{2}$, Lagrange's mean value theorem on $[0, x]$ provides a point $c \in (0, x)$ with $F(x) = F(x) - F(0) = x F'(c)$. Since $0 < c < x \leq \frac{1}{2}$, equation (4) gives $F'(c) = 2F(2c)$, and F is nondecreasing with $2c \leq 2x$, so $F'(c) \leq 2F(2x)$. $\square$

Theorem 6.6 (Effective flatness, mean value form). *Let $n \geq 0$. If $x \geq 0$ and $2^n x \leq 1$, then*

$$F(x) \leq 2^{\binom{n+1}{2}} x^n. \quad (71)$$

Proof. Induction on n . For $n = 0$ the claim is $F \leq 1$. Assume it for n , and let $x \geq 0$ satisfy $2^{n+1} x \leq 1$. Then $x \leq \frac{1}{2}$, because $2^{n+1} \geq 2$, so [theorem 6.5](#) applies; and $2^n(2x) = 2^{n+1}x \leq 1$, so the inductive hypothesis applies at $2x$. Therefore

$$F(x) \leq 2x F(2x) \leq 2x \cdot 2^{\binom{n+1}{2}} (2x)^n = 2^{\binom{n+1}{2} + n + 1} x^{n+1} = 2^{\binom{n+2}{2}} x^{n+1},$$

the last step being the Pascal identity $\binom{n+2}{2} = \binom{n+1}{2} + (n+1)$. $\square$

Replacing the mean value theorem by the fundamental theorem of calculus recovers a factorial that the pointwise argument throws away. The reason is structural: the mean value theorem evaluates the integrand at a single unknown point and then bounds it by its largest value on the interval, whereas integrating the same bound keeps the whole profile.

Theorem 6.7 (Effective flatness, integral form). *Let $n \geq 0$. If $x \geq 0$ and $2^n x \leq 1$, then*

$$F(x) \leq \frac{2^{\binom{n+1}{2}}}{n!} x^n. \quad (72)$$

Proof. Since $F(0) = 0$ and $F'(t) = 2F(2t)$ for $0 \leq t \leq \frac{1}{2}$ by (4),

$$F(x) = \int_0^x 2F(2t) dt \quad (0 \leq x \leq \tfrac{1}{2}). \quad (73)$$

Write $a_n = 2^{\binom{n+1}{2}}/n!$ and induct. For $n = 0$ the claim is $F \leq 1$. Assume $F(y) \leq a_n y^n$ whenever $y \geq 0$ and $2^n y \leq 1$, and let $x \geq 0$ satisfy $2^{n+1} x \leq 1$; then $x \leq \frac{1}{2}$. For $0 \leq t \leq x$ we have $2^n(2t) \leq 2^{n+1}x \leq 1$, so the inductive hypothesis gives $2F(2t) \leq 2a_n(2t)^n = 2^{n+1}a_n t^n$. Inserting this into (73) and integrating,

$$F(x) \leq 2^{n+1}a_n \int_0^x t^n dt = \frac{2^{n+1}a_n}{n+1} x^{n+1} = \frac{2^{\binom{n+1}{2} + n + 1}}{(n+1)!} x^{n+1} = a_{n+1} x^{n+1},$$

again by the Pascal identity. $\square$

Both bounds are stated on the dyadic range $2^n x \leq 1$, and the natural point to test them is the right endpoint $x = 2^{-n}$ of that range, which is also the point where the n th derivative attains its maximum in (70). There $\binom{n+1}{2} - n^2 = -\binom{n}{2}$, so (71) and (72) read

$$F(2^{-n}) \leq 2^{-\binom{n}{2}} \quad \text{and} \quad F(2^{-n}) \leq \frac{2^{-\binom{n}{2}}}{n!}, \quad (74)$$

so the entire gain of the integral argument over the mean value argument is the factorial. The second bound is now within an explicit factor of the truth. Indeed (102) gives the exact value $F(2^{-n}) = d_n / (n! 2^{\binom{n}{2}})$ with d_n the half moment (84), so the right-hand side of (72) at $x = 2^{-n}$ overshoots the true value by exactly the factor $1/d_n$. This is consistent, and it also shows how little room is left: from (84) and $0 \leq \text{up} \leq 1$,

$$d_n = n \int_0^1 t^{n-1} \text{up}(t) dt \leq n \int_0^1 t^{n-1} dt = 1,$$

with equality only for $n = 0$. At $n = 1$ the estimate reads $F(\frac{1}{2}) \leq 1$ against the true value $F(\frac{1}{2}) = \frac{1}{2} = d_1$.

The fold (8) transports the estimate to both ends of the support of the bump: if $|x| \leq 1$ and $2^n(1 - |x|) \leq 1$, then

$$\text{up}(x) \leq \frac{2^{\binom{n+1}{2}}}{n!} (1 - |x|)^n. \quad (75)$$

For $n = 1$ this is the linear majorant (20) restricted to the two end zones. Both effective forms are contributions of the formal development; the mean value derivation is retained there alongside the sharper one because it uses nothing but the differential equation and monotonicity.

6.5 Polynomial behavior near dyadic points

Let $x_0 = a/2^n$ be a dyadic point in the interior of the unit interval. For $m > n$, the argument $2^m x_0$ in (67) is an even integer. The local translate formula (12) and flatness of up at its endpoints imply

$$F^{(m)}(x_0) = 0 \quad (m > n). \quad (76)$$

Thus the Taylor series of F at a dyadic point terminates.

Proposition 6.8 (Terminating Taylor series). *If $x_0 = a/2^n \in (0, 1)$ with a odd, then the formal Taylor series of F at x_0 is a polynomial of degree at most n . Nevertheless F does not agree with that polynomial on any neighborhood of x_0 .*

Proof. Equation (76) proves termination. If F agreed with the Taylor polynomial on a nonempty interval, analyticity of the polynomial and the differential scaling relation would propagate polynomial behavior across successively enlarged dyadic intervals. Ultimately F would be polynomial on $(0, 1)$, which is incompatible with its flatness at 0 and its nonconstant value $F(1/2) = 1/2$. Equivalently, a polynomial with all positive-order derivatives zero at 0 is constant, whereas F is not. $\square$

6.5.1 The exact degree of the terminating polynomial

The proposition on terminating Taylor series bounds the degree of that polynomial by n . The bound is always attained, because the n th derivative at a level- n dyadic point with odd numerator can be computed exactly and is never zero.

Proposition 6.9 (Exact dyadic derivative). *Let $x_0 = a/2^n \in (0, 1)$ with a odd, say $a = 2b + 1$; necessarily $n \geq 1$. Then*

$$F^{(n)}(x_0) = \varepsilon_b 2^{\binom{n+1}{2}} = -\varepsilon_a 2^{\binom{n+1}{2}}, \quad (77)$$

with ε as in (10). In particular $|F^{(n)}(x_0)| = 2^{n(n+1)/2} \neq 0$.

Proof. The signed extension $\mathcal{F}$ agrees with F on $[0, 1]$, hence all their derivatives agree at the interior point x_0 . By the derivative scaling identity (67),

$$F^{(n)}(x_0) = \mathcal{F}^{(n)}(x_0) = 2^{\binom{n+1}{2}} \mathcal{F}(2^n x_0) = 2^{\binom{n+1}{2}} \mathcal{F}(a).$$

The point $a = 2b + 1$ lies in the block $[2b, 2b + 2]$, so the single-block formula (12) applies and gives

$$\mathcal{F}(a) = \varepsilon_b \text{up}(a - 2b - 1) = \varepsilon_b \text{up}(0) = \varepsilon_b,$$

since $\text{up}(0) = 1$. This is the first equality in (77). For the second, (64) gives $\text{wt}_2(a) = \text{wt}_2(2b + 1) = \text{wt}_2(b) + 1$, hence $\varepsilon_b = -\varepsilon_a$. Finally $\binom{n+1}{2} = n(n+1)/2$ and $|\varepsilon_b| = 1$. $\square$

The computation isolates exactly what makes $m = n$ the last surviving order. In (67) the argument $2^m x_0 = 2^{m-n} a$ is an even integer for every $m > n$, and $\mathcal{F}$ vanishes at even integers; this is (76). For $m = n$ that argument is the odd integer a , and $\mathcal{F}$ takes the value ± 1 at every odd integer. The truncation of the Taylor series is therefore governed by a parity, and the last nonzero coefficient is forced.

Proposition 6.10 (Exact degree). *Let $x_0 = a/2^n \in (0, 1)$ with a odd, and let*

$$T_{x_0}(x) = \sum_{k=0}^n \frac{F^{(k)}(x_0)}{k!} (x - x_0)^k \quad (78)$$

be the Taylor polynomial of F at x_0 . Then T_{x_0} is the whole formal Taylor series of F at x_0 and has degree exactly n , with leading coefficient $-\varepsilon_a 2^{\binom{n+1}{2}}/n!$.

Proof. By (76) every coefficient of order $m > n$ in the formal Taylor series of F at x_0 vanishes, so that series is the polynomial (78) and $\deg T_{x_0} \leq n$. Its coefficient of $(x - x_0)^n$ is $F^{(n)}(x_0)/n!$, which by (77) equals $-\varepsilon_a 2^{\binom{n+1}{2}}/n!$ and is nonzero. Hence $\deg T_{x_0} = n$. $\square$

Remark 6.11. The same computation applies to Rvachev's bump on all of $(-1, 1)$, and not merely on the half where up is a translate of F . Let $n \geq 0$ and let $y_0 = a/2^n - 1$ with a odd and $0 < a < 2^{n+1}$, so that $y_0 \in (-1, 1)$. For $m \geq n$ the identity (68) gives $\text{up}^{(m)}(y_0) = 2^{\binom{m+1}{2}} \mathcal{F}(2^{m-n} a)$, whose argument is an even integer when $m > n$ and the odd integer a when $m = n$. Hence $\text{up}^{(m)}(y_0) = 0$ for $m > n$ while $\text{up}^{(n)}(y_0) = -\varepsilon_a 2^{\binom{n+1}{2}} \neq 0$, so the Taylor polynomial of up at y_0 has degree exactly n . This sharpening is what makes the nowhere-analyticity corollary effective rather than merely soft: a convergent power series centered at a level- n odd dyadic point would force up to be a polynomial of degree n on a whole interval, while every neighborhood of that point contains odd dyadic points of arbitrarily large level m , at which the m th derivative is nonzero by (77).

Corollary 6.12 (Nowhere analytic). *Neither F on $(0, 1)$ nor up on $(-1, 1)$ is real analytic at any point.*

Proof. Dyadic points are dense. If F were analytic at one point, the differential equations and translation/reflection identities would extend analyticity to an interval and then to dyadic points in that interval, contradicting the preceding proposition. The statement for up follows from the fold relation. $\square$

6.6 The exact analytic locus

The corollary above can be sharpened to an exact description of the analytic locus of each of the three functions. Two points are worth isolating in advance. The endpoints are not neutral: both 0 and 1 belong to the non-analytic set of F , and both -1 and 1 belong to the non-analytic set of up , even though each function is analytic arbitrarily close to them on the outside. And the right endpoint of the unit interval must be treated separately, because the fold identity that handles every other point degenerates there.

Theorem 6.13 (Analytic locus of F). *Let $x \in \mathbb{R}$. The bounded Fabius function is real analytic at x if and only if $x \notin [0, 1]$.*

Proof. If $x < 0$, then F vanishes identically on the neighbourhood $(-\infty, 0)$ of x by (1); if $x > 1$, then F equals 1 identically on the neighbourhood $(1, \infty)$ of x by (2). A function that is constant near a point is analytic at that point.

Conversely, let $x \in [0, 1]$. For $0 < x < 1$ this is the corollary above.

Suppose F were analytic at 0. Its Taylor coefficients at 0 are $F(0) = 0$ and $F^{(m)}(0)/m! = 0$ for $m \geq 1$ by (6), so the Taylor series at 0 is identically zero and analyticity would force F to vanish on a neighbourhood of the origin. That contradicts theorem 2.5(i), which gives $F(t) > 0$ for every $t > 0$.

Suppose F were analytic at 1. Its Taylor coefficients at 1 are $F(1) = 1$ and, again by (6), $F^{(m)}(1)/m! = 0$ for $m \geq 1$, so analyticity would force F to equal 1 on a neighbourhood of 1. That contradicts theorem 2.5(ii), which gives $F(t) < 1$ for every $t < 1$. $\square$

Remark 6.14. The right endpoint really does need its own argument. The fold (8) gives $F(y) = \text{up}(1 - y)$ for every $y \leq 1$, and for no $y > 1$: there $\text{up}(1 - y) = F(2 - y)$, which is smaller than $F(y) = 1$. So the reflection is an identity of germs at every point $y < 1$, and at $y = 1$ it is only a one-sided identity; non-analyticity of up at the origin therefore cannot be transported to F at 1 along it. The flatness argument above sidesteps the difficulty. The formal development takes the other available route and transports analyticity at 1 to analyticity at 0 along the global symmetry (15), which, unlike the fold, is a genuine identity of germs at every point of the line.

Corollary 6.15 (Analytic locus of up). *Rvachev's function is real analytic at x if and only if $|x| > 1$. In particular it is not analytic at either endpoint of its support.*

Proof. If $|x| > 1$, then up vanishes identically near x by theorem 2.6(iii). If $|x| < 1$, this is the corollary above. At $x = 1$, use that $\text{up}(y) = F(1 - y)$ for all $y > 0$, hence on a neighbourhood of 1; therefore $\text{up}(1) = F(0) = 0$ and $\text{up}^{(m)}(1) = (-1)^m F^{(m)}(0) = 0$ for $m \geq 1$ by (6). Analyticity at 1 would force up to vanish on a neighbourhood of 1, contradicting theorem 2.6(iii). Evenness gives the same conclusion at $x = -1$. $\square$

The statement of the corollary on the closed interval $[-1, 1]$ is the unnumbered corollary following equation (24) of [8]; non-analyticity in the interior is Fabius's original point [2].

Theorem 6.16 (Analytic locus of the signed extension). *The signed extension $\mathcal{F}$ is real analytic at x if and only if $x < 0$. Equivalently, it is analytic at no point of $[0, \infty)$.*

Proof. For $x < 0$ the extension vanishes identically on $(-\infty, 0)$, hence is analytic at x .

Let $x \geq 0$ and suppose first that x is not an even integer. Put $b = \lfloor x/2 \rfloor$, so that $2b < x < 2b + 2$. On that open block, (12) gives $\mathcal{F}(y) = \varepsilon_b \text{up}(y - 2b - 1)$, an identity of germs at x . If $\mathcal{F}$ were analytic at x , then up would be analytic at $x - 2b - 1 \in (-1, 1)$, contradicting theorem 6.15.

Suppose now that $x = 2b$ with b a nonnegative integer. Evaluating (12) at the left endpoint of the block $[2a, 2a + 2]$ shows that $\mathcal{F}(2a) = \varepsilon_a \text{up}(-1) = 0$ for every nonnegative integer a ; that is, $\mathcal{F}$ vanishes at every nonnegative even integer. By (67),

$$\mathcal{F}^{(m)}(2b) = 2^{\binom{m+1}{2}} \mathcal{F}(2^m \cdot 2b) = 2^{\binom{m+1}{2}} \mathcal{F}(2^{m+1}b) = 0$$

for every $m \geq 0$, because $2^{m+1}b$ is again a nonnegative even integer. So all Taylor coefficients of $\mathcal{F}$ at $2b$ vanish, and analyticity would force $\mathcal{F}$ to vanish on a neighbourhood of $2b$. But (12) and theorem 2.6(iii) give $\mathcal{F}(y) = \pm \text{up}(y - 2b - 1) \neq 0$ for every y with $2b < y < 2b + 2$. $\square$

Warning 6.17. It is tempting to compress the last theorem into the slogan that the signed extension is nowhere analytic. That is false. The extension vanishes identically on $(-\infty, 0]$ and is therefore analytic at every negative point. What is true is that its analytic locus is exactly the open half-line $(-\infty, 0)$, so that, in contrast with F , whose non-analytic set is the compact interval $[0, 1]$, the extension fails to be analytic on an unbounded set. The origin lies in the non-analytic set of both.

The three loci therefore are $\mathbb{R} \setminus [0, 1]$ for F , $\mathbb{R} \setminus [-1, 1]$ for up , and $(-\infty, 0)$ for $\mathcal{F}$. In every case the boundary points of the interval belong to the non-analytic set, and in every case the reason is the same: the function is flat to infinite order there while being nonconstant on one side. This is precisely the mechanism by which a C^∞ function fails to be analytic, and the Fabius function is the historical example of it [2]. The formal development records the analytic locus of F exactly, the non-analyticity of up at every point of the closed interval $[-1, 1]$, and the extension case on the first block $[0, 2)$; the block induction just given carries the last statement to the whole half-line.

7 Moments and generating functions

Exact arithmetic enters through two rational moment sequences. One is attached to the even moments of up ; the other records its one-sided moments and, at the same time, the values of F at reciprocal powers of two.

7.1 The even moments c_n

Recall

$$c_n = \int_{-1}^1 x^{2n} \text{up}(x) dx, \quad c_0 = 1.$$

Theorem 7.1 (Moment recurrence). *For $n \geq 1$,*

$$(2n+1)(2^{2n}-1)c_n = \sum_{k=0}^{n-1} \binom{2n+1}{2k} c_k. \quad (79)$$

Consequently every c_n is rational and positive.

Proof. Insert the power series (63) into (56). Since

$$\text{sinc}(\pi z) = \sum_{j \geq 0} \frac{(-1)^j (\pi z)^{2j}}{(2j+1)!}$$

and

$$\widehat{\text{up}}(z/2) = \sum_{k \geq 0} \frac{(-1)^k c_k (\pi z)^{2k}}{(2k)!},$$

comparison of the coefficient of $(-1)^n (\pi z)^{2n}$ gives

$$\frac{2^{2n} c_n}{(2n)!} = \sum_{k=0}^n \frac{c_k}{(2k)!(2n-2k+1)!}.$$

Multiplying by $(2n+1)!$ yields

$$(2n+1)2^{2n} c_n = \sum_{k=0}^n \binom{2n+1}{2k} c_k.$$

The term $k = n$ is $(2n+1)c_n$; moving it to the left gives (79). Positivity follows either from the integral definition or by induction from the recurrence. $\square$

The first values are

$$1, \quad \frac{1}{9}, \quad \frac{19}{675}, \quad \frac{583}{59535}, \quad \frac{132809}{32531625}, \dots \quad (80)$$

A useful denominator clearing is

$$c_n = \frac{C_n}{(2n+1)!! \prod_{j=1}^n (2^{2j}-1)}, \quad (81)$$

where C_n is a positive integer. It can be defined without division by

$$C_0 = 1, \quad (82)$$

$$C_{n+1} = \sum_{k=0}^n C_k \binom{2n+3}{2k} \prod_{j=k+1}^n (2j+1) \prod_{j=k+1}^n (2^{2j}-1). \quad (83)$$

The formal development proves directly that every C_n is odd. Modulo 2, all factors in the two products are 1, and Lucas's theorem [13] shows that among the coefficients $\binom{2n+3}{2k}$ with $0 \leq k \leq n$ an odd number are odd. This closes the induction.

7.2 The half moments d_n

Define

$$d_0 = 1, \quad d_n = n \int_0^1 t^{n-1} \text{up}(t) dt \quad (n \geq 1). \quad (84)$$

Introduce the entire exponential generating function

$$G(z) = 1 + z \int_0^1 \text{up}(t) e^{zt} dt = \sum_{n=0}^{\infty} d_n \frac{z^n}{n!}. \quad (85)$$

Theorem 7.2 (Generating-function equation). *For every $z \in \mathbb{C}$,*

$$G(2z) = \frac{e^z - 1}{z} G(z), \quad (86)$$

where the quotient has its removable value 1 at $z = 0$.

Proof. One may derive the identity directly from (28) by integration by parts. An especially transparent route uses Fourier analysis. From evenness and the fold relation,

$$G(z) = e^{z/2} \widehat{\text{up}}\left(\frac{iz}{4\pi}\right). \quad (87)$$

Applying (56) at $iz/(2\pi)$ and simplifying $\text{sinc}(iz/2) = 2 \sinh(z/2)/z$ gives

$$G(2z) = e^z \frac{2 \sinh(z/2)}{z} \widehat{\text{up}}\left(\frac{iz}{4\pi}\right) = \frac{e^z - 1}{z} G(z).$$

□

Corollary 7.3 (Half-moment recurrence). *For $n \geq 1$,*

$$(n+1)(2^n - 1)d_n = \sum_{k=0}^{n-1} \binom{n+1}{k} d_k. \quad (88)$$

In particular $d_n \in \mathbb{Q}$.

Proof. Expand $(e^z - 1)/z = \sum_{j \geq 0} z^j/(j+1)!$ in (86). Comparison of the coefficient of $z^n/n!$ gives

$$2^n d_n = \frac{1}{n+1} \sum_{k=0}^n \binom{n+1}{k} d_k.$$

Isolating the $k = n$ term proves the recurrence. □

The first values are

$$1, \quad \frac{1}{2}, \quad \frac{5}{18}, \quad \frac{1}{6}, \quad \frac{143}{1350}, \quad \frac{19}{270}, \quad \frac{1153}{23814}, \dots \quad (89)$$

As for c_n , one may clear denominators without division:

$$d_n = \frac{D_n}{(n+1)! \prod_{j=1}^n (2^j - 1)}, \quad (90)$$

where $D_n \in \mathbb{N}$. A division-free recurrence follows by substituting (90) into (88). The normalized parity theorem says that, for $n \geq 1$, the reduced numerator and denominator of $2d_n$ are both odd.

7.3 Relation between the two moment sequences

Expanding (87) in powers of z gives a finite binomial transform at each order.

Proposition 7.4. *For every $n \geq 0$,*

$$d_n = 2^{-n} \sum_{0 \leq k \leq n/2} \binom{n}{2k} c_k. \quad (91)$$

Proof. By (63),

$$\widehat{\text{up}}\left(\frac{iz}{4\pi}\right) = \sum_{k \geq 0} c_k \frac{(z/2)^{2k}}{(2k)!}.$$

Multiplying by $e^{z/2}$ and extracting the coefficient of $z^n/n!$ gives exactly (91). $\square$

7.3.1 Integer numerators and a divisibility bridge

The normalizations (81) and (90) produce two integer sequences, and the relation (91) between the moments becomes an exact divisibility statement between them. Throughout, empty products equal 1 and $(-1)!! = 1$.

Re-indexing (83) by $n \mapsto n-1$ puts the division-free recurrence for C_n in the form used below: $C_0 = 1$ and, for $n \geq 1$,

$$C_n = \sum_{k=0}^{n-1} C_k \binom{2n+1}{2k} \prod_{j=k+1}^{n-1} (2j+1) \prod_{j=k+1}^{n-1} (2^{2j} - 1). \quad (92)$$

The primary text records only that a companion recurrence for D_n follows by substitution. It is the following.

Proposition 7.5 (Division-free recurrence for D_n). *Set $D_0 = 1$. For every $n \geq 1$,*

$$D_n = \sum_{k=0}^{n-1} D_k \binom{n+1}{k} \frac{n!}{(k+1)!} \prod_{j=k+1}^{n-1} (2^j - 1). \quad (93)$$

Every factor on the right is a positive integer, so $D_n \in \mathbb{N}$ for all n .

Proof. Substitute (90) into (88). The left side becomes

$$(n+1)(2^n - 1) \frac{D_n}{(n+1)! \prod_{j=1}^n (2^j - 1)} = \frac{D_n}{n! \prod_{j=1}^{n-1} (2^j - 1)},$$

and the right side becomes $\sum_{k=0}^{n-1} \binom{n+1}{k} D_k [(k+1)! \prod_{j=1}^k (2^j - 1)]^{-1}$. Multiplying through by $n! \prod_{j=1}^{n-1} (2^j - 1)$ gives (93). Since $k \leq n-1$, the quotient $n!/(k+1)!$ is a product of consecutive integers and hence an integer, and the remaining product is an integer as well. Induction from $D_0 = 1$ proves integrality and positivity. $\square$

The first values are

n	C_n	D_n
0	1	1
1	1	1
2	19	5
3	2915	84
4	2788989	4004
5	14754820185	494760
6	402830065455939	150120600
7	—	107969547840

Substituting (90) into (102) clears all denominators of the reciprocal-power values at once:

$$F(2^{-n}) = \frac{2^{-\binom{n}{2}} D_n}{n! (n+1)! \prod_{j=1}^n (2^j - 1)}. \quad (94)$$

This is the exact closed form behind (109): the entire dependence on n beyond elementary factors is carried by the single integer D_n .

The two integer sequences are linked through the odd half moments. The identity responsible is never displayed in the primary text, so we record it.

Lemma 7.6. *For every $k \geq 0$,*

$$d_{2k+1} = \frac{2k+1}{2} c_k. \quad (95)$$

Proof. By (84), $d_{2k+1} = (2k+1) \int_0^1 t^{2k} \text{up}(t) dt$. The integrand is even, so $\int_0^1 t^{2k} \text{up}(t) dt = \frac{1}{2} \int_{-1}^1 t^{2k} \text{up}(t) dt = \frac{1}{2} c_k$ by (62). $\square$

Theorem 7.7 (Bridge between the integer numerators). *For every $n \geq 0$,*

$$\frac{D_{2n+1}}{2n+1} = 2^n (n+1)! C_n \prod_{k=0}^n (2^{2k+1} - 1). \quad (96)$$

In particular $(2n+1)C_n$ divides D_{2n+1} .

Proof. By (90) and (95),

$$D_{2n+1} = d_{2n+1} (2n+2)! \prod_{j=1}^{2n+1} (2^j - 1) = \frac{2n+1}{2} c_n (2n+2)! \prod_{j=1}^{2n+1} (2^j - 1).$$

Insert (81) and divide by $2n+1$:

$$\frac{D_{2n+1}}{2n+1} = \frac{C_n}{2} \cdot \frac{(2n+2)!}{(2n+1)!!} \cdot \frac{\prod_{j=1}^{2n+1} (2^j - 1)}{\prod_{j=1}^n (2^{2j} - 1)}.$$

The last quotient retains exactly the factors with odd index, that is $\prod_{k=0}^n (2^{2k+1} - 1)$, since the even indices $j = 2k$ with $1 \leq k \leq n$ are precisely those canceled by the denominator. For the factorial quotient, separate the even factors of $(2n+2)!$:

$$(2n+2)! = [2 \cdot 4 \cdots (2n+2)] [1 \cdot 3 \cdots (2n+1)] = 2^{n+1} (n+1)! (2n+1)!!,$$

so $(2n+2)!/(2n+1)!! = 2^{n+1}(n+1)!$. Combining the two evaluations gives

$$\frac{D_{2n+1}}{2n+1} = \frac{C_n}{2} \cdot 2^{n+1}(n+1)! \prod_{k=0}^n (2^{2k+1} - 1),$$

which is (96). The right side is an integer multiple of C_n , so $D_{2n+1} = (2n+1)C_n \cdot 2^n(n+1)! \prod_{k=0}^n (2^{2k+1} - 1)$ and the divisibility follows. $\square$

Remark 7.8 (Independent numerical verification). Every numeric value printed in this subsection was recomputed independently in exact rational arithmetic: the moments were generated from (79) and (88), the numerators were obtained both by clearing denominators in (81) and (90) and by running the division-free recurrences (92) and (93), and the two computations were compared entry by entry. The closed form (94), the identity (95), the bridge (96), and the divisibility $(2n+1)C_n \mid D_{2n+1}$ were checked for all indices $n \leq 20$. For instance, at $n = 2$, $D_5/5 = 98952 = 2^2 \cdot 3! \cdot C_2 \cdot (2-1)(2^3-1)(2^5-1)$ with $C_2 = 19$, and $5 \cdot 19 = 95$ divides $D_5 = 494760$.

7.4 Bernoulli-number recurrences

The two refinement equations can be inverted before coefficients are compared. Doing so replaces the binomial convolutions of (79) and (88) by recurrences whose coefficients are Bernoulli numbers.

Throughout this subsection B_k denotes the Bernoulli numbers in the convention

$$\frac{z}{e^z - 1} = \sum_{k \geq 0} B_k \frac{z^k}{k!}, \quad B_1 = -\frac{1}{2}, \quad (97)$$

so that $B_0 = 1$, $B_2 = \frac{1}{6}$, $B_4 = -\frac{1}{30}$, $B_6 = \frac{1}{42}$, and $B_{2m+1} = 0$ for $m \geq 1$. The sign of B_1 is not cosmetic here. It is what produces the isolated first-order term in (101); the opposite convention $B_1 = +\frac{1}{2}$ reverses that term's sign and destroys the recurrence.

Rewrite (56) and (86) by moving the refinement factor to the other side:

$$\frac{\pi z}{\sin \pi z} \widehat{\text{up}}(z) = \widehat{\text{up}}(z/2), \quad \frac{z}{e^z - 1} G(2z) = G(z). \quad (98)$$

Both reciprocal factors have classical Bernoulli expansions. The second is (97) itself, and the first comes from

$$\frac{w}{\sin w} = \sum_{k \geq 0} (-1)^k (2 - 2^{2k}) B_{2k} \frac{w^{2k}}{(2k)!}, \quad |w| < \pi, \quad (99)$$

evaluated at $w = \pi z$. The first two terms of (99) are $1 + w^2/6$, as they must be.

Theorem 7.9 (Bernoulli recurrences). *For every $n \geq 1$,*

$$c_n = \frac{1}{2^{2n} - 1} \sum_{k=1}^n 2^{2n-2k} (2^{2k} - 2) \binom{2n}{2k} B_{2k} c_{n-k}, \quad (100)$$

and

$$d_n = \frac{n 2^{n-2}}{2^n - 1} d_{n-1} - \frac{1}{2^n - 1} \sum_{k=1}^{\lfloor n/2 \rfloor} \binom{n}{2k} 2^{n-2k} B_{2k} d_{n-2k}. \quad (101)$$

In (101) the sum is empty for $n = 1$.

Proof. For the even moments, put $w = \pi z$. By (63),

$$\widehat{\text{up}}(z) = \sum_{n \geq 0} (-1)^n c_n \frac{(2w)^{2n}}{(2n)!}, \quad \widehat{\text{up}}(z/2) = \sum_{n \geq 0} (-1)^n c_n \frac{w^{2n}}{(2n)!}.$$

Multiply the first series by (99) and use the left identity in (98). Both sides are entire in w , so their coefficients agree. The coefficient of w^{2n} on the left is

$$\begin{aligned} \sum_{k=0}^n (-1)^k (2 - 2^{2k}) B_{2k} \frac{1}{(2k)!} \cdot (-1)^{n-k} c_{n-k} \frac{2^{2n-2k}}{(2n-2k)!} \\ = \frac{(-1)^n}{(2n)!} \sum_{k=0}^n \binom{2n}{2k} (2 - 2^{2k}) 2^{2n-2k} B_{2k} c_{n-k}, \end{aligned}$$

and on the right it is $(-1)^n c_n / (2n)!$. Canceling the common factor $(-1)^n / (2n)!$ gives

$$\sum_{k=0}^n \binom{2n}{2k} (2 - 2^{2k}) 2^{2n-2k} B_{2k} c_{n-k} = c_n.$$

The term $k = 0$ equals $2^{2n} c_n$, because $B_0 = 1$ and $2 - 2^0 = 1$. Moving it to the right gives $-(2^{2n} - 1) c_n$ there, and reversing the sign of the bracket $2 - 2^{2k}$ in the remaining terms yields (100).

For the half moments, expand $G(2z) = \sum_{n \geq 0} 2^n d_n z^n / n!$ from (85) and multiply by (97). The right identity in (98) and comparison of the coefficient of $z^n / n!$ give

$$\sum_{m=0}^n \binom{n}{m} B_m 2^{n-m} d_{n-m} = d_n.$$

The term $m = 0$ is $2^n d_n$. The term $m = 1$ is $n B_1 2^{n-1} d_{n-1} = -n 2^{n-2} d_{n-1}$, by the convention (97). Every term with m odd and $m \geq 3$ vanishes, so the remaining terms are those with $m = 2k$ and $1 \leq k \leq \lfloor n/2 \rfloor$. Isolating d_n and dividing by $2^n - 1$ produces (101). $\square$

The two forms are equivalent to (79) and (88), since each determines its sequence uniquely from $c_0 = d_0 = 1$. The binomial recurrences are better for exact computation, because they involve only integers and one division; the Bernoulli forms are better for asymptotic estimates, because the size of B_{2k} is known explicitly.

Remark 7.10 (The first values, checked against (80) and (89)). For $n = 1$, (100) reads $3c_1 = \binom{2}{2} 2^0 (2^2 - 2) B_2 c_0 = 2B_2 = \frac{1}{3}$, so $c_1 = \frac{1}{9}$. For $n = 2$,

$$15c_2 = \binom{4}{2} 2^2 (2^2 - 2) B_2 c_1 + \binom{4}{4} 2^0 (2^4 - 2) B_4 c_0 = \frac{8}{9} - \frac{7}{15} = \frac{19}{45},$$

so $c_2 = \frac{19}{675}$. For $n = 1$ the sum in (101) is empty and $d_1 = \frac{1}{2}$. For $n = 2$, $d_2 = \frac{2}{3} d_1 - \frac{1}{3} \binom{2}{2} 2^0 B_2 d_0 = \frac{1}{3} - \frac{1}{18} = \frac{5}{18}$. For $n = 3$, $d_3 = \frac{6}{7} d_2 - \frac{1}{7} \binom{3}{2} 2^1 B_2 d_1 = \frac{5}{21} - \frac{1}{14} = \frac{1}{6}$. All five values agree with (80) and (89). An independent computation in exact rational arithmetic confirms that (100)–(101) reproduce (79)–(88) term by term for all $n \leq 20$.

8 Exact values at reciprocal powers of two

The half moments are already endpoint values in disguise.

Theorem 8.1 (Reciprocal-power values). *For every $n \geq 0$,*

$$F(2^{-n}) = \frac{d_n}{n! 2^{\binom{n}{2}}}. \quad (102)$$

Equivalently,

$$d_n = n! 2^{\binom{n}{2}} F(2^{-n}). \quad (103)$$

Proof. The case $n = 0$ is $F(1) = d_0 = 1$, so assume $n \geq 1$. The whole argument is a single chain of primitives supplied by (28). For $r \geq 0$ set

$$f_r(t) = 2^{\binom{r}{2}} \text{up}\left(\frac{t}{2^r} - \frac{1}{2^r} - \frac{1}{2^{r-1}} - \cdots - \frac{1}{2}\right), \quad (104)$$

where the subtracted sum is empty for $r = 0$, so that $f_0 = \text{up}$. Since $\sum_{j=1}^r 2^{-j} = 1 - 2^{-r}$, the shift collapses and

$$f_r(t) = 2^{\binom{r}{2}} \text{up}((t+1)2^{-r} - 1). \quad (105)$$

The chain has two properties on $[-1, 1]$. First, $f_r(-1) = 2^{\binom{r}{2}} \text{up}(-1) = 0$ for every r . Second,

$$f'_{r+1} = f_r \quad \text{on } [-1, 1]. \quad (106)$$

To see (106), differentiate (105) at index $r+1$ and write $x = (t+1)2^{-(r+1)} - 1$, so that $2x+1 = (t+1)2^{-r} - 1$ and $2x-1 = (t+1)2^{-r} - 3$. For $t \in [-1, 1]$ and $r \geq 0$ we have $0 \leq (t+1)2^{-r} \leq 2$, hence $2x-1 \leq -1$; since up is supported on $[-1, 1]$ and $\text{up}(-1) = 0$, the second term of (28) vanishes and $\text{up}'(x) = 2 \text{up}(2x+1)$. Therefore

$$f'_{r+1}(t) = 2^{\binom{r+1}{2}} 2^{-(r+1)} \text{up}'(x) = 2^{\binom{r+1}{2}-r} \text{up}((t+1)2^{-r} - 1) = f_r(t),$$

because $\binom{r+1}{2} - r = \binom{r}{2}$. The exponent bookkeeping is exactly what makes the chain close.

Put $J_n = \int_{-1}^0 t^{n-1} \text{up}(t) dt$. Substituting $t \mapsto -t$ and using evenness of up gives

$$\int_0^1 t^{n-1} \text{up}(t) dt = (-1)^{n-1} J_n. \quad (107)$$

Integrate J_n by parts $n-1$ times, replacing f_j by f'_{j+1} at each step. The boundary contribution at $t = -1$ vanishes at every step because $f_{j+1}(-1) = 0$, and the contribution at $t = 0$ vanishes because the surviving polynomial factor t^{n-1-j} still has positive degree for $0 \leq j \leq n-2$. Hence

$$J_n = (-1)^{n-1} (n-1)! \int_{-1}^0 f_{n-1}(t) dt = (-1)^{n-1} (n-1)! (f_n(0) - f_n(-1)) = (-1)^{n-1} (n-1)! f_n(0),$$

where the last integral was evaluated by (106) once more. By (105), evenness, and (8),

$$f_n(0) = 2^{\binom{n}{2}} \text{up}(2^{-n} - 1) = 2^{\binom{n}{2}} \text{up}(1 - 2^{-n}) = 2^{\binom{n}{2}} F(2^{-n}).$$

Combining this with (107) makes the two signs cancel and yields

$$\int_0^1 t^{n-1} \text{up}(t) dt = (n-1)! 2^{\binom{n}{2}} F(2^{-n}). \quad (108)$$

Multiplying by n and using (84) gives (103), and dividing gives (102). $\square$

The first values are therefore

n	$F(2^{-n})$	
0	1	
1	$\frac{1}{2}$	
2	$\frac{5}{72}$	
3	$\frac{288}{143}$	
4	$\frac{2073600}{19}$	
5	$\frac{33177600}{1153}$	
6	$\frac{561842749440}{583}$	
7	$\frac{179789679820800}{1}$	(109)

The denominators grow far faster than exponentially, in agreement with the quadratic leading term in the logarithmic asymptotic derived later.

8.1 A triangular recurrence and its nonrecursive solution

The values in (109) may be generated without first computing the moments. Write

$$V_n = F(2^{-n}). \quad (110)$$

Eliminating d_n between (88) and (102) gives, for $n \geq 1$,

$$V_n = \frac{2^{-\binom{n}{2}}}{2^n - 1} \sum_{k=0}^{n-1} \frac{2^{\binom{k}{2}}}{(n-k+1)!} V_k, \quad V_0 = 1. \quad (111)$$

The restriction $n \geq 1$ is essential: at $n = 0$ the denominator $2^n - 1$ is zero and the sum is empty, whereas $V_0 = F(1) = 1$. Thus the initial value is part of the recurrence data, not a removable endpoint convention.

The triangular powers of two disappear after the normalization

$$A_n = 2^{\binom{n}{2}} V_n. \quad (112)$$

Indeed, $A_0 = 1$ and

$$A_n = \sum_{k=0}^{n-1} \frac{A_k}{(2^n - 1)(n - k + 1)!}. \quad (113)$$

This is a weighted-path recurrence on the directed graph with one edge $k \rightarrow n$ whenever $k < n$. Iterating it terminates, and the resulting paths can be indexed by ordered compositions.

The recurrence is triangular: once $A_0 = 1$ is fixed, each A_n is a linear combination of earlier values. Induction therefore gives uniqueness. In particular, the argument below is purely algebraic and applies to any sequence satisfying (113); no further analytic property of F is used.

An ordered composition of n is a tuple $\rho = (r_1, \dots, r_m)$ of positive integers with sum n . Put

$$s_j = r_1 + \dots + r_j, \quad W(\rho) = \prod_{j=1}^m \frac{1}{(2^{s_j} - 1)(r_j + 1)!}. \quad (114)$$

For $n = 0$ there is one empty composition; its empty product is 1.

Theorem 8.2 (Finite composition formula). *For every $n \geq 0$,*

$$F(2^{-n}) = 2^{-\binom{n}{2}} \sum_{\rho \models n} \prod_{j=1}^{\text{len}(\rho)} \frac{1}{(2^{s_j} - 1)(r_j + 1)!} \quad (115)$$

where $\rho = (r_1, \dots, r_m)$ and $s_j = r_1 + \dots + r_j$. For $n = 0$ the unique empty composition makes the right side equal to 1.

Proof. Let $\Sigma_n = \sum_{\rho \models n} W(\rho)$. The empty-composition convention gives $\Sigma_0 = 1$. Fix $n \geq 1$ and split every composition of n immediately before its last block. If the preceding blocks sum to k , then $0 \leq k < n$, the last block is $n - k$, and the last factor of (114) is

$$\frac{1}{(2^n - 1)(n - k + 1)!}.$$

Conversely, appending the block $n - k$ to any composition of k gives exactly one composition of n with penultimate partial sum k . Therefore

$$\Sigma_n = \sum_{k=0}^{n-1} \frac{\Sigma_k}{(2^n - 1)(n - k + 1)!}.$$

Thus (Σ_n) and (A_n) have the same initial value and the same triangular recurrence. Induction gives $\Sigma_n = A_n$, and multiplying by $2^{-\binom{n}{2}}$ proves (115). $\square$

Every point 2^{-n} lies in $[0, 1]$, where the bounded Fabius function and the signed global extension agree. Consequently the same formula holds with $F(2^{-n})$ replaced by $\mathcal{F}(2^{-n})$.

Equivalently, put $i_0 = 0$ and $i_j = s_j$. Compositions of length m correspond bijectively to chains

$$0 = i_0 < i_1 < \dots < i_m = n, \quad r_j = i_j - i_{j-1}.$$

Hence, for $n \geq 1$,

$$F(2^{-n}) = 2^{-\binom{n}{2}} \sum_{m=1}^n \sum_{0=i_0 < i_1 < \dots < i_m=n} \prod_{j=1}^m \frac{1}{(2^{i_j} - 1)(i_j - i_{j-1} + 1)!}. \quad (116)$$

There are 2^{n-1} compositions of a positive integer n , so this is finite and nonrecursive, although exponential in size. For fixed m , the inner sum may instead be written over $1 \leq i_1 < \dots < i_{m-1} \leq n - 1$, with $i_0 = 0$ and $i_m = n$; when $m = 1$ there are no internal indices and there is one direct path $0 \rightarrow n$.

The common factor $2^{-\binom{n}{2}}$ has a direct path interpretation. Before normalization, the edge $k \rightarrow r$ has weight

$$U_{r,k} = \frac{2^{\binom{k}{2} - \binom{r}{2}}}{(2^r - 1)(r - k + 1)!}. \quad (117)$$

Along a chain $0 = i_0 < i_1 < \dots < i_m = n$, the powers of two telescope:

$$\prod_{j=1}^m 2^{\binom{i_{j-1}}{2} - \binom{i_j}{2}} = 2^{\binom{i_0}{2} - \binom{i_m}{2}} = 2^{-\binom{n}{2}}. \quad (118)$$

The remaining factors are precisely the normalized chain weight in (116). Thus normalization extracts an endpoint factor common to every path, rather than merely shortening the recurrence.

Example 8.3 (The first nontrivial path sums). For $n = 2$, the compositions are (2) and $(1, 1)$, with weights

$$W(2) = \frac{1}{(2^2 - 1)3!} = \frac{1}{18}, \quad W(1, 1) = \frac{1}{(2^1 - 1)2!} \frac{1}{(2^2 - 1)2!} = \frac{1}{12}.$$

Hence $A_2 = 5/36$ and $F(2^{-2}) = 2^{-1}A_2 = 5/72$. For $n = 3$, the four ordered compositions (3) , $(1, 2)$, $(2, 1)$, $(1, 1, 1)$ have weights

$$\frac{1}{168}, \quad \frac{1}{84}, \quad \frac{1}{252}, \quad \frac{1}{168}.$$

The unequal weights of $(1, 2)$ and $(2, 1)$ record their different intermediate partial sums. Their total is $A_3 = 1/36$, giving $F(2^{-3}) = 2^{-3}A_3 = 1/288$.

For a precise matrix form, fix N and define a matrix indexed by $0, \dots, N$ by

$$T_{i,j}^{[N]} = \begin{cases} \frac{1}{(2^i - 1)(i - j + 1)!}, & j < i, \\ 0, & j \geq i. \end{cases}$$

With $\mathbf{A} = (A_0, \dots, A_N)^\top$ and $\mathbf{e}_0 = (1, 0, \dots, 0)^\top$, the initial value and recurrence become

$$\mathbf{A} = \mathbf{e}_0 + T^{[N]} \mathbf{A}.$$

Since $T^{[N]}$ is strictly lower triangular, $(T^{[N]})^{N+1} = 0$ and

$$\mathbf{A} = (I - T^{[N]})^{-1} \mathbf{e}_0 = (I + T^{[N]} + \dots + (T^{[N]})^N) \mathbf{e}_0.$$

The entry $(T^{[N]})_{n,0}^m$ is the sum of products of edge weights over all length- m increasing paths from 0 to n , so its n th component is exactly (116). This is a finite geometric identity for a nilpotent matrix, not an analytic convergence argument. Finally, if

$$D = \text{diag}(2^{\binom{0}{2}}, 2^{\binom{1}{2}}, \dots, 2^{\binom{N}{2}}),$$

then the matrix of the original recurrence is $U = D^{-1}T^{[N]}D$. This diagonal similarity is the matrix form of (118).

Corollary 8.4 (Positivity). *For every $n \geq 0$, $F(2^{-n}) > 0$; equivalently $A_n > 0$.*

Proof. For $n = 0$ the single empty composition contributes the empty product 1. Let $n \geq 1$. Every partial sum s_j of a composition of n satisfies $s_j \geq 1$, so $2^{s_j} - 1 \geq 1 > 0$, and every factorial $(r_j + 1)!$ is positive. Hence each weight (114) is a positive rational, the sum (115) has at least the one term $\rho = (n)$, and the factor $2^{-\binom{n}{2}}$ is positive. $\square$

The argument uses no property of F beyond the recurrence: no integral representation, no probabilistic interpretation, no monotonicity, no continuity. Positivity of the reciprocal-power values is a statement about a triangular linear system with positive coefficients. Combined with monotonicity of F it recovers $F(x) > 0$ for $0 < x \leq 1$, but the corollary itself is purely arithmetic, and it is the step that shows the numerators in (109) cannot vanish.

The worked example above stops at $n = 3$. The first case in which compositions of three distinct lengths occur is $n = 4$.

Example 8.5 (The eight compositions of 4). There are $2^3 = 8$ ordered compositions of 4. Their partial sums and weights (114) are

ρ	$(s_1, \dots, s_m)$	$W(\rho)$
(4)	(4)	$\frac{1}{(2^4 - 1)5!} = \frac{1}{1800}$
(1, 3)	(1, 4)	$\frac{1}{(2^1 - 1)2!} \cdot \frac{1}{(2^4 - 1)4!} = \frac{1}{720}$
(2, 2)	(2, 4)	$\frac{1}{(2^2 - 1)3!} \cdot \frac{1}{(2^4 - 1)3!} = \frac{1}{1620}$
(3, 1)	(3, 4)	$\frac{1}{(2^3 - 1)4!} \cdot \frac{1}{(2^4 - 1)2!} = \frac{1}{5040}$
(1, 1, 2)	(1, 2, 4)	$\frac{1}{(2^1 - 1)2!} \cdot \frac{1}{(2^2 - 1)2!} \cdot \frac{1}{(2^4 - 1)3!} = \frac{1}{1080}$
(1, 2, 1)	(1, 3, 4)	$\frac{1}{(2^1 - 1)2!} \cdot \frac{1}{(2^3 - 1)3!} \cdot \frac{1}{(2^4 - 1)2!} = \frac{1}{2520}$
(2, 1, 1)	(2, 3, 4)	$\frac{1}{(2^2 - 1)3!} \cdot \frac{1}{(2^3 - 1)2!} \cdot \frac{1}{(2^4 - 1)2!} = \frac{1}{7560}$
(1, 1, 1, 1)	(1, 2, 3, 4)	$\frac{1}{(2^1 - 1)2!} \cdot \frac{1}{(2^2 - 1)2!} \cdot \frac{1}{(2^3 - 1)2!} \cdot \frac{1}{(2^4 - 1)2!} = \frac{1}{5040}$

Their total is

$$A_4 = \frac{1}{1800} + \frac{1}{720} + \frac{1}{1620} + \frac{1}{5040} + \frac{1}{1080} + \frac{1}{2520} + \frac{1}{7560} + \frac{1}{5040} = \frac{143}{32400}, \quad (119)$$

and since $\binom{4}{2} = 6$,

$$F(2^{-4}) = 2^{-6} A_4 = \frac{143}{2073600},$$

which is the entry for $n = 4$ in (109). Note that (3, 1) and (1, 1, 1, 1) happen to share the weight $1/5040$; the coincidence is numerical, and the two compositions have different lengths and different chains. As at $n = 3$, the reversed pairs (1, 3) and (3, 1), and (1, 1, 2) and (2, 1, 1), carry different weights, because the Mersenne factors depend on the partial sums and not on the multiset of block sizes.

Both the eight individual weights and the total (119) were recomputed in exact rational arithmetic directly from (114), with the assignment of each weight to its composition checked term by term; the composition sum was also verified against the recurrence (113) for $0 \leq n \leq 7$.

Finally, the three presentations given in this subsection are not three theorems but one.

Proposition 8.6 (One object, three presentations). *Let $(V_n)_{n \geq 0}$ be a sequence in $\mathbb{Q}$ with $V_0 = 1$, and put $A_n = 2^{\binom{n}{2}} V_n$. The following are equivalent.*

- (i) (V_n) satisfies the triangular recurrence (111) for every $n \geq 1$.
- (ii) (V_n) is given by the composition sum (115) for every $n \geq 0$.
- (iii) For every $N \geq 0$, the vector $\mathbf{A} = (A_0, \dots, A_N)^\top$ satisfies $\mathbf{A} = (I - T^{[N]})^{-1} \mathbf{e}_0 = (I + T^{[N]} + \dots + (T^{[N]})^N) \mathbf{e}_0$.

Each of the three determines (V_n) uniquely, and the common value is $V_n = F(2^{-n})$.

Proof. Statement (i) is equivalent to the normalized recurrence (113), since $A_n = 2^{\binom{n}{2}} V_n$ and the powers of two cancel. Splitting a composition before its last block turns the composition sum into (113), so (ii) implies (i); conversely the recurrence is triangular, hence has a unique solution with the given initial value, and the composition sum is one such solution, so (i) implies (ii). For (iii), the identity $\mathbf{A} = \mathbf{e}_0 + T^{[N]} \mathbf{A}$ is exactly (113) written for the indices $0, \dots, N$ together with $A_0 = 1$; since $T^{[N]}$ is strictly lower triangular, $(T^{[N]})^{N+1} = 0$ and $I - T^{[N]}$ is invertible with the displayed nilpotent geometric inverse. Expanding the matrix powers gives the chain sum (116), which the

bijection between chains and compositions carries to (ii). The identification $V_n = F(2^{-n})$ is (102) together with (88). $\square$

This settles what “closed form” means here, and what it does not mean. Formula (115) is closed in the sense that it is nonrecursive: each summand is determined by the blocks of a single composition and by n alone, with no reference to earlier values of the sequence, and the matrix presentation is a finite polynomial in a nilpotent matrix rather than a limit. It is not closed in the sense of being cheap: the sum has 2^{n-1} terms, so as an evaluation procedure it is exponentially worse than the $\mathcal{O}(n^2)$ rational operations of the recurrence itself. Eliminating recursion and reducing arithmetic cost are different goals, and only the first is achieved here. The value of the composition form is structural — it exhibits positivity termwise, it displays the telescoping of the powers of two along paths as in (118), and it makes the edge weights (117) visible — while the recurrence and the bit recursion remain the instruments of computation.

8.2 An exact two-adic valuation

8.2.1 An exact count of odd binomial coefficients

The parity statements used above and below rest on a single counting lemma, which Lucas's theorem makes exact rather than merely qualitative.

Lemma 8.7 (Odd entries in row $2n + 1$). *Let $n \geq 1$. Among the coefficients $\binom{2n+1}{2k}$ with $0 \leq k \leq n$, exactly $2^{\text{wt}_2(n)}$ are odd. Among all the coefficients $\binom{2n+1}{k}$ with $0 \leq k \leq 2n + 1$, exactly $2^{\text{wt}_2(n)+1}$ are odd.*

Proof. Lucas's theorem modulo 2 [13] states that $\binom{N}{K}$ is odd exactly when every binary digit of K is at most the corresponding binary digit of N ; equivalently, when the set of positions of the 1-bits of K is contained in that of N .

The binary expansion of $N = 2n + 1$ is the expansion of n shifted up by one position, with a final digit 1 appended. Hence K satisfies the Lucas condition precisely when its lowest digit is arbitrary and its remaining digits, read as the binary expansion of $\lfloor K/2 \rfloor$, select a subset of the 1-bits of n . There are $2^{\text{wt}_2(n)}$ such subsets and two choices of the lowest digit, giving $2^{\text{wt}_2(n)+1}$ odd entries in the whole row. Restricting to even $K = 2k$ fixes the lowest digit to 0 and leaves $2^{\text{wt}_2(n)}$ odd entries; as K runs over the even integers in $[0, 2n + 1]$, the index k runs over $[0, n]$. $\square$

Corollary 8.8. *Let $n \geq 1$. The number of odd coefficients $\binom{2n+1}{2k}$ with $0 \leq k \leq n - 1$ is $2^{\text{wt}_2(n)} - 1$, and the number of odd coefficients $\binom{2n+1}{k}$ with $1 \leq k \leq 2n - 1$ is $2^{\text{wt}_2(n)+1} - 3$. Both numbers are odd.*

Proof. The omitted coefficient in the first count is $\binom{2n+1}{2n} = 2n + 1$, which is odd. The omitted coefficients in the second count are $\binom{2n+1}{0} = 1$, $\binom{2n+1}{2n} = 2n + 1$ and $\binom{2n+1}{2n+1} = 1$, all odd. Subtract from the two totals of Theorem 8.7. Since $n \geq 1$ forces $\text{wt}_2(n) \geq 1$, both $2^{\text{wt}_2(n)}$ and $2^{\text{wt}_2(n)+1}$ are even, so $2^{\text{wt}_2(n)} - 1$ and $2^{\text{wt}_2(n)+1} - 3$ are odd. $\square$

Proposition 8.9 (Oddness of the even-moment numerators). *Every C_n is odd.*

Proof. Induct on n , the case $C_0 = 1$ being immediate. Let $n \geq 1$ and assume $C_0, \dots, C_{n-1}$ are odd. In (92) every factor other than the binomial coefficient is odd: the entries $2j + 1$ are odd, each $2^{2j} - 1$ is odd, and each C_k is odd by hypothesis. Hence, modulo 2, the sum counts the odd coefficients $\binom{2n+1}{2k}$ with $0 \leq k \leq n - 1$. By Theorem 8.8 that count is $2^{\text{wt}_2(n)} - 1$, an odd number. Therefore C_n is odd. $\square$

We can now prove the parity statement for the half moments that was asserted after (90). Work in the local ring $\mathbb{Z}_{(2)}$ of rational numbers with odd denominator; its residue field is $\mathbb{Z}/2\mathbb{Z}$, and a rational number x lies in $\mathbb{Z}_{(2)}$ with $v_2(x) = 0$ exactly when its reduced numerator and denominator are both odd.

Theorem 8.10 (Normalized parity). *For every $n \geq 1$,*

$$v_2(2d_n) = 0. \quad (120)$$

Equivalently, the reduced numerator and denominator of $2d_n$ are both odd, and $v_2(d_n) = -1$.

Proof. For an odd index $n = 2k + 1$, combine (95) with (81):

$$2d_{2k+1} = (2k + 1)c_k = \frac{(2k + 1) C_k}{(2k + 1)!! \prod_{j=1}^k (2^{2j} - 1)}.$$

The numerator is odd by Theorem 8.9, and the denominator is a product of odd integers. Hence $v_2(2d_{2k+1}) = 0$. This covers every odd index, including $n = 1$.

For an even index, write (88) at $2n$ and multiply by 2:

$$(2n + 1)(2^{2n} - 1) 2d_{2n} = \sum_{k=0}^{2n-1} \binom{2n+1}{k} 2d_k. \quad (121)$$

Proceed by strong induction on $n \geq 1$, assuming $v_2(2d_k) = 0$ for $1 \leq k \leq 2n - 1$; the odd indices in that range are already settled unconditionally, and the even ones are covered by the induction hypothesis. Every term on the right of (121) then lies in $\mathbb{Z}_{(2)}$, so the right side does, and the prefactor $(2n + 1)(2^{2n} - 1)$ on the left is an odd integer and hence a unit of $\mathbb{Z}_{(2)}$. Solving for $2d_{2n}$ shows $2d_{2n} \in \mathbb{Z}_{(2)}$.

Reduce (121) modulo 2 in $\mathbb{Z}_{(2)}$. The prefactor is congruent to 1. The term $k = 0$ contributes $2d_0 = 2 \equiv 0$. Each term with $1 \leq k \leq 2n - 1$ has $2d_k$ a unit, so it is congruent to $\binom{2n+1}{k}$. Therefore

$$2d_{2n} \equiv \#\{1 \leq k \leq 2n - 1 : \binom{2n+1}{k} \text{ odd}\} = 2^{\text{wt}_2(n)+1} - 3 \equiv 1 \pmod{2},$$

using Theorem 8.8. Hence $v_2(2d_{2n}) = 0$, completing the induction. The final assertion is the definition of v_2 on $\mathbb{Q}^\times$ together with $v_2(d_n) = v_2(2d_n) - 1$. $\square$

Theorem 8.10 is exactly the input required by the valuation theorem that follows, which is therefore now proved outright rather than conditionally.

The parity theorem for d_n immediately determines the power of two in the reduced denominator.

Theorem 8.11 (Exact 2-adic valuation). *For $n \geq 1$,*

$$v_2(F(2^{-n})) = -\binom{n}{2} - 1 - v_2(n!). \quad (122)$$

Proof. The reduced numerator and denominator of $2d_n$ are odd, so $v_2(d_n) = -1$. Apply (102) and add valuations:

$$v_2(F(2^{-n})) = v_2(d_n) - v_2(n!) - \binom{n}{2}.$$

$\square$

Legendre's formula

$$v_2(n!) = n - \text{wt}_2(n) \quad (123)$$

then makes the binary dependence explicit.

8.3 A normalized odd integer

For $n \geq 1$ define

$$R_n = 2^{\binom{n-1}{2}} (2n)! F(2^{-n}) \prod_{j=1}^{\lfloor n/2 \rfloor} (2^{2j} - 1). \quad (124)$$

The exponent of two in (124) is *positive*; a negative sign that appears in an abstract of the source paper is a typographical error.

That these numbers are integers at all was not obvious. The question was raised by Reshetnikov [20] on the basis of numerical evidence, and answered affirmatively by Arias de Reyna [8], whose paper devotes a section to it.

Theorem 8.12 (Reshetnikov's integers). *For every $n \geq 1$ the number R_n of (124) is an odd positive integer. It has the half-moment form*

$$R_n = 2d_n (2n-1)!! \prod_{m=1}^{\lfloor n/2 \rfloor} (2^{2m} - 1), \quad (125)$$

the odd-index closed form

$$R_{2n+1} = C_n \frac{(4n+1)!!}{(2n-1)!!} \quad (n \geq 0), \quad (126)$$

and the even-index expansion

$$R_{2n} = \sum_{k=0}^n \frac{2C_k}{2^{2n}} \binom{2n}{2k} \frac{(4n-1)!!}{(2k+1)!!} \prod_{l=k+1}^n (2^{2l} - 1) \quad (n \geq 1). \quad (127)$$

Empty products equal 1 and $(-1)!! = 1$.

Proof. The half-moment form. Substitute (102) into (124):

$$R_n = 2^{\binom{n-1}{2} - \binom{n}{2}} \frac{(2n)!}{n!} d_n \prod_{m=1}^{\lfloor n/2 \rfloor} (2^{2m} - 1).$$

Now $\binom{n}{2} - \binom{n-1}{2} = n-1$, and separating the even factors of $(2n)!$ gives $(2n)! = 2^n n! (2n-1)!!$, hence $(2n)!/n! = 2^n (2n-1)!!$. The two powers of two combine into $2^{-(n-1)} \cdot 2^n = 2$, which is (125). Every factor after $2d_n$ in (125) is odd, so

$$v_2(R_n) = v_2(2d_n) = 0 \quad (128)$$

by Theorem 8.10.

Odd indices. Take $n \mapsto 2n+1$ in (125), note $\lfloor (2n+1)/2 \rfloor = n$, and use (95) and (81):

$$R_{2n+1} = (2n+1)c_n(4n+1)!! \prod_{m=1}^n (2^{2m} - 1) = (2n+1) \frac{C_n(4n+1)!!}{(2n+1)!!} = C_n \frac{(4n+1)!!}{(2n-1)!!},$$

since $(2n+1)!! = (2n+1)(2n-1)!!$. This is (126). The quotient $(4n+1)!!/(2n-1)!!$ is the product of the odd integers $2n+1, 2n+3, \dots, 4n+1$, so R_{2n+1} is a product of odd integers and is therefore an odd integer.

Even indices. Take $n \mapsto 2n$ in (125), insert the binomial transform (91), and replace each c_k by (81):

$$\begin{aligned} R_{2n} &= 2 \cdot 2^{-2n} \sum_{k=0}^n \binom{2n}{2k} c_k (4n-1)!! \prod_{m=1}^n (2^{2m} - 1) \\ &= \sum_{k=0}^n \frac{2C_k}{2^{2n}} \binom{2n}{2k} \frac{(4n-1)!!}{(2k+1)!!} \prod_{l=k+1}^n (2^{2l} - 1), \end{aligned}$$

because the product $\prod_{m=1}^k (2^{2m} - 1)$ in the denominator of c_k cancels the initial segment of $\prod_{m=1}^n (2^{2m} - 1)$. This is (127). Since $k \leq n \leq 2n - 1$ for $n \geq 1$, each quotient $(4n - 1)!! / (2k + 1)!!$ is a product of consecutive odd integers and hence an integer; the binomial coefficients, the C_k and the remaining products are integers as well. Consequently $2^{2n-1} R_{2n}$ is an integer, so the only prime that can occur in the denominator of R_{2n} is 2. By (128) that denominator is odd, so it equals 1 and R_{2n} is an integer, necessarily odd.

Positivity. Every factor in (124) is positive, and $F(2^{-n}) > 0$ because F is strictly increasing on $[0, 1]$. $\square$

Remark 8.13 (Extended value list). The first nine values are

$$1, 5, 15, 1001, 5985, 2853675, 26261235, 72808620885, 915304354965, \quad (129)$$

extending the table of Section A by $R_8 = 72808620885$ and $R_9 = 915304354965$. All nine were recomputed independently in exact rational arithmetic from the definition (124), from the half-moment form (125), and, according to the parity of the index, from (126) or (127); the three routes agree, and every value is odd. The check was carried out for all $n \leq 24$.

9 Exact evaluation at arbitrary dyadic rationals

The rationality theorem for dyadic arguments is much stronger than the statement “ F happens to take rational values there.” It comes with two complementary exact algorithms:

1. a closed finite formula involving Thue–Morse signs and the rational moments c_k ;
2. a bit-recursive Taylor algorithm whose number of recursive calls is the number of set bits in the numerator.

The first is best for proofs; the second is best for computation.

9.1 The closed Thue–Morse formula

Theorem 9.1 (Exact dyadic formula). *Let $n, a \in \mathbb{N}$ with $0 \leq a \leq 2^n$. Then*

$$F\left(\frac{a}{2^n}\right) = \frac{2^{-\binom{n+1}{2}}}{n!} \sum_{h=0}^{a-1} (-1)^{\text{wt}_2(h)} \sum_{k=0}^{\lfloor n/2 \rfloor} \binom{n}{2k} c_k (2a - 2h - 1)^{n-2k}. \quad (130)$$

An empty outer sum has value 0.

Proof. Write $x = a/2^n$ and use $F(x) = \text{up}(x - 1)$. Since up and all its derivatives vanish at -1 , Taylor's theorem with integral remainder gives

$$\text{up}(x - 1) = \frac{1}{n!} \int_{-1}^{x-1} (x - 1 - t)^n \text{up}^{(n+1)}(t) dt. \quad (131)$$

Insert the exact derivative identity (68):

$$\text{up}^{(n+1)}(t) = 2^{\binom{n+2}{2}} \mathcal{F}(2^{n+1}(t + 1)).$$

Put $s = 2^{n+1}(t + 1)$. As t runs from -1 to $x - 1$, s runs from 0 to $2a$. Split this interval into the blocks $[2h, 2h + 2]$, $0 \leq h < a$. On the h th block, (12) gives

$$\mathcal{F}(s) = (-1)^{\text{wt}_2(h)} \text{up}(s - 2h - 1).$$

The substitution $u = s - 2h - 1$ maps the block to $[-1, 1]$. Gathering the powers of two gives

$$F(a/2^n) = \frac{2^{-(\frac{n+1}{2})}}{n!} \sum_{h=0}^{a-1} (-1)^{\text{wt}_2(h)} \int_{-1}^1 (2a - 2h - 1 - u)^n \text{up}(u) \, du. \quad (132)$$

Expand the n th power. Odd powers of u integrate to zero because up is even, while

$$\int_{-1}^1 u^{2k} \text{up}(u) \, du = c_k.$$

The surviving even terms are exactly the inner sum in (130). $\square$

Every quantity on the right of (130) is rational, so dyadic rationality is immediate. More importantly, the formula lives entirely in $\mathbb{Q}$ and can be evaluated without roundoff.

Theorem 9.2 (Global dyadic formula). *For all $n, a \in \mathbb{N}$ the signed global extension satisfies*

$$\mathcal{F}\left(\frac{a}{2^n}\right) = \frac{2^{-(\frac{n+1}{2})}}{n!} \sum_{h=0}^{a-1} \varepsilon_h \sum_{k=0}^{\lfloor n/2 \rfloor} \binom{n}{2k} c_k (2a - 2h - 1)^{n-2k}. \quad (133)$$

No upper bound on a is imposed. An empty outer sum has value 0.

Proof. By (67) the m th derivative of $\mathcal{F}$ is $2^{(\frac{m+1}{2})} \mathcal{F}(2^m x)$, and $\mathcal{F}$ vanishes on $(-\infty, 0]$. Hence $\mathcal{F}^{(m)}(0) = 0$ for every $m \geq 0$, and Taylor's theorem with integral remainder at the base point 0, applied through order n , gives for $x \geq 0$

$$\mathcal{F}(x) = \frac{1}{n!} \int_0^x (x-t)^n \mathcal{F}^{(n+1)}(t) \, dt. \quad (134)$$

Put $x = a/2^n$ and insert $\mathcal{F}^{(n+1)}(t) = 2^{(\frac{n+2}{2})} \mathcal{F}(2^{n+1}t)$. The substitution $u = 2^{n+1}t$ carries $[0, x]$ onto $[0, 2a]$ and produces $(x-t)^n = 2^{-n(n+1)}(2a-u)^n$ together with $dt = 2^{-(n+1)} du$. Since

$$\binom{n+2}{2} - n(n+1) - (n+1) = -\binom{n+1}{2},$$

this is exactly

$$\mathcal{F}\left(\frac{a}{2^n}\right) = \frac{2^{-(\frac{n+1}{2})}}{n!} \int_0^{2a} (2a-u)^n \mathcal{F}(u) \, du. \quad (135)$$

Split $[0, 2a]$ into the a blocks $[2h, 2h+2]$ with $0 \leq h < a$. On the h th block (12) gives $\mathcal{F}(u) = \varepsilon_h \text{up}(u - 2h - 1)$, and the substitution $v = u - 2h - 1$ maps that block onto $[-1, 1]$:

$$\int_{2h}^{2h+2} (2a-u)^n \mathcal{F}(u) \, du = \varepsilon_h \int_{-1}^1 (2a - 2h - 1 - v)^n \text{up}(v) \, dv.$$

Expand the n th power by the binomial theorem. The function up is even, so every odd moment vanishes, while $\int_{-1}^1 v^{2k} \text{up}(v) \, dv = c_k$ and the surviving sign $(-1)^{2k}$ is $+1$. Summing over h gives (133). $\square$

Two special cases are the ones used in practice.

Corollary 9.3 (Bounded case). *If $0 \leq a \leq 2^n$ then $\mathcal{F}(a/2^n) = F(a/2^n)$, so (133) reduces to (130).*

Proof. The two functions agree on $[0, 1]$ and $a/2^n \in [0, 1]$. $\square$

Corollary 9.4 (Bump case). *Let $n \in \mathbb{N}$ and $q \in \mathbb{Z}$. If $|q| < 2^n$, then*

$$\text{up}\left(\frac{q}{2^n}\right) = \frac{2^{-\binom{n+1}{2}}}{n!} \sum_{h=0}^{q+2^n-1} \varepsilon_h \sum_{k=0}^{\lfloor n/2 \rfloor} \binom{n}{2k} c_k (2q + 2^{n+1} - 2h - 1)^{n-2k}, \quad (136)$$

that is, (133) evaluated at $a = q + 2^n$. If $|q| \geq 2^n$, then $\text{up}(q/2^n) = 0$.

Proof. Taking $b = 0$ in (12) gives $\mathcal{F}(x) = \text{up}(x-1)$ for $0 \leq x \leq 2$. With $x = 1 + q/2^n = (q + 2^n)/2^n$ and $|q| < 2^n$ we have $0 < x < 2$, so $\text{up}(q/2^n) = \mathcal{F}((q + 2^n)/2^n)$, and (133) with $a = q + 2^n$ applies. The last statement is the support condition $\text{supp up} = [-1, 1]$. $\square$

Corollary 9.4 is the global form invoked in the proof of the uniform denominator multiplier theorem below. The point is that it genuinely requires Theorem 9.2 and not (130): for $0 < q < 2^n$ the numerator $a = q + 2^n$ exceeds 2^n , so the bounded statement does not apply. At the endpoints $|q| = 2^n$ the two clauses agree, both giving 0; for $q = -2^n$ the outer sum is empty and for $q = 2^n$ the value is $\mathcal{F}(2) = 0$.

Remark 9.5 (Why the sum is finite). Finiteness has two independent sources, and neither involves a limit. Horizontally, only the a complete translated bumps indexed by $0 \leq h < a$ meet the integration range in (135); the locally finite sum (11) therefore contributes finitely many terms. Vertically, expanding $(2a - 2h - 1 - v)^n$ produces only powers v^j with $j \leq n$, and all odd j integrate to zero, leaving the $\lfloor n/2 \rfloor + 1$ even moments $c_0, \dots, c_{\lfloor n/2 \rfloor}$. Every one of these is an explicit rational number produced by (79), so (133) is an exact algorithm over $\mathbb{Q}$ with no roundoff and no truncation. It is not, however, an efficient one: the outer range has length a , whereas the bit recursion of Algorithm 9.11 needs only $\text{wt}_2(a)$ steps.

9.2 Thue–Morse cancellation

The mechanism that makes the dyadic formulas simplify is the Prouhet cancellation carried by the Thue–Morse signs.

Lemma 9.6 (Affine power cancellation). *If $d < b$, then for all $X, Y \in \mathbb{Q}$,*

$$\sum_{h=0}^{2^b-1} (-1)^{\text{wt}_2(h)} (X + Yh)^d = 0. \quad (137)$$

Proof. It is enough to prove the claim for h^j , $0 \leq j \leq d$, and then expand the affine power. Let

$$S_{b,j} = \sum_{h=0}^{2^b-1} (-1)^{\text{wt}_2(h)} h^j.$$

Split the sum for $S_{b+1,j}$ at 2^b and use $\varepsilon_{2^b+h} = -\varepsilon_h$:

$$\begin{aligned} S_{b+1,j} &= \sum_{h < 2^b} \varepsilon_h (h^j - (2^b + h)^j) \\ &= - \sum_{r=0}^{j-1} \binom{j}{r} 2^{b(j-r)} S_{b,r}. \end{aligned}$$

If $j < b + 1$, every index r on the right satisfies $r < b$, so the induction hypothesis makes all terms zero. The base case is immediate. $\square$

A compact generating-function version is

$$\sum_{h < 2^b} (-1)^{\text{wt}_2(h)} e^{ht} = \prod_{j=0}^{b-1} (1 - e^{2^j t}), \quad (138)$$

whose right side has a zero of order b at $t = 0$. Differentiating fewer than b times at zero gives the same cancellation.

This lemma explains why apparently large translated sums in (130) collapse to short Taylor blocks when the numerator crosses a binary boundary.

The cancellation lemma states that the first b moments of the signed block vanish. The first surviving moment has an equally clean closed value, and that value is the constant that governs every leading coefficient in the q -binomial analysis of Section 9.8.

Proposition 9.7 (The first surviving Thue–Morse moment). *For every $k \geq 0$,*

$$\boxed{\sum_{r=0}^{2^k-1} \varepsilon_r r^k = (-1)^k 2^{\binom{k}{2}} k!}. \quad (139)$$

Proof. By (138),

$$\sum_{r < 2^k} \varepsilon_r e^{rt} = \prod_{j=0}^{k-1} (1 - e^{2^j t}). \quad (140)$$

Each factor is an entire function with a simple zero at $t = 0$, and

$$1 - e^{2^j t} = -2^j t + \mathcal{O}(t^2) \quad (t \rightarrow 0),$$

so the product has a zero of order exactly k at the origin, with

$$\prod_{j=0}^{k-1} (1 - e^{2^j t}) = \left(\prod_{j=0}^{k-1} (-2^j) \right) t^k + \mathcal{O}(t^{k+1}) = (-1)^k 2^{\binom{k}{2}} t^k + \mathcal{O}(t^{k+1}),$$

because $\sum_{j=0}^{k-1} j = \binom{k}{2}$. The left side of (140) is the exponential generating function

$$\sum_{d \geq 0} \left(\sum_{r < 2^k} \varepsilon_r r^d \right) \frac{t^d}{d!}.$$

Comparing coefficients of t^d for $d < k$ recovers (137); comparing coefficients of t^k gives

$$\frac{1}{k!} \sum_{r < 2^k} \varepsilon_r r^k = (-1)^k 2^{\binom{k}{2}}.$$

The case $k = 0$ is the empty product, both sides being 1. □

Corollary 9.8 (Translation invariance). *For every $k \geq 0$ and every Q in a field of characteristic zero,*

$$\sum_{r=0}^{2^k-1} \varepsilon_r (r + Q)^k = (-1)^k 2^{\binom{k}{2}} k!. \quad (141)$$

Proof. Expand $(r + Q)^k = \sum_{d=0}^k \binom{k}{d} Q^{k-d} r^d$ and interchange the two finite sums. For $d < k$ the inner sum $\sum_{r < 2^k} \varepsilon_r r^d$ vanishes by (137), so only $d = k$ survives, with coefficient 1. Apply (139). □

Thus the value is unchanged if r is replaced by $r + Q$ for any constant Q , precisely because all the lower moments vanish: a shift can only alter the coefficients of degrees below the top, and those coefficients are already zero. The centered choice $Q = -2^k$ gives

$$\sum_{r=0}^{2^k-1} \varepsilon_r (r - 2^k)^k = (-1)^k 2^{\binom{k}{2}} k!, \quad (142)$$

which is exactly the constant term of the block factorization (173) established later: dividing the centered block sum $C_k(X)$ by X^k leaves the value $(-1)^k 2^{\binom{k}{2}}$ at $X = 0$, because the remaining exponential factors there all tend to 1. This also shows that the order of vanishing in (171) is exactly k and not more, which is what makes the removal of the forced zero legitimate.

Remark 9.9 (Numerical check). The Thue–Morse signs for $0 \leq r \leq 7$ are $+, -, -, +, -, +, +, -$. Reading off the first three cases of (139):

$$\begin{aligned} k = 1 : \quad & 0 - 1 = -1, & (-1)^1 2^0 1! &= -1; \\ k = 2 : \quad & 0 - 1 - 4 + 9 = 4, & (-1)^2 2^1 2! &= 4; \\ k = 3 : \quad & 0 - 1 - 8 + 27 - 64 + 125 + 216 - 343 = -48, & (-1)^3 2^3 3! &= -48. \end{aligned}$$

These were recomputed in exact integer arithmetic, together with $k = 4, \dots, 7$, whose values are 1536, $-122\,880$, $23\,592\,960$, and $-10\,569\,646\,080$, in agreement with (139). The shift invariance (141) was checked in exact rational arithmetic for $k \leq 5$ at $Q = -2^k$, $Q = 1/2$, and $Q = 7$, and the vanishing of all lower moments for $k \leq 6$.

9.3 A Taylor block that removes the leading bit

Suppose $0 < x < 1$, and let r be the unique integer such that

$$2^{-r} \leq x < 2^{-r+1}. \quad (143)$$

Put $y = x - 2^{-r}$. Then $0 \leq y < 2^{-r}$.

Theorem 9.10 (Exact Taylor-block recursion). *With the notation above,*

$$F(x) = -F(y) + \sum_{j=0}^r \frac{2^{\binom{j+1}{2}} F(2^{j-r})}{j!} y^j. \quad (144)$$

Proof. Apply Taylor's theorem at the point 2^{-r} through degree r :

$$F(2^{-r} + y) = \sum_{j=0}^r \frac{F^{(j)}(2^{-r})}{j!} y^j + R_r(y). \quad (145)$$

The derivative scaling formula gives

$$F^{(j)}(2^{-r}) = 2^{\binom{j+1}{2}} F(2^{j-r}), \quad 0 \leq j \leq r. \quad (146)$$

It remains to identify the integral remainder. The scaling equation carries the interval $[2^{-r}, 2^{-r} + y]$ into the adjacent signed translate of the interval $[0, y]$; on that translate the Thue–Morse sign is negative. Reversing the substitutions in the derivative identity shows

$$R_r(y) = -\frac{1}{r!} \int_0^y (y-t)^r F^{(r+1)}(t) dt = -F(y),$$

where the last equality is Taylor's integral formula at the flat point 0. Substitution in (145) proves (144). $\square$

Using (102), the coefficients may be written entirely in terms of the half moments:

$$\frac{2^{\binom{j+1}{2}} F(2^{j-r})}{j!} = \frac{2^{\binom{j+1}{2} - \binom{r-j}{2}} d_{r-j}}{j!(r-j)!}. \quad (147)$$

This is the form that appears in the original arithmetic recurrence.

9.4 The bit-recursive evaluator

Let $x = a/2^e$ with $0 < a < 2^e$. Write

$$b = \lfloor \log_2 a \rfloor, \quad r = e - b, \quad q = a - 2^b. \quad (148)$$

Then 2^{-r} is the leading binary term of x and

$$y = \frac{q}{2^e}. \quad (149)$$

Applying (144) reduces $F(a/2^e)$ to $F(q/2^e)$; the highest set bit of the numerator has disappeared.

Algorithm 9.11 (Exact dyadic evaluator). Precompute $V_j = F(2^{-j})$ for $0 \leq j \leq e$ from (102). Define $\text{Eval}(e, a)$ recursively by

$$\text{Eval}(e, a) = \begin{cases} 0, & a \leq 0, \\ 1, & a \geq 2^e, \\ T_r(q/2^e) - \text{Eval}(e, q), & 0 < a < 2^e, \end{cases}$$

where b, r, q are given by (148) and

$$T_r(z) = \sum_{j=0}^r \frac{2^{\binom{j+1}{2}} V_{r-j}}{j!} z^j. \quad (150)$$

Evaluate T_r by Horner's rule.

A literal pseudocode version is:

```

Eval(e, a, V):
  if a <= 0:      return 0
  if a >= 2^e:    return 1

  b = floor_log2(a)
  r = e - b
  q = a - 2^b
  y = q / 2^e

  p = V[0]                // highest Taylor coefficient, j = r
  for m = 1 .. r:
    j = r - m
    p = V[m] + (2^(j+1) * y / (j+1)) * p

  return p - Eval(e, q, V)

```

The displayed loop is one convenient Horner ordering; any exact evaluation of (150) is equivalent.

Theorem 9.12 (Correctness and termination). *For every $e \in \mathbb{N}$ and $a \in \mathbb{Z}$,*

$$\text{Eval}(e, a) = F(a/2^e) \quad (151)$$

under the bounded convention $F = 0$ to the left and $F = 1$ to the right. The recursion makes exactly $\text{wt}_2(a)$ nontrivial calls when $0 \leq a < 2^e$.

Proof. The boundary clauses agree with (1) and (2). In the interior, (144) is precisely the recursive clause. Since $0 \leq q < a$, strong induction proves correctness and termination. Replacing a by q clears its highest set bit and leaves all lower bits unchanged, so the number of nontrivial steps is the popcount $\text{wt}_2(a)$. $\square$

Precomputing the reciprocal-power table costs $O(e^2)$ rational operations using the triangular recurrence for d_n . A single value then costs $O(e \text{wt}_2(a))$ arithmetic operations in a straightforward implementation. This is much smaller than the a outer summands in (130) when a is large.

Example 9.13 (The value at $5/16$). Here $a = 5 = 2^2 + 1$, $e = 4$, so $r = 2$, $q = 1$, and $y = 1/16$. Formula (144) gives

$$\begin{aligned} F(5/16) &= F(1/4) + 2F(1/2)y + \frac{2^3 F(1)}{2} y^2 - F(1/16) \\ &= \frac{5}{72} + \frac{1}{16} + \frac{1}{64} - \frac{143}{2073600} \\ &= \boxed{\frac{305857}{2073600}}. \end{aligned}$$

Only two recursive calls are needed, corresponding to the two set bits of 5.

9.5 Reduction of an arbitrary dyadic argument

Algorithm 9.11 evaluates the bounded function F on $[0, 1]$. The signed global extension is defined on all of $\mathbb{R}$ and is unbounded there, so a total evaluator for $\mathcal{F}$ needs one preliminary step: fold the argument back into the unit interval and record the sign that the fold costs. The fold is triangular in the sense that it is a reduction modulo 2 followed by a reflection.

Theorem 9.14 (Triangular reduction). *Let $n \in \mathbb{N}$ and put $s = 2^n$. Let $a \in \mathbb{N}$ and write*

$$a = 2sB + R, \quad B = \left\lfloor \frac{a}{2s} \right\rfloor, \quad 0 \leq R < 2s, \quad (152)$$

and set

$$C = \begin{cases} R, & R \leq s, \\ 2s - R, & R > s. \end{cases} \quad (153)$$

Then $0 \leq C \leq s$ and

$$\boxed{\mathcal{F}\left(\frac{a}{2^n}\right) = \varepsilon_B F\left(\frac{C}{2^n}\right)}. \quad (154)$$

For $a \in \mathbb{Z}$ with $a < 0$ one has $\mathcal{F}(a/2^n) = 0$.

Proof. Write $x = a/2^n = 2B + R/s$. Since $0 \leq R < 2s$ we have $2B \leq x < 2B + 2$, so x lies in the B th block and (12) gives

$$\mathcal{F}(x) = \varepsilon_B \text{up}(x - 2B - 1) = \varepsilon_B \text{up}\left(\frac{R}{s} - 1\right).$$

Put $t = R/s - 1$, so that $-1 \leq t < 1$. The fold relation (7) says $\text{up}(t) = F(t + 1)$ for $t \leq 0$ and $\text{up}(t) = F(1 - t)$ for $t > 0$; on $[-1, 1]$ both clauses read $\text{up}(t) = F(1 - |t|)$. If $R \leq s$ then $|t| = 1 - R/s$ and $F(1 - |t|) = F(R/s)$. If $R > s$ then $|t| = R/s - 1$ and $F(1 - |t|) = F(2 - R/s) = F((2s - R)/2^n)$. In both cases the argument is $C/2^n$ with C as in (153), and $0 \leq C \leq s$ is immediate from $0 \leq R < 2s$. The final statement is the vanishing of $\mathcal{F}$ on the negative half-line. $\square$

Combining (154) with Algorithm 9.11 gives a total exact evaluator for $\mathcal{F}$ at every dyadic rational: reduce with (152)–(153), compute the block sign $\varepsilon_B = (-1)^{\text{wt}_2(B)}$ from the binary digits of $B = \lfloor a/2^{n+1} \rfloor$, and call $\text{Eval}(n, C)$. The cost of the reduction is one shift, one mask, and one popcount.

The reduction is compatible with unreduced presentations. Replacing (n, a) by $(n+1, 2a)$ doubles s and R , leaves B unchanged, and doubles C ; the right side of (154) is therefore unchanged, as it must be.

Remark 9.15 (Recognizing dyadic arguments exactly). A reduced rational p/q with $q > 0$ is dyadic exactly when q is a power of two, and in that case $n = \log_2 q$ and $a = p$ are read off directly. An exact rational interface to F , up , and $\mathcal{F}$ therefore tests a single integer condition on the reduced denominator, returns “not dyadic” when it fails, and otherwise invokes the signed integer evaluator above. No floating-point recognition of powers of two is involved anywhere, so the interface is exact on every input and cannot be defeated by rounding in the recognition step.

9.6 A complete level-4 table

The exact values at denominator 16 are shown below. The second half follows from $F(1-x) = 1 - F(x)$.

x	$F(x)$	x	$F(x)$
0	0	1/2	1/2
1/16	143/2073600	9/16	1295857/2073600
1/8	1/288	5/8	215/288
3/16	46657/2073600	11/16	1767743/2073600
1/4	5/72	3/4	67/72
5/16	305857/2073600	13/16	2026943/2073600
3/8	73/288	7/8	287/288
7/16	777743/2073600	15/16	2073457/2073600

9.7 Denominator bounds

Let

$$M_m = \prod_{j=1}^m (2^{2j} - 1), \quad (2m+1)!! = 1 \cdot 3 \cdots (2m+1). \quad (155)$$

Theorem 9.16 (Uniform denominator multiplier). *Let $n \geq 1$ and $a \in \mathbb{Z}$ with $|a| < 2^n$. Then*

$$\text{up}(a/2^n) \, n! \, 2^{\binom{n+1}{2}} (2 \lfloor n/2 \rfloor + 1)!! \, M_{\lfloor n/2 \rfloor} \quad (156)$$

is an integer.

Proof. Apply the global dyadic form of (130). Replace each c_k by (81). For $k \leq \lfloor n/2 \rfloor$, both the odd double factorial in the denominator of c_k and its even-Mersenne product divide the two common products in (156). The remaining factors are integers. Thus every summand becomes integral after multiplication by the displayed common factor. $\square$

Let Δ_n be the least common multiple of the reduced denominators of the values $\text{up}(a/2^n)$ with odd a in the support. The theorem gives the explicit divisibility bound

$$\Delta_n \mid n! \, 2^{\binom{n+1}{2}} (2 \lfloor n/2 \rfloor + 1)!! \, M_{\lfloor n/2 \rfloor}. \quad (157)$$

The exact closed form of Δ_n is subtler. The source paper records a conjectural factorization; the formal development correctly retains it as a conjecture rather than a theorem.

9.7.1 Numerical values of the level denominators

The divisibility bound (157) is easy to prove and far from sharp. It is worth seeing the actual denominators. For $0 \leq n \leq 9$,

$$\begin{aligned} \Delta_0, \dots, \Delta_9 = & 1, \quad 2, \quad 72, \quad 288, \quad 2\,073\,600, \quad 33\,177\,600, \\ & 2\,809\,213\,747\,200, \quad 179\,789\,679\,820\,800, \\ & 704\,200\,217\,922\,109\,440\,000, \\ & 180\,275\,255\,788\,060\,016\,640\,000. \end{aligned} \quad (158)$$

The next two levels continue

$$\begin{aligned} \Delta_{10} &= 6\,231\,974\,256\,792\,696\,936\,191\,754\,240\,000, \\ \Delta_{11} &= 6\,381\,541\,638\,955\,721\,662\,660\,356\,341\,760\,000. \end{aligned} \quad (159)$$

Write B_n for the proved multiplier of (157),

$$B_n = n! 2^{\binom{n+1}{2}} (2\lfloor n/2 \rfloor + 1)!! M_{\lfloor n/2 \rfloor}, \quad (160)$$

with M_m as in (155). The quotients B_n/Δ_n are

n	B_n/Δ_n	n	B_n/Δ_n
0	1	6	160
1	1	7	2 240
2	2	8	2 688
3	12	9	48 384
4	8	10	161 280
5	80	11	3 548 160

The bound is exact only at $n = 0, 1$. Its slack is not bounded: it already exceeds 3.5×10^6 at $n = 11$ and continues to grow. The bound is therefore useful as a divisibility certificate and as a source of the exact power of two through (122), but it does not identify Δ_n .

Remark 9.17 (Recomputation). Every value in (158) and (159) was recomputed independently in exact rational arithmetic for this article, not copied from the source paper. The reciprocal-power table $V_j = F(2^{-j})$ was generated from the triangular recurrence (111) and cross-checked against the composition formula (115); every level- n value $\text{up}(a/2^n)$ with odd a was then produced by the bit recursion (144), reduced to lowest terms, and the least common multiple of the resulting denominators taken. As a control, the recomputed level-4 values reproduce the table of (109) and the complete level-4 table above. All ten published values agree with the recomputation, and the divisibility (157) was verified numerically for $0 \leq n \leq 11$.

9.7.2 The denominator conjecture

The list (158) has visible structure: the terms come in equal consecutive pairs after a power of two is removed. The right normalization is

$$\mathfrak{a}_n = 2^{-\binom{n}{2}} \Delta_n, \quad (161)$$

which is not to be confused with the composition normalization A_n of Section 8.1. Its first values are

$$\begin{aligned} \mathfrak{a}_0, \dots, \mathfrak{a}_{11} = & 1, \quad 2, \quad 36, \quad 36, \quad 32\,400, \quad 32\,400, \\ & 85\,730\,400, \quad 85\,730\,400, \\ & 2\,623\,350\,240\,000, \quad 2\,623\,350\,240\,000, \\ & 177\,123\,361\,504\,320\,000, \quad 177\,123\,361\,504\,320\,000. \end{aligned} \quad (162)$$

Conjecture 9.18 (Arias de Reyna denominator pattern). *With $\mathfrak{a}_n$ as in (161):*

- (i) $\mathfrak{a}_{2n} = \mathfrak{a}_{2n+1}$ for every $n \geq 1$;
- (ii) $2(2n-1)!$ divides $K_n := \mathfrak{a}_{2n-1}$ for every $n \geq 1$;
- (iii) the quotient

$$H_n = \frac{K_n}{2(2n-1)!} \quad (163)$$

is an odd integer for every $n \geq 1$.

Equivalently, if the conjecture holds then for every $n \geq 1$

$$\Delta_{2n-1} = 2^{1+\binom{2n-1}{2}}(2n-1)! H_n, \quad \Delta_{2n} = 2^{1+\binom{2n}{2}}(2n+1)! H_{n+1}, \quad (164)$$

so that the entire sequence (Δ_n) would be determined by the single odd sequence (H_n) together with explicit powers of two and factorials. The first values are

$$H_1, \dots, H_6 = 1, \quad 3, \quad 135, \quad 8\,505, \quad 3\,614\,625, \quad 2\,218\,656\,825. \quad (165)$$

The restriction $n \geq 1$ in part (i) is essential and not a convention: $\mathfrak{a}_0 = 1$ while $\mathfrak{a}_1 = 2$, so the pairing fails at the bottom of the range.

The conjecture is due to Arias de Reyna [8]. The proved bound (157) and the exact 2-adic valuation (122) constrain it – the latter pins down the power of two in each reduced denominator, which is why the normalization (161) is the natural one – but neither settles the odd part, and no proof of (i), (ii), or (iii) is known. The formal development records the statement as a conjecture and derives nothing from it.

Remark 9.19 (Verification). The values (162) and (165) were computed for this article from the recomputed denominators of Remark 9.17, in exact integer arithmetic. Part (i) was verified for $1 \leq n \leq 5$, that is, for the five pairs $(\mathfrak{a}_2, \mathfrak{a}_3)$ through $(\mathfrak{a}_{10}, \mathfrak{a}_{11})$. Part (ii) was verified for $1 \leq n \leq 6$, the largest case being $2 \cdot 11! = 79\,833\,600$ dividing $K_6 = \mathfrak{a}_{11} = 177\,123\,361\,504\,320\,000$. Every H_n in (165) is an odd integer, as (iii) requires, and the two presentations in (164) were checked against (158) for $1 \leq n \leq 5$.

9.8 A q -binomial form, double cancellation, and scalar translation

The formal development derives a second exact dyadic formula organized by Gaussian binomial coefficients. Put

$$(a; q)_n = \prod_{j=0}^{n-1} (1 - aq^j), \quad (166)$$

and write $\begin{bmatrix} n \\ k \end{bmatrix}_q$ for the Gaussian or q -binomial coefficient [15, 16, 17, 18]. The base $q = 1/2$ is forced by the dilation grid $1, 2, 4, \dots$: the corresponding Pochhammer product is the node polynomial for reciprocal dyadic scales.

The finite half-base q -binomial theorem is

$$(z; 1/2)_n = \sum_{k=0}^n (-1)^k 2^{-\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_{1/2} z^k. \quad (167)$$

To see its interpolation content, put

$$\mathfrak{m}_n = \prod_{\ell=1}^n (2^\ell - 1), \quad x_k = -2^k, \quad w_{n,k} = (-1)^k 2^{-\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_{1/2}.$$

Then

$$\sum_{k=0}^n w_{n,k} x_k^d = \begin{cases} 0, & d < n, \\ \mathbf{m}_n, & d = n, \end{cases} \quad \frac{w_{n,k}}{\mathbf{m}_n} = \frac{1}{\prod_{\ell \neq k} (x_k - x_\ell)}. \quad (168)$$

Thus the normalized Gaussian row is exactly the leading-coefficient functional for the Lagrange interpolant through $-1, -2, \dots, -2^n$. The normalization also satisfies

$$(1/2; 1/2)_n = 2^{-\binom{n+1}{2}} \mathbf{m}_n. \quad (169)$$

Likewise $\begin{bmatrix} n \\ k \end{bmatrix}_{1/2} = 2^{-k(n-k)} \begin{bmatrix} n \\ k \end{bmatrix}_2$, so the half-base coefficients are power-of-two normalizations of the integer Gaussian coefficients that count k -planes in $\mathbb{F}_2^n$.

Theorem 9.20 (Global q -binomial dyadic formula). *For $m, n \in \mathbb{N}$ and every scalar translation $Q \in \mathbb{C}$,*

$$\begin{aligned} \mathcal{F}(m/2^n) &= \frac{1}{2^{n^2} (1/2; 1/2)_n} \sum_{k=0}^n \frac{\begin{bmatrix} n \\ k \end{bmatrix}_{1/2}}{4^{\binom{k}{2}} (n+k)!} \\ &\quad \times \sum_{r=0}^{m2^k-1} (-1)^{\text{wt}_2(r)} (r - m2^k + Q)^{n+k}. \end{aligned} \quad (170)$$

Here the real value on the left is embedded in $\mathbb{C}$. The right side is independent of Q , so the identity also holds for real, rational, and Gaussian-rational shifts. If $0 \leq m \leq 2^n$, then $\mathcal{F}(m/2^n) = F(m/2^n)$, giving the bounded formula.

The shape (170) was first found experimentally, with the particular shift $Q = 1/2$, and posted as a conjecture in 2019 [19]. The formal development proves it for all $m, n \in \mathbb{N}$, for every complex Q , and for every unreduced representation $m/2^n$ of a dyadic rational.

Proof architecture. There are two nested finite differences. Inside a block of length 2^k , the generating polynomial factors as

$$\sum_{r=0}^{2^k-1} (-1)^{\text{wt}_2(r)} z^r = \prod_{j=0}^{k-1} (1 - z^{2^j}). \quad (171)$$

It has a zero of order k at $z = 1$, so the inner power sum annihilates every polynomial of degree below k . After this forced zero is removed, its degree- n coefficient is a polynomial of degree at most n in the geometric scale node -2^k .

Here is the coefficient calculation explicitly. Normalize the half moments by

$$a_j = \frac{d_j}{j!}, \quad \mathcal{A}(X) = \sum_{j \geq 0} a_j X^j, \quad E(X) = \frac{e^X - 1}{X}.$$

The half-moment recurrence is equivalent, coefficient by coefficient, to

$$\mathcal{A}(2X) = E(X) \mathcal{A}(X). \quad (172)$$

For the centered complete block set

$$C_k(X) = \sum_{r < 2^k} (-1)^{\text{wt}_2(r)} e^{(r-2^k)X}.$$

Binary factorization gives

$$\frac{C_k(X)}{X^k} = (-1)^k 2^{\binom{k}{2}} e^{-X} \prod_{j=0}^{k-1} E(-2^j X). \quad (173)$$

Iterating (172) along the negative dyadic orbit gives the same product:

$$\mathcal{A}(-2^k X) = \left(\prod_{j < k} E(-2^j X) \right) \mathcal{A}(-X).$$

Consequently the coefficient of X^n after the forced X^k is removed is, up to the displayed sign and power of two, the value at $z = -2^k$ of

$$[X^n] \mathcal{R}(z; X), \quad \mathcal{R}(z; X) = \frac{e^{-X} \mathcal{A}(zX)}{\mathcal{A}(-X)}.$$

For fixed n this is a polynomial in z of degree at most n , and its leading coefficient is a_n .

Across the scales, the weights in (168) form the divided-difference stencil on the nodes $-1, -2, \dots, -2^n$. The finite q -binomial theorem says that this stencil annihilates degrees below n and extracts the leading coefficient in degree n . It therefore returns $m_n a_n$. Using (169) and $F(2^{-n}) = 2^{-\binom{n}{2}} a_n$ gives the one-block formula. For a general numerator, the finite prefix series

$$B_m(X) = \sum_{h=0}^{m-1} (-1)^{\text{wt}_2(h)} e^{(m-1-h)X}$$

enters because writing $r = h2^k + s$ factors the long block as $B_m(-2^k X) C_k(X)$. The same divided difference then extracts the coefficient $[X^n](B_m(X) \mathcal{A}(X))$, which is exactly the signed dyadic value $2^{\binom{n}{2}} \mathcal{F}(m/2^n)$.

For translation invariance, package the entire numerator as a polynomial $N_{m,n}(X)$ over $\mathbb{Q}$. Multiplying every block exponential by e^{QX} cannot change its degree- n coefficient, because the aggregated coefficients of all lower degrees have vanished. Thus $N_{m,n}$ is a constant polynomial. Evaluation in any characteristic-zero field proves the rational, real, and complex statements algebraically; no density or continuity argument is involved. Decomposing the long prefix into m blocks of length 2^k supplies the arbitrary numerator m . $\square$

The endpoint cases are literal finite identities. When $m = 0$, every inner range is empty, so both sides are zero. When $n = 0$, the formula is the scale-zero signed block identity. A nonreduced presentation $m/2^n = 2m/2^{n+1}$ changes every finite sum but not its value, because both are identified with the same global function value.

Binary complementation gives a useful raw-coordinate version. Reflection $r \mapsto 2^k - 1 - r$ changes the Thue–Morse sign by $(-1)^k$. Once the exponent $n + k$ is included, the remaining sign is independent of k and is absorbed by $(-2)^{n^2}$. Consequently, for every $n \in \mathbb{N}$ and $Q \in \mathbb{C}$,

$$\mathcal{F}(2^{-n}) = \frac{1}{(-2)^{n^2} (1/2; 1/2)_n} \sum_{k=0}^n \frac{\begin{bmatrix} n \\ k \end{bmatrix}_{1/2}}{4^{\binom{k}{2}} (n+k)!} \sum_{r=0}^{2^k-1} (-1)^{\text{wt}_2(r)} (r+Q)^{n+k}. \quad (174)$$

Again the left side is embedded in $\mathbb{C}$, and the nested expression is independent of Q .

The q -binomial formula is not competitive with the bit recursion as an evaluator. Its value is structural: the inner difference is adapted to binary blocks, the outer Gaussian difference is adapted to geometric scales, and the same identity survives arbitrary complex scalar translation.

10 The global binary-reduction telescope

The Taylor-block recursion also applies to the signed global extension $\mathcal{F}$. It can therefore be iterated along the binary expansion of an arbitrary nonnegative real number, rather than only along the

finite expansion of a dyadic rational. The reductions need not terminate, but their residuals tend geometrically to zero.

Define the finite reduction polynomial

$$\mathcal{T}_m(y) = \sum_{j=0}^m 2^{\binom{j+1}{2} - \binom{m-j}{2}} \frac{d_{m-j}}{(m-j)!} \frac{y^j}{j!}. \quad (175)$$

All powers of 2 here are integer powers. In particular, negative exponents are allowed, and $\mathcal{T}_0(y) = 1$. The signed Taylor-block identity is

$$\mathcal{F}(2^{-m} + y) = -\mathcal{F}(y) + \mathcal{T}_m(y), \quad 0 \leq y < 2^{-m}. \quad (176)$$

Fix $x \geq 0$. At scale m , put

$$p_m = \lfloor 2^m x \rfloor, \quad y_m = x - \frac{p_m}{2^m}, \quad 0 \leq y_m < 2^{-m}. \quad (177)$$

The previous prefix requires a genuine half-scale value at $m = 0$:

$$p_{m-1}^* = \begin{cases} \lfloor x/2 \rfloor, & m = 0, \\ p_{m-1}, & m \geq 1. \end{cases} \quad (178)$$

Uniformly, $p_{m-1}^* = \lfloor p_m/2 \rfloor$. Thus $p_m = 2p_{m-1}^* + \delta_m$ with $\delta_m \in \{0, 1\}$. Define

$$\alpha_m(x) = (-1)^{\text{wt}_2(p_m)} (2p_{m-1}^* - p_m), \quad R_m(x) = (-1)^{\text{wt}_2(p_m)} \mathcal{F}(y_m). \quad (179)$$

The coefficient α_m is zero or a sign, and it is nonzero precisely when the newly exposed binary digit is 1.

Lemma 10.1 (One-step residual identity). *For $m \geq 1$,*

$$\alpha_m(x) \mathcal{T}_m(y_m) = R_{m-1}(x) - R_m(x). \quad (180)$$

Proof. If $p_m = 2p_{m-1}$, then $y_m = y_{m-1}$, the Thue–Morse signs agree, and $\alpha_m = 0$, so both sides vanish. If $p_m = 2p_{m-1} + 1$, then

$$y_{m-1} = 2^{-m} + y_m, \quad (-1)^{\text{wt}_2(p_m)} = -(-1)^{\text{wt}_2(p_{m-1})}, \quad \alpha_m = (-1)^{\text{wt}_2(p_{m-1})}.$$

Apply (176) to y_{m-1} and multiply by the old prefix sign; the two copies of $\mathcal{F}(y_m)$ combine exactly into $R_{m-1} - R_m$. $\square$

The scale-zero identity is separate and retains the parity of the integer block:

$$\mathcal{F}(x) = \alpha_0(x) \mathcal{T}_0(y_0) + R_0(x). \quad (181)$$

Indeed, write $p_0 = 2p_{-1}^* + \delta_0$ and use the two-unit translation law (12). When $\delta_0 = 0$, the polynomial term vanishes. When $\delta_0 = 1$, the identity $\mathcal{F}(1 + y_0) = -\mathcal{F}(y_0) + 1$ supplies it.

Theorem 10.2 (Exact finite binary telescope). *For $x \geq 0$ and $N \in \mathbb{N}$,*

$$\boxed{\mathcal{F}(x) = \sum_{m=0}^N \alpha_m(x) \mathcal{T}_m(y_m(x)) + R_N(x).} \quad (182)$$

Proof. Start with (181). Summing (180) for $m = 1, \dots, N$ telescopes all intermediate residuals. The case $N = 0$ is exactly the scale-zero identity. $\square$

For $N \geq 1$, $0 \leq y_N < 2^{-N} \leq 1/2$. On this interval $0 \leq \mathcal{F}(y) = F(y) \leq 2y$, so

$$|R_N(x)| \leq 2^{1-N}. \quad (183)$$

The one-step identity then gives, for $m \geq 2$,

$$|\alpha_m(x)\mathcal{T}_m(y_m)| \leq |R_{m-1}| + |R_m| \leq 4 \cdot 2^{-(m-1)}. \quad (184)$$

Thus the limiting series is absolutely, not merely conditionally, convergent.

Theorem 10.3 (Absolutely convergent global binary series). *For every $x \geq 0$,*

$$\boxed{\mathcal{F}(x) = \sum_{m=0}^{\infty} \alpha_m(x)\mathcal{T}_m(y_m(x))}. \quad (185)$$

On $0 \leq x \leq 1$, the sum is equivalently $F(x)$.

The index $m = 0$ is indispensable globally. For $0 \leq x < 1$ it vanishes, so a series starting at $m = 1$ happens to be valid there. At $x = 1$, however, the zeroth term is 1 and all later coefficients vanish. More generally it is the zeroth term that distinguishes even and odd integer blocks of the signed extension.

10.1 The corrected parity-power presentation

The same summands have a more literal presentation in terms of prefix parity and inverse dyadic values. Define, using integer exponents,

$$\kappa_j = \frac{((j : \mathbb{Z}) + 2)((j : \mathbb{Z}) - 1)}{2} = \binom{j+1}{2} - 1. \quad (186)$$

The integer interpretation matters: $\kappa_0 = -1$, hence $2^{\kappa_0} = 1/2$; natural subtraction would silently give the wrong term. Put

$$\mathcal{I}_m(y) = \sum_{j=0}^m 2^{\kappa_j} \mathcal{F}(2^{(j:\mathbb{Z})-(m:\mathbb{Z})}) \frac{y^j}{j!}. \quad (187)$$

Lemma 10.4. *For every m and y ,*

$$\mathcal{I}_m(y) = \frac{1}{2} \mathcal{T}_m(y). \quad (188)$$

Proof. For $j \leq m$, set $r = m - j$. The inverse-dyadic identity (102) gives

$$\mathcal{F}(2^{j-m}) = \frac{d_{m-j}}{(m-j)! 2^{\binom{m-j}{2}}}.$$

Together with $\kappa_j = \binom{j+1}{2} - 1$, this identifies each summand with one half of the corresponding summand in (175). $\square$

Let $\tau(a) = \text{wt}_2(a) \bmod 2$. Since

$$\frac{(-1)^a - 1}{2} = 2 \left\lfloor \frac{a}{2} \right\rfloor - a,$$

the literal summand

$$\begin{aligned} \Lambda_m(x) &:= ((-1)^{\lfloor 2^m x \rfloor} - 1)(-1)^{\tau(\lfloor 2^m x \rfloor)} \\ &\quad \times \sum_{j=0}^m 2^{((j+2)(j-1))/2} \mathcal{F}(2^{j-m}) \frac{(x - \lfloor 2^m x \rfloor / 2^m)^j}{j!} \end{aligned} \quad (189)$$

is exactly $\alpha_m(x)\mathcal{T}_m(y_m)$. Both exponents $((j+2)(j-1))/2$ and $j-m$ in this display are integer exponents, not truncated natural subtractions.

Theorem 10.5 (Corrected parity-power series). *For every $x \geq 0$,*

$$\boxed{\mathcal{F}(x) = \sum_{m=0}^{\infty} \Lambda_m(x)}, \quad (190)$$

and the series converges absolutely. On the half-open unit interval $0 \leq x < 1$, the zeroth summand vanishes and one may start at $m = 1$.

Proof. The elementary parity identity and (188) prove $\Lambda_m = \alpha_m \mathcal{T}_m(y_m)$ term by term. The claim follows from (185). The half-open corollary follows from $\lfloor x \rfloor = 0$; the closed endpoint $x = 1$ requires the restored zeroth term. $\square$

10.2 The global complex q -binomial series

The finite constant-polynomial identity from section 9.8 can be inserted at every scale of the binary telescope. For $Q \in \mathbb{C}$, define

$$\mathcal{N}_n^{(Q)} = \sum_{k=0}^n \frac{\lfloor n \rfloor_{1/2}}{4^{\binom{k}{2}} (n+k)!} \sum_{r=0}^{2^k-1} (-1)^{\text{wt}_2(r)} (r - 2^k + Q)^{n+k}. \quad (191)$$

Translation invariance and the inverse-dyadic normalization give

$$\frac{\mathcal{N}_n^{(Q)}}{2^{\binom{n}{2}} (1/2; 1/2)_n} = \frac{2^n d_n}{n!}. \quad (192)$$

The normalization differs from the 2^{n^2} in (170): here it produces a Taylor coefficient rather than a function value.

For $m \in \mathbb{N}$, put

$$\mathcal{Q}_m^{(Q)}(y) = 2^{-\binom{m+1}{2}} \sum_{n=0}^m \frac{(2^{m+1}y)^{m-n}}{(m-n)!} \frac{\mathcal{N}_n^{(Q)}}{2^{\binom{n}{2}} (1/2; 1/2)_n}. \quad (193)$$

Theorem 10.6 (q -binomial Taylor identity). *For every m , every $y \in \mathbb{R}$, and every $Q \in \mathbb{C}$,*

$$\mathcal{Q}_m^{(Q)}(y) = \mathcal{T}_m(y), \quad (194)$$

where the real polynomial on the right is embedded in $\mathbb{C}$. In particular, each finite summand is pointwise independent of Q before any infinite sum is taken.

Proof. Insert (192) into (193) and set $j = m - n$. Collecting the powers of 2 gives

$$2^{-\binom{m+1}{2}} 2^{(m+1)j} 2^{m-j} = 2^{\binom{j+1}{2} - \binom{m-j}{2}}.$$

The remaining factorials and moments are exactly those in (175). $\square$

Theorem 10.7 (Global complex q -binomial series). *For fixed $Q \in \mathbb{C}$ and every $x \geq 0$,*

$$\boxed{\mathcal{F}(x) = \sum_{m=0}^{\infty} \alpha_m(x) \mathcal{Q}_m^{(Q)}(y_m(x))}, \quad (195)$$

with the left side embedded in $\mathbb{C}$. The series converges absolutely.

This is not a new infinite basic-hypergeometric identity. The outer sum is the analytic binary telescope, while at each scale a finite q -binomial identity presents the same Taylor polynomial. Expanding one summand makes the architecture explicit:

$$\begin{aligned} \alpha_m(x) 2^{-\binom{m+1}{2}} \sum_{n=0}^m \frac{(2^{m+1}x - 2p_m(x))^{m-n}}{(m-n)!} \frac{1}{2^{\binom{n}{2}}(1/2; 1/2)_n} \\ \times \sum_{k=0}^n \frac{[n]_{1/2}}{4^{\binom{k}{2}}(n+k)!} \sum_{r=0}^{2^k-1} (-1)^{\text{wt}_2(r)} (r - 2^k + Q)^{n+k}. \end{aligned} \quad (196)$$

Convergence comes entirely from the binary-tail majorant (184); the special-function manipulations are finite and algebraic.

11 A complex q -binomial discrete limit

There is a second way in which the same Gaussian row produces $\mathcal{F}$. It is a genuine limit of finite shifted splines rather than a sequence of partial sums of (195). Reuse the safe midpoint cutoff $N_p(x)$ from (45). For $x \geq 0$ it is equivalent to the inclusive source range

$$0 \leq r \leq \left\lfloor 2^p x - \frac{1}{2} \right\rfloor, \quad (197)$$

including the empty case where the upper endpoint is -1 .

For fixed $Q \in \mathbb{C}$ and $x \in \mathbb{R}$, define

$$\begin{aligned} \mathcal{D}_n^{(Q)}(x) &= \frac{1}{2^{n^2}(1/2; 1/2)_n} \sum_{k=0}^n \frac{[n]_{1/2}}{4^{\binom{k}{2}}(n+k)!} \\ &\times \sum_{r=0}^{N_{n+k}(x)-1} (-1)^{\text{wt}_2(r)} (r - 2^{n+k}x + Q)^{n+k}. \end{aligned} \quad (198)$$

Theorem 11.1 (Complex discrete-limit formula). *For every fixed $Q \in \mathbb{C}$ and every $x \in \mathbb{R}$,*

$$\boxed{\mathcal{D}_n^{(Q)}(x) \longrightarrow \mathcal{F}(x)} \quad (199)$$

in $\mathbb{C}$. If $0 \leq x \leq 1$, the limit is equivalently $F(x)$.

Proof. If $x \leq 0$, every safe cutoff $N_p(x)$ is zero. Hence every shifted spline and every approximant (198) is zero, as is $\mathcal{F}(x)$. It remains to treat $x \geq 0$.

For a degree p and complex shift Q , introduce the finite shifted spline

$$S_p^{(Q)}(x) = \frac{(-1)^p}{2^{\binom{p}{2}} p!} \sum_{r=0}^{N_p(x)-1} (-1)^{\text{wt}_2(r)} (r - 2^p x + Q)^p. \quad (200)$$

At $Q = 1/2$ this is exactly the centered spline (46), so by (53),

$$S_p^{(1/2)}(x) \longrightarrow \mathcal{F}(x).$$

Expanding a fixed cutoff branch about $Q = 1/2$ gives an exact finite Taylor combination of lower-degree centered branches. The normalized Thue–Morse bounds then give, for $p \geq 1$,

$$\left| S_p^{(Q)}(x) - S_p^{(1/2)}(x) \right| \leq 2^{-(p-1)} \exp(|Q - \tfrac{1}{2}|). \quad (201)$$

Thus every fixed complex shift has the same spline limit.

Now reindex the outer sum in (198) by $j = n - k$. Direct collection of signs, powers, and factorials gives the exact finite identity

$$\mathcal{D}_n^{(Q)}(x) = \sum_{j=0}^n \omega_{n,j} S_{2n-j}^{(Q)}(x), \quad (202)$$

where

$$\omega_{n,j} = \frac{(-1)^j \begin{bmatrix} n \\ j \end{bmatrix}_{1/2} 2^{-(j+1)}}{(1/2; 1/2)_n}. \quad (203)$$

The finite q -binomial theorem gives the row normalization

$$\sum_{j=0}^n \omega_{n,j} = 1. \quad (204)$$

Applying it with the opposite sign also gives the uniform variation bound

$$\sum_{j=0}^n |\omega_{n,j}| = \frac{(-1/2; 1/2)_n}{(1/2; 1/2)_n} \leq 16. \quad (205)$$

Finally, every sampled degree satisfies $2n - j \geq n$. A finite-row Toeplitz lemma therefore carries the convergence of $S_p^{(Q)}(x)$ to the weighted rows (202). Explicitly, given $\epsilon > 0$, choose P so that $|S_p^{(Q)}(x) - \mathcal{F}(x)| < \epsilon/16$ for $p \geq P$. If $n \geq P$, then every $2n - j \geq n \geq P$, and the row-sum identity together with (205) gives

$$\left| \sum_{j=0}^n \omega_{n,j} S_{2n-j}^{(Q)}(x) - \mathcal{F}(x) \right| \leq \sum_{j=0}^n |\omega_{n,j}| |S_{2n-j}^{(Q)}(x) - \mathcal{F}(x)| < \epsilon.$$

This proves (199). □

Two distinctions prevent common misreadings of this theorem. First, unlike the complete finite dyadic numerator and each summand of the global q -series, a finite value $\mathcal{D}_n^{(Q)}(x)$ can depend on Q ; it is the limit that is independent of a fixed shift. Second, the Toeplitz row (202) is not a finite binary telescope or a partial sum of (195). These finite expressions share a limit but are not equal term by term.

12 Iterated prefix sums of the signed Thue–Morse sequence

The signed Thue–Morse sequence $\varepsilon_n = (-1)^{\text{wt}_2(n)}$ has already appeared twice in this article: once as the coefficient sequence of the locally finite global expansion (11), and once as the sign pattern that makes the finite dyadic sums of section 9 collapse. There is a third, entirely discrete, way in which the same signs produce the Fabius function. Sum the sequence, sum the result, and iterate. After k summations the sequence has been convolved with a discrete B-spline kernel of order k , and after a dyadic normalization the resulting integer grid reproduces, in the limit, the Rvachev bump and hence F .

This section develops that construction: the convolution law and its explicit binomial kernel, the exact zero runs forced by Prouhet cancellation, the two generating-series identities, the discrete functional equation satisfied by the normalized grid, and the precise sense in which the normalized grid converges. The convergence statement requires one shift of the prefix order and the centered reading of the grid cells; the literal normalization does not converge, and that failure is recorded separately in section 21.6. Two further pieces of foundational material are attached at the end: the lower branch of Lambert's W -function, which the asymptotic sections of this article use but never construct, and an exact binomial–logarithm formula for the Thue–Morse bit.

12.1 Iterated prefix sums and the discrete B-spline law

Definition 12.1 (Iterated prefix sums). Set $\sigma_0(n) = \varepsilon_n$ and, for $k \geq 0$,

$$\sigma_{k+1}(n) = \sum_{j=0}^n \sigma_k(j). \quad (206)$$

All sums are *inclusive*: the index n itself is summed.

Every $\sigma_k(n)$ is an integer, and $\sigma_k(0) = 1$ for all k . The inclusive convention has one consequence that must be kept in view throughout, because it is the source of an index shift later:

$$\sigma_{k+1}(n+1) - \sigma_{k+1}(n) = \sigma_k(n+1). \quad (207)$$

The increment of a prefix sum is the lower-order value at the *new* index, not at the old one.

Proposition 12.2 (Odd zeros of the first prefix sum). For every $m \geq 0$, $\sigma_1(2m+1) = 0$.

Proof. By (65), $\varepsilon_{2j} + \varepsilon_{2j+1} = 0$ for every j . The range $0 \leq n \leq 2m+1$ is the disjoint union of the pairs $\{2j, 2j+1\}$ for $0 \leq j \leq m$, so the sum of ε_n over that range vanishes. $\square$

The general structure is a discrete convolution. Write

$$\mathcal{B}_k(i) = \binom{i+k-1}{k-1} \quad (k \geq 1, i \geq 0) \quad (208)$$

for the number of ways of writing i as an ordered sum of k nonnegative integers. The family $\mathcal{B}_k$ is the k -fold discrete convolution power of the constant kernel $\mathcal{B}_1 \equiv 1$; it is the discrete B-spline of order k . Because the argument $i = n - m$ is automatically nonnegative in the sums below, no indicator function is needed.

Theorem 12.3 (Discrete B-spline convolution law). For every $k \geq 1$ and every $n \geq 0$,

$$\sigma_k(n) = \sum_{m=0}^n \mathcal{B}_k(n-m) \varepsilon_m = \sum_{m=0}^n \binom{n-m+k-1}{k-1} \varepsilon_m. \quad (209)$$

Equivalently, in a form valid for every $k \geq 0$,

$$\sigma_{k+1}(n) = \sum_{m=0}^n \binom{n-m+k}{k} \varepsilon_m. \quad (210)$$

Proof. Induct on k in the form (210). For $k = 0$ the binomial coefficient is 1 and the assertion is the definition of σ_1 . Assume (210) for k . Then

$$\begin{aligned} \sigma_{k+2}(n) &= \sum_{j=0}^n \sigma_{k+1}(j) = \sum_{j=0}^n \sum_{m=0}^j \binom{j-m+k}{k} \varepsilon_m \\ &= \sum_{m=0}^n \varepsilon_m \sum_{i=0}^{n-m} \binom{i+k}{k} = \sum_{m=0}^n \binom{n-m+k+1}{k+1} \varepsilon_m, \end{aligned}$$

where the interchange is the triangular reindexing $i = j - m$ and the inner evaluation is the hockey-stick identity $\sum_{i=0}^N \binom{i+k}{k} = \binom{N+k+1}{k+1}$. $\square$

Remark 12.4. The kernel in (209) is exactly the coefficient sequence of $(1-z)^{-k}$. [Theorem 12.3](#) is therefore the coefficient form of the series identity proved in [theorem 12.10](#) below, and either statement may be taken as the definition of the other. Both are recorded here because the coefficient form is what one actually evaluates, while the series form is what one manipulates.

12.2 Prouhet annihilation and exact dyadic zero runs

Equation (137) already records that a complete dyadic block of Thue–Morse signs annihilates every polynomial of degree below the block exponent. For the prefix sums we also need the sharp boundary case, which identifies the first power that survives.

Lemma 12.5 (Sharp Prouhet extraction). *Write $S_{r,d} = \sum_{h < 2^r} \varepsilon_h h^d$. Then $S_{r,d} = 0$ for $0 \leq d < r$, and*

$$S_{r,r} = (-1)^r 2^{\binom{r}{2}} r!. \quad (211)$$

Consequently, for every $P \in \mathbb{Q}[X]$ with $\deg P \leq r$,

$$\sum_{h < 2^r} \varepsilon_h P(h) = [X^r] P \cdot (-1)^r 2^{\binom{r}{2}} r!, \quad (212)$$

and the sum vanishes whenever $\deg P < r$.

Proof. The vanishing statement is (137) with $X = 0, Y = 1$. For the boundary case, split the block by parity rather than by halves. Since $\varepsilon_{2j} = \varepsilon_j$ and $\varepsilon_{2j+1} = -\varepsilon_j$ by (65),

$$S_{r+1,d} = \sum_{j < 2^r} \varepsilon_j [(2j)^d - (2j+1)^d] = - \sum_{c=0}^{d-1} \binom{d}{c} 2^c S_{r,c},$$

because $(2j+1)^d - (2j)^d = \sum_{c < d} \binom{d}{c} (2j)^c$. Take $d = r+1$. All terms with $c < r$ vanish by the first assertion, leaving only $c = r$:

$$S_{r+1,r+1} = - \binom{r+1}{r} 2^r S_{r,r} = -(r+1) 2^r S_{r,r}.$$

Since $S_{0,0} = \varepsilon_0 = 1$, induction gives $S_{r,r} = (-1)^r r! 2^{0+1+\dots+(r-1)} = (-1)^r r! 2^{\binom{r}{2}}$, which is (211). Formula (212) follows by expanding P in the monomial basis: every monomial of degree below r contributes zero, and the top monomial contributes (211) times its coefficient. $\square$

Corollary 12.6 (Affine sharp case). *For all $X, Y \in \mathbb{Q}$,*

$$\sum_{h < 2^r} \varepsilon_h (X + Yh)^r = (-1)^r Y^r 2^{\binom{r}{2}} r!.$$

In particular the sum does not depend on the translation X .

Proof. Apply (212) to $P(H) = (X + YH)^r$, whose degree is at most r and whose coefficient of H^r is Y^r . $\square$

Equivalently, (212) is the statement that the block polynomial of (171) has a zero of exact order r at $z = 1$: the m th derivative of $\prod_{j < r} (1 - z^{2^j})$ at $z = 1$ vanishes for $m < r$, while the r th derivative equals $(-1)^r r! \prod_{j < r} 2^j = (-1)^r r! 2^{\binom{r}{2}}$. We use the polynomial form because it applies directly to the binomial kernel (208), which is a polynomial in the summation index.

Theorem 12.7 (Exact dyadic zero runs). *Let $1 \leq k \leq r$. Then*

$$\sigma_k(n) = 0 \quad \text{for } 2^r - k \leq n \leq 2^r - 1, \quad (213)$$

$$\sigma_k(2^r - k - 1) = (-1)^{r-k}, \quad (214)$$

$$\sigma_k(2^r) = -1. \quad (215)$$

Formula (214) also holds for $k = 0$, where it reduces to $\varepsilon_{2^r-1} = (-1)^r$, and (215) also holds for $k = 0$. Hence the zero run terminating at $2^r - 1$ has length exactly k : it is bounded on the left by a value of modulus one and on the right by the value -1 .

Proof. Endpoint. Put $q = k - 1$, so $0 \leq q < r$. By (209),

$$\sigma_k(2^r - 1) = \sum_{m=0}^{2^r-1} \binom{2^r - 1 - m + q}{q} \varepsilon_m = \frac{1}{q!} \sum_{m < 2^r} \varepsilon_m \prod_{i=1}^q (2^r - 1 - m + i).$$

The product is a polynomial in m of degree $q < r$, so the sum vanishes by (212).

Rest of the run. We prove, by induction on d , that $\sigma_k(2^r - 1 - d) = 0$ whenever $1 \leq k \leq r$ and $0 \leq d < k$. The case $d = 0$ is the endpoint. Suppose the claim holds for d and that $d + 1 < k$, so $k \geq 2$ and $1 \leq k - 1 \leq r$. The induction hypothesis applied at order k gives $\sigma_k(2^r - 1 - d) = 0$, and applied at order $k - 1$, which is legitimate because $d < k - 1$, gives $\sigma_{k-1}(2^r - 1 - d) = 0$. Now use (207) with $n = 2^r - 1 - (d + 1)$, so that $n + 1 = 2^r - 1 - d$:

$$0 - \sigma_k(n) = \sigma_k(n + 1) - \sigma_k(n) = \sigma_{k-1}(n + 1) = 0.$$

This proves (213), since the indices $2^r - k, \dots, 2^r - 1$ are exactly the $2^r - 1 - d$ with $0 \leq d < k$.

Left boundary. Induct on k . For $k = 0$ the number $2^r - 1$ has r binary digits, all equal to one, so $\varepsilon_{2^r-1} = (-1)^r$. Assume (214) at order k and let $k + 1 \leq r$. Put $n = 2^r - (k + 1) - 1$, so $n + 1 = 2^r - k - 1$. By the induction hypothesis $\sigma_k(n + 1) = (-1)^{r-k}$, and by (213) at order $k + 1$ we have $\sigma_{k+1}(n + 1) = \sigma_{k+1}(2^r - (k + 1)) = 0$. Then (207) gives $-\sigma_{k+1}(n) = (-1)^{r-k}$, that is, $\sigma_{k+1}(n) = (-1)^{r-(k+1)}$.

Right boundary. Induct on k . For $k = 0$, $\text{wt}_2(2^r) = 1$, so $\varepsilon_{2^r} = -1$. If (215) holds at order k and $k + 1 \leq r$, then by (207) and (213), $\sigma_{k+1}(2^r) = \sigma_{k+1}(2^r - 1) + \sigma_k(2^r) = 0 - 1 = -1$. $\square$

Remark 12.8. The right boundary value is not $+1$. Every iterated prefix sum, at every order $k \leq r$, takes the value -1 at the index 2^r . This single sign is what makes the literal normalization of section 12.4 fail at the right endpoint; see section 21.6.

12.3 Two generating-series identities

All series in this subsection are formal power series over $\mathbb{Z}$; no convergence hypothesis on a complex variable is used or needed.

Let

$$\mathcal{K}_r(z) = \sum_{n < 2^r} \varepsilon_n z^n, \quad \mathcal{K}(z) = \sum_{n \geq 0} \varepsilon_n z^n, \quad \mathcal{S}_k(z) = \sum_{n \geq 0} \sigma_k(n) z^n. \quad (216)$$

Proposition 12.9 (Block product and coefficient stabilization). *For every $r \geq 0$,*

$$\mathcal{K}_r(z) = \prod_{j=0}^{r-1} (1 - z^{2^j}), \quad (217)$$

which is (171). Moreover, for every $n < 2^r$ the coefficient of z^n in the finite product (217) equals ε_n and is therefore independent of r . Consequently the infinite product $\prod_{j \geq 0} (1 - z^{2^j})$ is a well-defined formal power series, and it equals $\mathcal{K}(z)$.

Proof. Induct on r . For $r = 0$ both sides are 1. For the step, split the block of length 2^{r+1} at 2^r and use $\varepsilon_{2^r+n} = -\varepsilon_n$ for $n < 2^r$:

$$\mathcal{K}_{r+1}(z) = \sum_{n < 2^r} \varepsilon_n z^n - z^{2^r} \sum_{n < 2^r} \varepsilon_n z^n = \mathcal{K}_r(z)(1 - z^{2^r}),$$

and the product formula follows. For the stabilization, $\mathcal{K}_r$ is a polynomial of degree $2^r - 1$ whose coefficient of z^n is ε_n for every $n < 2^r$; adjoining the factor $1 - z^{2^r}$ changes only coefficients of index $\geq 2^r$. Thus for fixed n the coefficient of z^n in $\prod_{j < r} (1 - z^{2^j})$ is constant once $2^r > n$, which is the definition of convergence of the infinite product in the formal topology, and the stable value is ε_n . $\square$

Theorem 12.10 (Prefix generating series). *For every $k \geq 0$,*

$$\mathcal{S}_k(z) = \frac{\mathcal{K}(z)}{(1-z)^k}, \quad (218)$$

equivalently, in denominator-cleared form,

$$(1-z)^k \mathcal{S}_k(z) = \mathcal{K}(z) = \prod_{j \geq 0} (1 - z^{2^j}). \quad (219)$$

Both identities have a finite verification: for every r and every $n < 2^r$, the coefficient of z^n on either side of (219) may be computed from the polynomial $\prod_{j < r} (1 - z^{2^j})$ alone.

Proof. For $k = 0$ both sides of (218) are $\mathcal{K}(z)$. For $k \geq 1$, expand $(1-z)^{-k} = \sum_{i \geq 0} \binom{i+k-1}{k-1} z^i = \sum_{i \geq 0} \mathcal{B}_k(i) z^i$. The coefficient of z^n in $\mathcal{K}(z)(1-z)^{-k}$ is then the Cauchy product $\sum_{m=0}^n \mathcal{B}_k(n-m) \varepsilon_m$, which is $\sigma_k(n)$ by (209). Since $(1-z)^k$ is a unit multiple of the invertible series $(1-z)^{-k}$, the cleared form (219) is equivalent, and the product expression is [theorem 12.9](#). The finite verification is the stabilization statement of that proposition: no coefficient of index below 2^r sees any factor $1 - z^{2^j}$ with $j \geq r$. $\square$

The two identities of [theorem 12.10](#) connect the prefix sums to the polynomial approximants used in [section 12.5](#). Put

$$p_n(z) = \prod_{i=1}^n (1 + z + \cdots + z^{2^i-1}), \quad \delta_n = \deg p_n = \sum_{i=1}^n (2^i - 1) = 2^{n+1} - n - 2. \quad (220)$$

Since $1 - z^{2^j} = (1-z)(1+z+\cdots+z^{2^j-1})$ and the factor with $j = 0$ is trivial, (217) factors as

$$\mathcal{K}_{n+1}(z) = (1-z)^{n+1} p_n(z). \quad (221)$$

Comparing (221) with (219) at order $k = n + 1$ and using stabilization below 2^{n+1} gives the exact coefficient identification

$$\sigma_{n+1}(m) = [z^m] p_n(z) \quad (0 \leq m < 2^{n+1}). \quad (222)$$

Note that (222) must fail at $m = 2^{n+1}$, because $\delta_n < 2^{n+1}$ makes the right side zero while the left side is -1 by (215). That single index is the exceptional right endpoint of [theorem 12.16](#).

12.4 The functional equation of the normalized prefix grid

The natural dyadic normalization of the prefix sums is

$$\mathcal{V}_k(j) = \frac{\sigma_k(j)}{2^{\binom{k}{2}}}, \quad x_{k,j} = \frac{j}{2^k}, \quad (223)$$

so that $\mathcal{V}_k$ is a rational-valued function on the level- k dyadic grid. The exponent $\binom{k}{2}$ is forced: it is the exponent that appears in the sharp Prouhet value (211), in the centered spline (46), and in the derivative scaling (67).

Theorem 12.11 (Discrete functional equation). *Let $q \geq 0$ and $0 \leq j < 2^q$, and put $h = 2^{-(q+1)}$ and $x = x_{q+1,j}$. Then*

$$\frac{\mathcal{V}_{q+1}(j+1) - \mathcal{V}_{q+1}(j)}{h} = 2 \mathcal{V}_q(j+1), \quad (224)$$

and the abscissa attached to the right-hand value is

$$x_{q,j+1} = 2(x+h) = 2x_{q+1,j+1} \in [0, 1]. \quad (225)$$

The same identity holds verbatim for the shifted grid $\tilde{\mathcal{V}}_k(j) = \sigma_{k+1}(j)/2^{\binom{k}{2}}$ of [section 12.5](#).

Proof. Because $\binom{q+1}{2} = \binom{q}{2} + q$, the forward difference (207) gives

$$\mathcal{V}_{q+1}(j+1) - \mathcal{V}_{q+1}(j) = \frac{\sigma_{q+1}(j+1) - \sigma_{q+1}(j)}{2^{\binom{q}{2}+q}} = \frac{\sigma_q(j+1)}{2^{\binom{q}{2}}} \cdot \frac{1}{2^q} = \frac{\mathcal{V}_q(j+1)}{2^q}.$$

Dividing by $h = 2^{-(q+1)}$ multiplies by 2^{q+1} and gives (224). For (225), $2(x+h) = 2(j+1)/2^{q+1} = (j+1)/2^q = x_{q,j+1}$, and the hypothesis $j < 2^q$ is exactly the condition $j+1 \leq 2^q$ that keeps this abscissa in the unit interval. The shifted grid satisfies the same computation with σ replaced by $\sigma_{\bullet+1}$ throughout. $\square$

Equation (224) is the discrete counterpart of the global differential equation (66): a difference quotient at level $q+1$ equals twice a level- q value at the doubled abscissa. Two features of the discrete statement have no continuous analogue and are easy to lose. First, the right-hand side is evaluated at the *forward* node $2(x+h)$ rather than at $2x$; this is the inclusive-prefix convention (207). Second, the identity is an assertion about a level- q value, so it is meaningful as a statement about F or up only when the doubled abscissa lies in the relevant interval, that is, only when $j < 2^q$. Both features are used in section 12.5.

12.5 Corrected normalization and pointwise convergence

We now identify the normalized prefix grid with the histogram approximants of the probabilistic construction, and deduce convergence. Two adjustments to the naive reading are required, and both are forced by identities already proved: the prefix order must be shifted by one, because (222) matches σ_{n+1} with p_n ; and the grid index m must be read as labelling the *center* of a cell of width 2^{-n} , not the left endpoint of a cell of width $2^{-(n+1)}$.

Recall from (39) and (40) that up is the density of $Y = \sum_{i \geq 0} V_i 2^{-i-1}$, with V_i independent and uniform on $[-1, 1]$. Replacing each uniform variable by a lattice uniform variable produces an exact finite discrete law. Concretely, let $D_1, \dots, D_n$ be independent with D_i uniform on $\{0, 1, \dots, 2^i - 1\}$, so that $D_1 + \dots + D_n$ has generating polynomial p_n of (220) and total mass $p_n(1) = 2^{\binom{n+1}{2}}$. Place the mass of the monomial z^m at

$$\xi_{n,m} = \frac{2m - \delta_n}{2^{n+1}}, \quad (226)$$

and spread it uniformly over the cell $I_{n,m}$ of width 2^{-n} centered at $\xi_{n,m}$. The consecutive centers are spaced 2^{-n} apart, so the cells tile $\mathbb{R}$. The resulting density is the step function

$$\Theta_n(x) = \frac{2^n}{2^{\binom{n+1}{2}}} \sum_{m=0}^{\delta_n} [z^m] p_n(z) \cdot \mathbf{1}_{I_{n,m}}(x), \quad (227)$$

where a point shared by two adjacent cells is given the weight $\frac{1}{2}$ in each, so that the two contributions add to the single value the formula is intended to have. This endpoint convention is not cosmetic: with ordinary closed-interval indicators the value at $x = 0$ would be counted twice already for $n = 1$.

Theorem 12.12 (Histogram approximants). *For every $n \geq 0$ the function Θ_n satisfies $0 \leq \Theta_n \leq 1$, is nondecreasing on $(-\infty, 0)$, is nonincreasing on $(0, \infty)$, and satisfies $\Theta_n(0) = 1$. Moreover $\Theta_n(x) \rightarrow \text{up}(x)$ for every real x .*

Proof architecture. Unimodality and symmetry are statements about the coefficient sequence of p_n . Each factor of (220) is palindromic with nonnegative coefficients and unimodal, and these two properties are preserved by multiplication; the placement (226) converts symmetry about $\delta_n/2$ into symmetry about $x = 0$. The normalization $\Theta_n(0) = 1$ asserts that the coefficient of p_n at the central index, averaged over the two central indices when δ_n is odd, equals $2^{\binom{n+1}{2}-n}$; this is the identity that forces the half-weight endpoint convention, as the case $p_1 = 1 + z$, $\delta_1 = 1$ already shows. For the

limit, note that the law behind (227) is that of $\sum_{i=1}^n (2D_i - 2^i + 1)2^{-n-1}$, a triangular array whose i th summand is, after the reindexing $j = n + 1 - i$, a lattice discretization with mesh 2^{-n} of the uniform law on $[-2^{-j}, 2^{-j}]$. Since $\delta_n = \sum_i (2^i - 1)$, these summands do add up to the variable placed at (226). Because the meshes tend to zero, these laws converge weakly to the law of Y , so all interval integrals of Θ_n converge to the corresponding integrals of up . A sequence of unimodal densities with a common peak location, whose interval integrals converge to those of a continuous unimodal limit, converges pointwise everywhere. This last step is the only place where continuity of up is used, and it supplies no rate. $\square$

Theorem 12.13 (Exact identification of the corrected grid). *Put $\tilde{\mathcal{V}}_n(m) = \sigma_{n+1}(m)/2^{\binom{n}{2}}$. For every $n \geq 0$ and every m with $0 \leq m < 2^{n+1}$,*

$$\tilde{\mathcal{V}}_n(m) = \Theta_n(\xi_{n,m}). \quad (228)$$

Proof. The point $\xi_{n,m}$ lies in the interior of the cell $I_{n,m}$ and in no other cell, so (227) evaluates there to

$$\frac{2^n}{2^{\binom{n+1}{2}}} [z^m] p_n(z),$$

including when $m > \delta_n$, where both sides vanish. Since $\binom{n+1}{2} = \binom{n}{2} + n$, that value is $[z^m] p_n / 2^{\binom{n}{2}}$, which is $\tilde{\mathcal{V}}_n(m)$ by (222). $\square$

Sampling at the dyadic floor now gives an approximation scheme. For $x \geq 0$ set

$$j_n(x) = \lfloor 2^{n+1}x \rfloor, \quad \Sigma_n(x) = \tilde{\mathcal{V}}_n(j_n(x)) = \frac{\sigma_{n+1}(j_n(x))}{2^{\binom{n}{2}}}. \quad (229)$$

Lemma 12.14 (Moving cell centers). *For every $x \in [0, 1]$, $\xi_{n,j_n(x)} \rightarrow 2x - 1$ as $n \rightarrow \infty$.*

Proof. By (226), $\xi_{n,j_n(x)} = 2 \cdot j_n(x)2^{-(n+1)} - \delta_n 2^{-(n+1)}$. The first term tends to $2x$ because $0 \leq 2^{n+1}x - j_n(x) < 1$, and $\delta_n 2^{-(n+1)} = 1 - (n+2)2^{-(n+1)} \rightarrow 1$. $\square$

Lemma 12.15 (Stability under moving arguments). *If $u_n \rightarrow y$ in $\mathbb{R}$, then $\Theta_n(u_n) \rightarrow \text{up}(y)$.*

Proof. Let $\varepsilon > 0$. Suppose first $y > 0$. By continuity of up choose $d \in (0, y/2]$ with $|\text{up}(y \pm d) - \text{up}(y)| < \varepsilon/2$. For all large n we have $u_n \in [y - d, y + d] \subset (0, \infty)$, so the monotonicity of theorem 12.12 on $(0, \infty)$ gives

$$\Theta_n(y + d) \leq \Theta_n(u_n) \leq \Theta_n(y - d).$$

By pointwise convergence at the two fixed points $y \pm d$, for all large n both outer terms lie within $\varepsilon/2$ of $\text{up}(y \pm d)$, hence within ε of $\text{up}(y)$. The case $y < 0$ is identical using monotonicity on $(-\infty, 0)$. For $y = 0$ the upper bound is $\Theta_n(u_n) \leq 1 = \text{up}(0)$, and the lower bound is $\Theta_n(-d)$, $\Theta_n(0) = 1$, or $\Theta_n(d)$ according to the sign of u_n ; each of the three converges to a value within $\varepsilon/2$ of $\text{up}(0)$. $\square$

Theorem 12.16 (Corrected prefix approximation). *For every $x \in [0, 1]$,*

$$\Sigma_n(x) \longrightarrow \text{up}(2x - 1), \quad (230)$$

and consequently, for every $x \in [0, 1]$,

$$\Sigma_n\left(\frac{x}{2}\right) \longrightarrow F(x). \quad (231)$$

Proof. Let $0 \leq x < 1$. Then $2^{n+1}x < 2^{n+1}$, so $j_n(x) < 2^{n+1}$ and [theorem 12.13](#) applies: $\Sigma_n(x) = \Theta_n(\xi_{n,j_n(x)})$. Combining [theorem 12.14](#) with [theorem 12.15](#) gives $\Sigma_n(x) \rightarrow \text{up}(2x - 1)$.

At $x = 1$ the identification fails, because $j_n(1) = 2^{n+1}$ is exactly the first index not covered by (222). Here the value is computed instead from (215) with $k = r = n + 1$:

$$\Sigma_n(1) = \frac{\sigma_{n+1}(2^{n+1})}{2^{\binom{n}{2}}} = -\frac{1}{2^{\binom{n}{2}}} \rightarrow 0 = \text{up}(1),$$

since $\binom{n}{2} \rightarrow \infty$ and $\text{up}(1) = F(0) = 0$. This proves (230) on all of $[0, 1]$. For (231), apply it at $x/2 \in [0, 1]$ and use $2(x/2) - 1 = x - 1 \leq 0$, so that $\text{up}(x - 1) = F(x)$ by (7). $\square$

Warning 12.17. [Theorem 12.16](#) is a pointwise statement and nothing more. No uniform rate is asserted, and none is proved. The proof passes through [theorem 12.15](#), whose only quantitative inputs are the shared monotonicity of the Θ_n and the modulus of continuity of the limit up at the single point $2x - 1$; the sandwich it uses degrades as that modulus degrades. A uniform bound of the kind supplied for the centered splines by (52) is available from [section 4.3](#), but it is a different approximation scheme, and the two should not be conflated.

Remark 12.18. The two corrections are independent of one another and both are necessary. Shifting the prefix order from k to $k + 1$ is what makes (222) available; reading the index m as the center (226) of a cell of width 2^{-n} rather than as an abscissa $m2^{-k}$ is what supplies the affine change of variable $x \mapsto 2x - 1$ between the unit interval and the support of up . In particular the corrected grid is a sample of the compactly supported bump, not of the distribution function; this is why its right-endpoint limit is 0 and not 1.

12.6 The lower Lambert branch

The asymptotic analysis of [section 14](#) is organized around the coordinate (258), where W_{-1} denotes the lower real branch of Lambert's W -function [11]. That section uses three facts about W_{-1} without proving any of them: that the branch exists on $(-e^{-1}, 0)$, that it is characterized by being the solution below -1 , and that it has the two-term expansion recorded below. We supply them here, since they are the foundational input for (260) and for every subsequent estimate in the Lambert variable.

Proposition 12.19 (Existence, defining equation, and uniqueness). *The map $u \mapsto u \log u$ is a strictly decreasing bijection from $(0, e^{-1})$ onto $(-e^{-1}, 0)$. For $z \in (-e^{-1}, 0)$ let $u(z)$ be its unique preimage and set $W_{-1}(z) = \log u(z)$. Then*

$$W_{-1}(z) < -1, \quad W_{-1}(z) e^{W_{-1}(z)} = z, \quad (232)$$

and $W_{-1}(z)$ is the unique real $w < -1$ with $we^w = z$.

Proof. On $(0, e^{-1})$ the derivative of $u \log u$ is $1 + \log u < 0$, so the map is strictly decreasing there. Its limit at $u \downarrow 0$ is 0 and its value at $u = e^{-1}$ is $-e^{-1}$; by the intermediate value theorem and strict monotonicity it is a bijection onto $(-e^{-1}, 0)$. Write $u = u(z)$ and $w = \log u$. Since $0 < u < e^{-1}$ we get $w < -1$, and $e^w = u$, so $we^w = u \log u = z$, which is (232). Conversely, if $w < -1$ and $we^w = z$, put $v = e^w$; then $v \in (0, e^{-1})$ and $v \log v = we^w = z$, so $v = u(z)$ by injectivity, hence $w = \log u(z) = W_{-1}(z)$. $\square$

Corollary 12.20 (Solution of the saddle equation). *Let $x > 0$ with $Lx < e^{-1}$, where $L = \log 2$ as in (237), and put $\lambda = -W_{-1}(-Lx)/L$ as in (259). Then $\lambda > 1/L$ and $\lambda 2^{-\lambda} = x$, which is (260).*

Proof. The number $z = -Lx$ lies in $(-e^{-1}, 0)$, so [theorem 12.19](#) applies. Write $W = W_{-1}(z)$, so $\lambda = -W/L$ and $2^{-\lambda} = e^{-L\lambda} = e^W$. Hence $\lambda 2^{-\lambda} = (-W/L)e^W = -z/L = x$. Finally $W < -1$ gives $\lambda > 1/L$. $\square$

Proposition 12.21 (Two-term expansion). *As $\epsilon \downarrow 0$,*

$$W_{-1}(-\epsilon) - (\log \epsilon - \log |\log \epsilon|) \longrightarrow 0. \quad (233)$$

Proof. For $0 < \epsilon < e^{-1}$ put $y = y(\epsilon) = -W_{-1}(-\epsilon) > 1$. Negating (232) gives the defining equation in the convenient form

$$y e^{-y} = \epsilon.$$

The map $t \mapsto t e^{-t}$ is strictly decreasing on $[1, \infty)$ with limit 0 at infinity, so $y(\epsilon) \rightarrow \infty$ as $\epsilon \downarrow 0$. Taking logarithms in the displayed equation gives $\log \epsilon = \log y - y$, hence

$$\frac{-\log \epsilon}{y} = 1 - \frac{\log y}{y} \longrightarrow 1,$$

because $\log y/y \rightarrow 0$. Finally, $\epsilon < 1$ makes $\log \epsilon < 0$, so $|\log \epsilon| = -\log \epsilon$, and

$$\begin{aligned} W_{-1}(-\epsilon) - (\log \epsilon - \log |\log \epsilon|) &= -y - (\log y - y) + \log(-\log \epsilon) \\ &= \log \frac{-\log \epsilon}{y} \longrightarrow \log 1 = 0. \end{aligned} \quad \square$$

Remark 12.22. [Theorem 12.21](#) is exactly the accuracy that the elementary log–log forms of [section 16](#) can inherit. It is an $o(1)$ statement, not an $\mathcal{O}$ statement with a rate, and substituting it into the main term (262) in place of the exact phase $\lambda(x)$ destroys the periodic correction at exactly the scale discussed in [section 21](#).

12.7 A binomial–logarithm formula for the Thue–Morse bit

The zero-one Thue–Morse bit $b(n) = \text{wt}_2(n) \bmod 2$ of (352) satisfies $\varepsilon_n = 1 - 2b(n)$. It admits a closed formula in which the entire binary structure is hidden inside a row of Pascal's triangle. Set

$$\mathcal{G}(n) = \sum_{k=0}^n (-1)^{\binom{n}{k}} \in \mathbb{Z}. \quad (234)$$

Lemma 12.23 (Glaisher count). *The number of odd entries in the n th row of Pascal's triangle is $2^{\text{wt}_2(n)}$.*

Proof. Let $\nu(n)$ denote that number, so that $\nu(n)$ is the number of nonzero coefficients of $(1+z)^n$ in $\mathbb{F}_2[z]$. There $(1+z)^2 = 1+z^2$, hence

$$(1+z)^{2n} = (1+z^2)^n, \quad (1+z)^{2n+1} = (1+z^2)^n(1+z).$$

The substitution $z \mapsto z^2$ does not change the number of nonzero coefficients, so $\nu(2n) = \nu(n)$; and multiplying a polynomial in z^2 by $1+z$ produces two summands with disjoint supports, one even and one odd, so $\nu(2n+1) = 2\nu(n)$. Since $\text{wt}_2(2n) = \text{wt}_2(n)$ and $\text{wt}_2(2n+1) = \text{wt}_2(n) + 1$ by (64), and $\nu(0) = 1 = 2^{\text{wt}_2(0)}$, induction on the binary representation gives $\nu(n) = 2^{\text{wt}_2(n)}$. $\square$

Proposition 12.24 (The logarithm is exact). *For every $n \geq 0$,*

$$n + 1 - \mathcal{G}(n) = 2^{\text{wt}_2(n)+1}. \quad (235)$$

In particular $n + 1 - \mathcal{G}(n)$ is a positive power of two, and $\log_2(n + 1 - \mathcal{G}(n)) = \text{wt}_2(n) + 1$ is an integer.

Proof. Each summand of (234) is -1 at an odd entry and $+1$ at an even entry, and the row has $n + 1$ entries. Hence $\mathcal{G}(n) = (n + 1) - 2\nu(n)$ with ν as in [theorem 12.23](#), so $n + 1 - \mathcal{G}(n) = 2\nu(n) = 2^{\text{wt}_2(n)+1}$. $\square$

Theorem 12.25 (Binomial–logarithm formula). *For every $n \geq 0$,*

$$b(n) = \frac{1 + (-1)^{\log_2 \left(n+1 - \sum_{k=0}^n (-1)^{\binom{n}{k}} \right)}}{2}, \quad \varepsilon_n = 1 - 2b(n). \quad (236)$$

Proof. By [theorem 12.24](#) the exponent equals $\text{wt}_2(n) + 1$. If $\text{wt}_2(n)$ is even then $b(n) = 0$ and $(-1)^{\text{wt}_2(n)+1} = -1$, so the right side is 0; if $\text{wt}_2(n)$ is odd then $b(n) = 1$ and $(-1)^{\text{wt}_2(n)+1} = 1$, so the right side is 1. The second identity is $\varepsilon_n = (-1)^{\text{wt}_2(n)} = 1 - 2(\text{wt}_2(n) \bmod 2)$. $\square$

Remark 12.26. The order of the two steps matters. As printed, (236) raises -1 to a logarithm, an operation that is meaningful only because the argument of that logarithm has first been proved to be exactly a power of two. The formal development therefore proves (235) before it interprets the outer expression at all, evaluates the logarithm on the natural numbers, and reads the final division in $\mathbb{Q}$. A floating-point evaluation of the same display, or an interpretation of the logarithm as a real-valued function whose output is then rounded, is an approximation to a formula rather than the formula.

13 Endpoint asymptotics: exact logarithmic decomposition

We now turn to the behavior of $F(x)$ as $x \downarrow 0$. This is the technically deepest part of the theory and the place where a very small, but mathematically unavoidable, log-periodic fluctuation appears.

Throughout the asymptotic sections, put

$$L = \log 2. \quad (237)$$

We use the standard Stieltjes convention

$$\zeta(1+s) = \frac{1}{s} + \gamma - \gamma_1 s + \mathcal{O}(s^2), \quad (238)$$

where γ is Euler's constant and γ_1 is the first Stieltjes constant.

13.1 The negative Laplace product

The function G from (85) is the moment generating function of the random variable X in (33):

$$G(z) = \mathbb{E}e^{zX}. \quad (239)$$

Indeed, this also follows from (87) and $Y = 2X - 1$. Therefore, for $s > 0$, $G(-s)$ is the ordinary Laplace transform of the probability law.

Iterating the functional equation (86) in the negative direction gives an exact product.

Proposition 13.1 (Negative Laplace product). *For $s > 0$,*

$$G(-s) = \prod_{j=1}^{\infty} \frac{1 - e^{-s/2^j}}{s/2^j}. \quad (240)$$

Consequently

$$\Lambda(s) := \log G(-s) = \sum_{j=1}^{\infty} \kappa(s/2^j), \quad \kappa(u) = \log \frac{1 - e^{-u}}{u}. \quad (241)$$

The series converges absolutely and locally uniformly on $(0, \infty)$.

Proof. Putting $z = -s/2$ in (86) gives

$$G(-s) = \frac{1 - e^{-s/2}}{s/2} G(-s/2).$$

After N iterations, the remaining factor is $G(-s/2^N)$, which tends to $G(0) = 1$. For small u ,

$$\kappa(u) = -\frac{u}{2} + \mathcal{O}(u^2),$$

so the logarithmic series is dominated on compact s -intervals by a geometric series. $\square$

The product has the exact dilation law

$$\Lambda(2s) = \kappa(s) + \Lambda(s). \quad (242)$$

This equation resembles a renewal equation on the logarithmic scale. Its homogeneous solutions are periodic functions of $\log_2 s$, which is the conceptual origin of the fluctuation in the final asymptotic.

13.2 Removing the quadratic drift

Define the forward tail

$$T(s) = \sum_{m=0}^{\infty} \log(1 - e^{-s2^m}) < 0 \quad (243)$$

and the positive tail error

$$E(s) = -T(s) \geq 0. \quad (244)$$

The series is exponentially small as $s \rightarrow \infty$.

Lemma 13.2 (Forward-tail bound). *For $s > 0$,*

$$0 \leq E(s) \leq \frac{e^{-s}}{(1 - e^{-s})^2}. \quad (245)$$

In particular $E(s) \leq 4e^{-s}$ for $s \geq \log 2$.

Proof. For $0 < q < 1$, $-\log(1 - q) \leq q/(1 - q)$. Hence

$$E(s) \leq \frac{1}{1 - e^{-s}} \sum_{m \geq 0} e^{-s2^m}.$$

Since $2^m \geq m + 1$, the last sum is at most $\sum_{m \geq 0} e^{-s(m+1)} = e^{-s}/(1 - e^{-s})$. $\square$

For $t \in \mathbb{R}$, set

$$P(t) = \Lambda(2^t) + \frac{L}{2}(t^2 - t) + T(2^t). \quad (246)$$

Theorem 13.3 (Exact periodic decomposition). *The function P is 1-periodic and smooth. For every $s > 0$, with $t = \log_2 s$, one has the exact identity*

$$\Lambda(s) = -\frac{(\log s)^2}{2L} + \frac{\log s}{2} + P(\log_2 s) + E(s). \quad (247)$$

Proof. Write $s = 2^t$. Using (242),

$$\Lambda(2s) = \Lambda(s) + \log(1 - e^{-s}) - \log s.$$

Also

$$T(2s) = T(s) - \log(1 - e^{-s}).$$

The change in the quadratic correction is

$$\frac{L}{2}((t+1)^2 - (t+1) - (t^2 - t)) = Lt = \log s.$$

The three changes cancel, proving $P(t+1) = P(t)$. Rearranging (246) and using $E = -T$ yields (247). Smoothness follows by differentiating locally uniformly convergent series; periodicity then makes every derivative bounded. $\square$

Let

$$\overline{P} = \int_0^1 P(t) dt, \quad \Psi(t) = P(t) - \overline{P}. \quad (248)$$

Then Ψ is smooth, real-valued, 1-periodic, and has mean zero.

13.3 Fourier coefficients of the periodic correction

The periodic term is not an unspecified bounded nuisance: all of its Fourier coefficients are explicit. Put

$$\chi_k = \frac{2\pi i k}{L} \quad (k \in \mathbb{Z}), \quad (249)$$

and use the convention

$$\widehat{\Psi}(k) = \int_0^1 \Psi(t) e^{-2\pi i k t} dt. \quad (250)$$

Theorem 13.4 (Gamma–zeta Fourier coefficients). *One has $\widehat{\Psi}(0) = 0$, and for $k \neq 0$,*

$$\boxed{\widehat{\Psi}(k) = -\frac{\Gamma(-\chi_k)\zeta(1-\chi_k)}{L}}. \quad (251)$$

Every nonzero Fourier coefficient is nonzero. In particular, Ψ is genuinely nonconstant.

Proof. The zero coefficient is zero by definition. For $k \neq 0$, change variables $s = 2^t$ in (250). Dyadic periodization turns the integral over one logarithmic period into a Mellin transform; this is the standard harmonic-sum framework for dyadic periodicities [12]. The basic Mellin identity is

$$\int_0^\infty s^{z-1} \log(1 - e^{-s}) ds = -\Gamma(z)\zeta(1+z), \quad \Re z > 0. \quad (252)$$

It follows by expanding $\log(1 - e^{-s}) = -\sum_{m \geq 1} e^{-ms}/m$ and integrating term by term. Near $s = 0$ the kernel contains logarithmic and linear divergences; the quadratic correction in (246) is exactly the finite-part subtraction that removes them. Analytic continuation of the regularized Mellin transform to the imaginary axis gives (251) at $z = -\chi_k$.

The Gamma function has no zeros. Moreover $1 - \chi_k$ lies on the line $\Re z = 1$ away from the pole at 1, and the zeta function has no zeros there. Hence the product in (251) is nonzero. $\square$

Conjugate symmetry $\widehat{\Psi}(-k) = \overline{\widehat{\Psi}(k)}$ reconstructs a real-valued Fourier series. Stirling's estimate for Γ on vertical lines implies exponential decay of the coefficients, so the Fourier series and all of its differentiated series converge rapidly.

The first coefficient is tiny:

$$\widehat{\Psi}(1) \approx (7.5586475410 - 7.4701035590i) \times 10^{-7}, \quad (253)$$

with magnitude approximately $1.0627116252 \times 10^{-6}$. The second is smaller by six orders of magnitude,

$$\widehat{\Psi}(2) \approx (3.2481234772 + 5.8517589871i) \times 10^{-13}, \quad (254)$$

of magnitude $6.6927863679 \times 10^{-13}$, so the first harmonic already describes Ψ to better than one part in a million of its own size. Thus the visible oscillation is on the scale of a few parts in a million in the logarithm. Tiny is not the same as zero: its nonvanishing changes the logical status of an asymptotic equivalent.

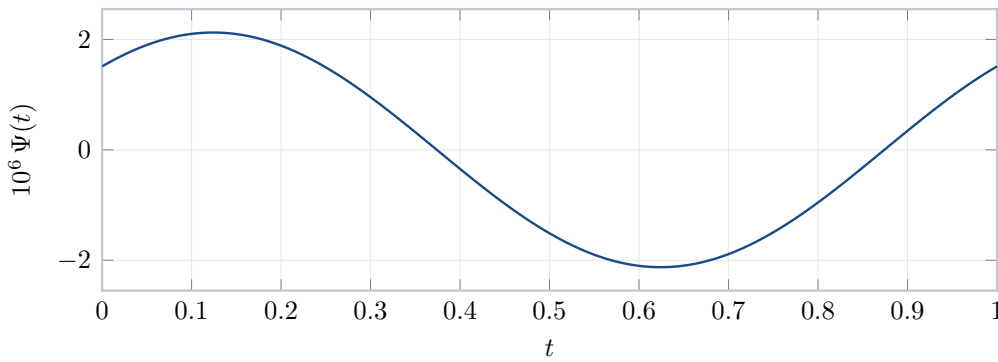


Figure 3: The periodic correction Ψ , magnified by 10^6 . The curve is indistinguishable from its first Fourier harmonic at this resolution, and the half-period antisymmetry $\Psi(t + \frac{1}{2}) \approx -\Psi(t)$ visible here is the statement that the even harmonics are negligible. The vertical scale is the point of the picture: the oscillation is a few parts in a million in the logarithm, but it does not tend to zero.

13.4 The finite part at the Mellin origin

The constant that enters the saddle asymptotic comes from the finite part of $-\Gamma(z)\zeta(1+z)$ at $z = 0$. Using

$$\begin{aligned} \Gamma(z) &= \frac{1}{z} - \gamma + \left(\frac{\gamma^2}{2} + \frac{\pi^2}{12}\right)z + \mathcal{O}(z^2), \\ \zeta(1+z) &= \frac{1}{z} + \gamma - \gamma_1 z + \mathcal{O}(z^2), \end{aligned}$$

one finds

$$-\Gamma(z)\zeta(1+z) + \frac{1}{z^2} = A_0 + \mathcal{O}(z), \quad (255)$$

where

$$A_0 = \frac{6\gamma^2 + 12\gamma_1 - \pi^2}{12}. \quad (256)$$

The final constant is

$$C_F = \frac{A_0}{L} - \frac{7L}{12} - \frac{1}{2} \log \pi \approx -2.02798389863885942397. \quad (257)$$

The terms $-7L/12$ and $-(1/2) \log \pi$ arise from the dyadic Euler–Maclaurin finite part and the leading Gaussian saddle mass, respectively.

14 The corrected sharp asymptotic

14.1 The lower Lambert coordinate

For sufficiently small $x > 0$, define

$$W = W_{-1}(-Lx), \quad (258)$$

where W_{-1} is the lower real branch of Lambert's W -function [11], and put

$$\lambda = \lambda(x) = -\frac{W}{L}. \quad (259)$$

Then $\lambda \rightarrow \infty$ as $x \downarrow 0$ and

$$\lambda 2^{-\lambda} = x. \quad (260)$$

Equivalently,

$$r = 2^\lambda = \frac{\lambda}{x}. \quad (261)$$

The real number r is the explicit Lambert saddle radius used in the contour analysis.

14.2 The main term

Define

$$\mathcal{M}(x) = -\frac{L}{2}\lambda^2 + \left(1 + \frac{L}{2}\right)\lambda - \frac{1}{2}\log \lambda + C_F + \Psi(\lambda), \quad (262)$$

where C_F is given by (257) and $\lambda = \lambda(x)$. The same expression can be written compactly in the Lambert variable W :

$$\mathcal{M}(x) = -\frac{7L}{12} - \frac{1}{2}\log(\pi x) + \frac{A_0 - W - \frac{1}{2}W^2}{L} + \Psi\left(-\frac{W}{L}\right). \quad (263)$$

The equivalence follows from $W = -L\lambda$ and $\log x = \log \lambda - L\lambda$.

Theorem 14.1 (Corrected sharp endpoint asymptotic). *As $x \downarrow 0$,*

$$\log F(x) = \mathcal{M}(x) + \mathcal{O}\left(\frac{1}{\lambda(x)}\right) = \mathcal{M}(x) + \mathcal{O}\left(\frac{1}{-\log x}\right). \quad (264)$$

Equivalently,

$$F(x) \sim \exp \mathcal{M}(x). \quad (265)$$

The periodic term is evaluated at the *exact* phase $\lambda(x)$, not at a truncated log-log approximation to it. This matters at the stated error scale.

14.3 Bromwich inversion

Because $G(-z) = \mathbb{E}e^{-zX}$ is entire and F is continuous, Laplace–Stieltjes inversion gives, for every $r > 0$ and $0 < x < 1$,

$$F(x) = \frac{1}{2\pi i} \int_{r-i\infty}^{r+i\infty} \frac{e^{zx} G(-z)}{z} dz. \quad (266)$$

Set $z = r(1 + i\theta)$ and choose $r = 2^\lambda = \lambda/x$. Then

$$F(x) = \frac{e^{\lambda+\Lambda(r)}}{2\pi} \int_{-\infty}^{\infty} e^{i\lambda\theta} \frac{G(-r(1+i\theta))}{G(-r)} \frac{d\theta}{1+i\theta}. \quad (267)$$

The exact real decomposition (247) at $r = 2^\lambda$ gives

$$\lambda + \Lambda(r) = -\frac{L}{2}\lambda^2 + \left(1 + \frac{L}{2}\right)\lambda + P(\lambda) + E(r). \quad (268)$$

Since

$$\bar{P} = \frac{A_0}{L} - \frac{L}{12}, \quad \bar{P} - \frac{1}{2}\log(2\pi) = C_F, \quad (269)$$

the missing factors in (262) are exactly the leading Gaussian mass $1/\sqrt{2\pi\lambda}$ and the centered fluctuation $\Psi = P - \bar{P}$.

14.4 The local phase

Let

$$u = \log(1 + i\theta), \quad (270)$$

with the principal logarithm near the saddle. Subtracting the value at $\theta = 0$ from the complexified version of (247), and including the factor $(1 + i\theta)^{-1}$ in (267), gives the local phase

$$\Phi_\lambda(\theta) = \lambda(i\theta - u) - \frac{u}{2} - \frac{u^2}{2L} + \Psi\left(\lambda + \frac{u}{L}\right) - \Psi(\lambda) + \mathcal{E}_r(\theta), \quad (271)$$

where $\mathcal{E}_r(0) = 0$ and, on a fixed neighborhood of zero, every derivative of $\mathcal{E}_r$ is exponentially small in r . This last term comes from the forward tail E and is far smaller than every power of $1/\lambda$.

The elementary expansion

$$\log(1 + i\theta) = i\theta + \frac{\theta^2}{2} - \frac{i\theta^3}{3} - \frac{\theta^4}{4} + \mathcal{O}(\theta^5) \quad (272)$$

shows that

$$\lambda(i\theta - u) = -\frac{\lambda\theta^2}{2} + \frac{i\lambda\theta^3}{3} + \frac{\lambda\theta^4}{4} + \mathcal{O}(\lambda|\theta|^5). \quad (273)$$

Thus the natural central scale is

$$\theta = \frac{v}{\sqrt{\lambda}}. \quad (274)$$

Uniformly for $|v|$ growing sufficiently slowly with λ ,

$$\Phi_\lambda(v/\sqrt{\lambda}) = -\frac{v^2}{2} + \lambda^{-1/2}P_1(v; \lambda) + \lambda^{-1}P_2(v; \lambda) + \mathcal{O}\left(\lambda^{-3/2}(1 + |v|^9)\right), \quad (275)$$

where

$$A(u) = \frac{\Psi'(u)}{L} - \frac{1}{2}, \quad (276)$$

$$B(u) = -\frac{1}{4} + \frac{1}{2L} + \frac{\Psi'(u)}{2L} - \frac{\Psi''(u)}{2L^2}, \quad (277)$$

$$P_1(v; u) = i\left(A(u)v + \frac{v^3}{3}\right), \quad (278)$$

$$P_2(v; u) = \frac{v^4}{4} + B(u)v^2. \quad (279)$$

In (275), the second argument is $u = \lambda$.

14.5 Central and tail estimates

For completeness, the quantitative saddle proof is organized by the following estimates. They are stated here in the form needed for the argument; the formal development proves them with explicit domains and constants.

Lemma 14.2 (Central expansion). *For every fixed truncation order N , there is $\delta > 0$ such that on $|\theta| \leq \delta$ the exponent in (267) admits a Taylor expansion through order N after the scaling (274). Its coefficients are smooth 1-periodic functions of λ , and the remainder is dominated by a polynomial in v times the next power of $\lambda^{-1/2}$.*

Proof. The logarithm in (270) is analytic for $|\theta| < 1$. The periodic function Ψ is smooth with bounded derivatives of every order. Taylor's theorem applied to each term of (271), together with the exponentially small bounds on $\mathcal{E}_r$, gives a uniform expansion. The dependence on λ occurs only through $\Psi^{(j)}(\lambda)$, hence is periodic. $\square$

Lemma 14.3 (Gaussian domination). *There are positive constants c, δ and λ_0 such that, for $\lambda \geq \lambda_0$ and $|\theta| \leq \delta$,*

$$\Re \Phi_\lambda(\theta) \leq -c\lambda\theta^2. \quad (280)$$

Proof architecture. The exact function $\Re(i\theta - \log(1 + i\theta)) = -(1/2) \log(1 + \theta^2)$ supplies the negative quadratic term. All remaining terms in (271) have bounded first and second derivatives on a fixed neighborhood. For large λ the $-\lambda\theta^2/2$ contribution dominates them. $\square$

Lemma 14.4 (Off-saddle decay). *For each N one may choose the central window so that the contribution to (267) outside that window is $O(\lambda^{-N})$ relative to the leading saddle mass.*

Proof architecture. Split the negative-Laplace product at the dyadic index nearest λ . In the intermediate range, a positive proportion of the factors have arguments of order one and their moduli are strictly smaller off the real axis, yielding exponential decay in $\lambda\theta^2$. In the far range, repeated use of the sinc/Laplace product bounds gives arbitrarily high polynomial decay. The two bounds overlap and cover the whole complement of the central window. The factor $(1 + i\theta)^{-1}$ is harmless. $\square$

These lemmas justify extending the scaled central integral to the entire real v -axis and integrating its expansion term by term against $e^{-v^2/2}$.

14.6 Completion of the leading proof

Keeping only the Gaussian term in (275) gives

$$\int_{\mathbb{R}} e^{\Phi_\lambda(\theta)} d\theta = \frac{1}{\sqrt{\lambda}} \int_{\mathbb{R}} e^{-v^2/2} dv (1 + \mathcal{O}(\lambda^{-1})) = \sqrt{\frac{2\pi}{\lambda}} (1 + \mathcal{O}(\lambda^{-1})). \quad (281)$$

The term of order $\lambda^{-1/2}$ integrates to zero because P_1 is odd in v and purely imaginary. Combining (268), (269), and (281) yields

$$\log F(x) = -\frac{L}{2}\lambda^2 + \left(1 + \frac{L}{2}\right)\lambda - \frac{1}{2}\log \lambda + C_F + \Psi(\lambda) + \mathcal{O}(\lambda^{-1}),$$

which is (264). Finally, $\lambda \asymp -\log x$ follows from (260), so the two displayed error scales are equivalent.

15 First correction and the all-orders saddle expansion

The leading theorem leaves an error of order $1/\lambda$. The same contour analysis can be continued to arbitrary order. The coefficient functions are periodic because every local coefficient is built from derivatives of Ψ evaluated at the exact phase λ .

15.1 The first correction

Expand the exponential in (275):

$$e^{\Phi_\lambda(v/\sqrt{\lambda})} = e^{-v^2/2} \left[1 + \lambda^{-1/2} P_1(v; \lambda) \right. \quad (282)$$

$$\left. + \lambda^{-1} \left(P_2(v; \lambda) + \frac{1}{2} P_1(v; \lambda)^2 \right) + \mathcal{O}(\lambda^{-3/2}(1 + |v|^{12})) \right]. \quad (283)$$

Let Z be a standard normal random variable. Since

$$\mathbb{E}Z^2 = 1, \quad \mathbb{E}Z^4 = 3, \quad \mathbb{E}Z^6 = 15, \quad (284)$$

the normalized coefficient of λ^{-1} in the saddle mass is

$$\begin{aligned} b_1(u) &= \mathbb{E} \left[P_2(Z; u) + \frac{1}{2} P_1(Z; u)^2 \right] \\ &= \frac{3}{4} + B(u) - \frac{1}{2} \left(A(u)^2 + 2A(u) + \frac{5}{3} \right). \end{aligned} \quad (285)$$

Substituting (276) and (277) simplifies the expression considerably.

Theorem 15.1 (First saddle correction). *Define*

$$A_1(u) = \frac{1}{24} + \frac{1}{2L} - \frac{\Psi''(u) + \Psi'(u)^2}{2L^2}. \quad (286)$$

Then

$$\boxed{\log F(x) = \mathcal{M}(x) + \frac{A_1(\lambda(x))}{\lambda(x)} + \mathcal{O}(\lambda(x)^{-2})}. \quad (287)$$

Proof. The order- $\lambda^{-1/2}$ term in (283) is odd and integrates to zero. Equation (285) gives the first nonzero relative correction to the integral. A direct simplification yields (286). If the normalized saddle mass is $1 + b_1/\lambda + \mathcal{O}(\lambda^{-2})$, then its logarithm is $b_1/\lambda + \mathcal{O}(\lambda^{-2})$, so the same coefficient appears in the logarithmic expansion. The off-saddle contribution is smaller than the stated remainder by the tail estimates of the preceding section. $\square$

The constant part $1/24 + 1/(2L)$ is the usual type of Gaussian/curvature correction. The terms involving Ψ' and Ψ'' record the fact that the log-periodic environment is not constant across the saddle width.

15.2 All orders

At order N , Taylor expansion of (271) produces finitely many polynomials in v whose coefficients are differential polynomials in

$$\Psi'(u), \Psi''(u), \dots, \Psi^{(2N)}(u).$$

After exponentiation, all odd powers of $\lambda^{-1/2}$ integrate to zero by parity. The remaining Gaussian moments are integers, so one obtains integer powers of $1/\lambda$.

Theorem 15.2 (Full logarithmic expansion). *There are smooth 1-periodic functions $A_j : \mathbb{R} \rightarrow \mathbb{R}$, $j \geq 0$, with $A_0 = 0$ and A_1 given by (286), such that for every $N \geq 1$,*

$$\boxed{\log F(x) = \mathcal{M}(x) + \sum_{j=1}^{N-1} \frac{A_j(\lambda(x))}{\lambda(x)^j} + \mathcal{O}(\lambda(x)^{-N})} \quad (288)$$

as $x \downarrow 0$.

Proof architecture. Fix N . Expand the analytic local phase through sufficiently high order in $\lambda^{-1/2}$, with a Gaussian-dominated remainder. Exponentiate using complete Bell polynomials, integrate term by term, and discard odd terms. The quantitative tail theorem makes the complement of the central region $O(\lambda^{-N})$. This gives a relative saddle mass expansion

$$1 + \sum_{j=1}^{N-1} \frac{b_j(\lambda)}{\lambda^j} + O(\lambda^{-N}), \quad (289)$$

where every b_j is smooth and periodic. Taking the formal logarithm defines the A_j . If

$$\log \left(1 + \sum_{j \geq 1} b_j z^j \right) = \sum_{j \geq 1} A_j z^j,$$

then the coefficients may be generated recursively by

$$A_n = b_n - \frac{1}{n} \sum_{k=1}^{n-1} k A_k b_{n-k}. \quad (290)$$

All operations preserve smooth periodic dependence on the phase. $\square$

This form of the expansion is preferable to a very long series in nested logarithms. The exact Lambert coordinate absorbs the dominant implicit inversion, while the exact periodic phase prevents phase errors from being magnified into false remainder claims.

16 Elementary log–log forms and the missing periodic term

Lambert W gives the cleanest formula, but an expansion in elementary logarithms is useful for comparison with published displays. It must, however, retain Ψ at the exact Lambert phase.

16.1 The literal small- x expansion

Put

$$S = -\log x, \quad m = \log S, \quad a = \log L. \quad (291)$$

Thus $S \rightarrow \infty$ as $x \downarrow 0$. Define the nonperiodic elementary expression

$$\begin{aligned} \mathcal{E}_0(x) = & -\frac{S^2}{2L} - \frac{Sm}{L} + \left(\frac{1}{2} + \frac{1+a}{L} \right) S \\ & - \frac{m^2}{2L} + \frac{am}{L} + \frac{6\gamma^2 + 12\gamma_1 - \pi^2 - 6a^2}{12L} - \frac{7L}{12} - \frac{1}{2} \log \pi \\ & - \frac{m^2}{2LS} + \frac{am}{LS}. \end{aligned} \quad (292)$$

Theorem 16.1 (Corrected elementary expansion). As $x \downarrow 0$,

$$\log F(x) = \mathcal{E}_0(x) + \Psi(\lambda(x)) + \mathcal{O}\left(\frac{1}{-\log x}\right). \quad (293)$$

Proof architecture. The standard lower-branch Lambert expansion solves $\lambda L - \log \lambda = S$ by successive substitution. Substituting that expansion into (262), and retaining enough phase terms to control the amplified quadratic contribution, gives (292). A coarse two-term approximation to λ is insufficient: because the main exponent contains $-L\lambda^2/2$, an error $\delta\lambda$ contributes about $-L\lambda\delta\lambda$, so the phase

must be known one order more accurately than a naive count suggests. The fourth Lambert phase term in the formal proof is exactly what reduces the residual to $O(1/S)$. $\square$

In the original variable $\ell = \log x < 0$, the same expression is

$$\begin{aligned} \mathcal{E}_0(x) = & -\frac{\ell^2}{2L} + \frac{\ell \log(-\ell)}{L} - \left(\frac{1}{2} + \frac{1 + \log L}{L} \right) \ell \\ & - \frac{\log^2(-\ell)}{2L} + \frac{\log L \log(-\ell)}{L} \\ & + \frac{6\gamma^2 + 12\gamma_1 - \pi^2 - 6 \log^2 L}{12L} - \frac{7L}{12} - \frac{1}{2} \log \pi \\ & + \frac{\log^2(-\ell)}{2L\ell} - \frac{\log L \log(-\ell)}{L\ell}. \end{aligned} \quad (294)$$

This is the literal elementary display formalized in the repository, with the periodic term added separately.

16.2 The dyadic logarithmic scale

Set $x = 2^{-t}$ and let

$$\lambda_t = -\frac{1}{L} W_{-1}(-L2^{-t}). \quad (295)$$

Then λ_t is the unique large solution of

$$\lambda_t - \log_2 \lambda_t = t. \quad (296)$$

Its first terms are

$$\lambda_t = t + \frac{\log t}{L} + \frac{\log t}{L^2 t} + \frac{\log t - \frac{1}{2} \log^2 t}{L^3 t^2} + \mathcal{O}\left(\frac{\log^3 t}{t^3}\right). \quad (297)$$

Define

$$\mathcal{D}_0(t) = -\frac{L}{2} t^2 - t \log t + \left(1 + \frac{L}{2}\right) t - \frac{\log^2 t}{2L} + C_F - \frac{\log^2 t}{2L^2 t}. \quad (298)$$

Theorem 16.2 (Corrected dyadic-scale expansion). *As $t \rightarrow \infty$,*

$$\boxed{\log F(2^{-t}) = \mathcal{D}_0(t) + \Psi(\lambda_t) + \mathcal{O}(t^{-1})}. \quad (299)$$

The literal substitution of $x = 2^{-t}$ into (292) differs from $\mathcal{D}_0(t)$ by the explicit term

$$\frac{\log^2 L}{2L^2 t}, \quad (300)$$

which is itself $O(1/t)$. Thus both nonperiodic displays are valid at the stated precision, but they are not algebraically identical.

16.3 Why the uncorrected formula is false at the claimed scale

Let $\mathcal{M}_0(x) = \mathcal{M}(x) - \Psi(\lambda(x))$ be the nonperiodic Lambert main. From (264),

$$\log F(x) - \mathcal{M}_0(x) = \Psi(\lambda(x)) + O(1/\lambda(x)). \quad (301)$$

Since Ψ is nonconstant, the right side does not tend to zero.

Proposition 16.3 (Failure without the periodic correction). *The relation*

$$\log F(x) = \mathcal{M}_0(x) + O(1/(-\log x)) \quad (302)$$

is false. In fact $F(x)/\exp(\mathcal{M}_0(x))$ has distinct subsequential limits.

Proof. Choose $u, v \in [0, 1)$ with $\Psi(u) \neq \Psi(v)$. Set

$$\lambda_n = n + u, \quad x_n = \lambda_n 2^{-\lambda_n}.$$

Then $x_n \downarrow 0$ and the exact Lambert phase of x_n is λ_n . Hence

$$\log F(x_n) - \mathcal{M}_0(x_n) \rightarrow \Psi(u).$$

The analogous sequence with phase $n + v$ tends to $\Psi(v)$. Since $1/(-\log x) \rightarrow 0$, the bound (302) is impossible. Exponentiating gives distinct ratio limits $e^{\Psi(u)}$ and $e^{\Psi(v)}$. $\square$

Numerically the discrepancy is minute because the first Fourier mode has size about 10^{-6} , but asymptotic equivalence is a limit statement, not a graphical tolerance. A nonzero bounded oscillation cannot be hidden inside an error term that tends to zero.

16.4 Symmetry at the right endpoint

The symmetry $F(1 - x) = 1 - F(x)$ immediately transfers the left-endpoint expansion to the right:

$$1 - F(1 - x) = F(x). \quad (303)$$

Thus the same Lambert phase, periodic fluctuation, and all-orders coefficients describe the survival probability near 1.

17 Consequences of the endpoint expansion

The full sharp formula immediately settles several qualitative questions that are otherwise awkward to read from the differential equation.

Theorem 17.1 (Quadratic logarithmic law). *As $x \downarrow 0$,*

$$\frac{\log F(x)}{(\log x)^2} \rightarrow -\frac{1}{2 \log 2}. \quad (304)$$

Equivalently, on the dyadic scale,

$$\log F(2^{-t}) \sim -\frac{\log 2}{2} t^2. \quad (305)$$

Proof. In (262), $\lambda \sim S/L$ with $S = -\log x$. The quadratic term $-L\lambda^2/2$ dominates the linear term, the logarithmic term, and the bounded periodic correction. Substitution gives (304). $\square$

Corollary 17.2 (Between powers and pure endpoint exponentials). *For every $N > 0$ and every $c > 0$,*

$$F(x) = o(x^N), \quad (306)$$

and

$$e^{-c/x} = o(F(x)) \quad (307)$$

as $x \downarrow 0$.

Proof. The logarithm of x^N is $-NS$, whereas $\log F(x) \sim -S^2/(2L)$, so $F(x)/x^N \rightarrow 0$. On the other hand, $\log e^{-c/x} = -ce^S$, which is much more negative than $-S^2/(2L)$; therefore $e^{-c/x}/F(x) \rightarrow 0$. $\square$

This places the endpoint decay in an intermediate regime. The function is flatter than any finite algebraic order, as every compactly supported smooth function must be at its support boundary, but it is vastly larger than the familiar model $e^{-1/x}$.

Corollary 17.3 (Asymptotics of reciprocal-power values). *For integers $n \rightarrow \infty$,*

$$\begin{aligned} \log F(2^{-n}) = & -\frac{L}{2}n^2 - n \log n + \left(1 + \frac{L}{2}\right)n - \frac{\log^2 n}{2L} + C_F \\ & - \frac{\log^2 n}{2L^2n} + \Psi(\lambda_n) + O(n^{-1}), \end{aligned} \quad (308)$$

where λ_n is defined by (295). Combining this with (102) gives an equally sharp expansion for the half moments d_n .

The formula shows why a fixed multiplicative constant cannot complete a recurrence-based ansatz for $F(2^{-n})$: the fractional parts of λ_n drift, and the factor $\exp \Psi(\lambda_n)$ is nonconstant.

18 The Fourier–Legendre expansion of up

A second exact spectral representation uses the ordinary Legendre polynomials P_n on $[-1, 1]$. Unlike the Fourier transform, this is a polynomial expansion on the natural support interval, and its coefficients can be expressed by the reciprocal-power dyadic values already computed.

18.1 Orthogonality and coefficients

Normalize the Legendre polynomials by $P_n(1) = 1$. They satisfy

$$-\frac{d}{dx}((1-x^2)P'_n(x)) = n(n+1)P_n(x), \quad (309)$$

with orthogonality

$$\int_{-1}^1 P_m(x)P_n(x) dx = \frac{2}{2n+1} \delta_{mn}. \quad (310)$$

For a continuous function f , define

$$\alpha_n(f) = \frac{2n+1}{2} \int_{-1}^1 f(x)P_n(x) dx. \quad (311)$$

Because up is even and $P_n(-x) = (-1)^n P_n(x)$,

$$\alpha_{2n+1}(\text{up}) = 0. \quad (312)$$

Put

$$u_n = \alpha_{2n}(\text{up}) = \frac{4n+1}{2} \int_{-1}^1 \text{up}(x)P_{2n}(x) dx. \quad (313)$$

18.2 Exact coefficient formula

The even Legendre polynomial has the finite power expansion

$$P_{2n}(x) = 4^{-n} \sum_{k=0}^n (-1)^{n+k} \binom{2n}{n+k} \binom{2n+2k}{2n} x^{2k}. \quad (314)$$

The moment-to-dyadic bridge is

$$c_k = 2(2k)! 2^{\binom{2k+1}{2}} F(2^{-2k-1}). \quad (315)$$

This is (108) with an even exponent, together with symmetry.

Theorem 18.1 (Exact Legendre coefficients). *For every $n \geq 0$,*

$$\begin{aligned} u_n &= 4^{-n} (4n+1) \sum_{k=0}^n (-1)^{n+k} \binom{2n}{n+k} \binom{2n+2k}{2n} \\ &\quad \times (2k)! 2^{\binom{2k+1}{2}} F(2^{-2k-1}). \end{aligned} \quad (316)$$

In particular every u_n is rational.

Proof. Insert (314) into (313) and interchange the finite sum with the integral. The integral of $x^{2k} \text{up}(x)$ is c_k . Substitution of (315) cancels the factor $1/2$ in (313) and gives (316). $\square$

Thus the entire polynomial spectral expansion is anchored in the same exact arithmetic table $F(2^{-m})$ that drives the bit-recursive evaluator.

Example 18.2 (The first two Legendre coefficients). Take $n = 0$ and $n = 1$ in (313). Since $P_0 = 1$ and $P_2(x) = (3x^2 - 1)/2$,

$$\begin{aligned} u_0 &= \frac{1}{2} \int_{-1}^1 \text{up}(x) dx = \frac{c_0}{2} = \frac{1}{2}, \\ u_1 &= \frac{5}{2} \int_{-1}^1 \text{up}(x) \frac{3x^2 - 1}{2} dx = \frac{5}{4} (3c_1 - c_0) = \frac{5}{4} \left(\frac{1}{3} - 1 \right) = -\frac{5}{6}. \end{aligned}$$

The exact formula (316) returns the same two numbers without reference to the moments. For $n = 0$ the sum has the single term $\binom{0}{0} \binom{0}{0} 0! 2^{\binom{1}{2}} F(2^{-1}) = \frac{1}{2}$, so $u_0 = \frac{1}{2}$. For $n = 1$,

$$u_1 = \frac{5}{4} \left[-\binom{2}{1} \binom{2}{2} 0! 2^{\binom{1}{2}} F(2^{-1}) + \binom{2}{2} \binom{4}{2} 2! 2^{\binom{3}{2}} F(2^{-3}) \right] = \frac{5}{4} \left[-1 + \frac{96}{288} \right] = -\frac{5}{6},$$

using $\binom{1}{2} = 0$, $\binom{3}{2} = 3$, $F(2^{-1}) = \frac{1}{2}$ and $F(2^{-3}) = \frac{1}{288}$ from (109). Hence the Fourier–Legendre expansion (320) begins

$$\text{up}(t) = \frac{1}{2} - \frac{5}{6} P_2(t) + \dots \quad (317)$$

Both values are stated in the normalization (313) used here, in which $u_n = \alpha_{2n}(\text{up})$ already carries the factor $(4n+1)/2$ of (311); the bare integrals are $\int_{-1}^1 \text{up} P_0 = 1$ and $\int_{-1}^1 \text{up} P_2 = -\frac{1}{3}$. The negative sign of u_1 is genuine and reflects the fact that up is more sharply peaked at the origin than the constant $\frac{1}{2}$.

18.3 Absolute and uniform convergence

The Legendre Sturm–Liouville operator

$$\mathcal{L}f = -\frac{d}{dx}((1-x^2)f'(x)) \quad (318)$$

is formally self-adjoint on $[-1, 1]$. Boundary terms vanish because of the factor $1 - x^2$. Repeated integration by parts therefore gives, for $n \geq 1$,

$$\alpha_n(f) = \frac{2n+1}{2[n(n+1)]^m} \int_{-1}^1 (\mathcal{L}^m f)(x) P_n(x) dx. \quad (319)$$

Since $|P_n(x)| \leq 1$ on $[-1, 1]$, any C^∞ function has coefficients that decay faster than every power of n .

Theorem 18.3 (Uniform Legendre series). *On the entire closed interval $[-1, 1]$,*

$$\boxed{\text{up}(x) = \sum_{n=0}^{\infty} u_n P_{2n}(x).} \quad (320)$$

The series converges absolutely and uniformly, including at both endpoints.

Proof. Apply (319) to $f = \text{up}$. Choosing $m \geq 2$ gives $|\alpha_n(\text{up})| \leq Cn^{-3}$, say. Since $|P_n| \leq 1$, the Weierstrass test gives absolute and uniform convergence. Let the uniform sum be g . Orthogonality shows that g and up have the same Legendre coefficients. Completeness of the Legendre polynomials in $L^2[-1, 1]$ implies $g = \text{up}$ almost everywhere, and continuity upgrades equality to every point. Odd coefficients vanish by (312), leaving (320). $\square$

The endpoint statement is worth emphasizing. Generic Fourier–Legendre convergence theorems are often quoted only on the open interval, or with an endpoint average. Here smoothness and absolute uniform convergence give the literal values $\text{up}(\pm 1) = 0$.

18.4 A practical polynomial approximation

Truncating (320) at $n = N$ gives a polynomial of degree $2N$ with exact rational coefficients after expanding the P_{2n} . The error is bounded by

$$\left\| \text{up} - \sum_{n=0}^N u_n P_{2n} \right\|_{\infty} \leq \sum_{n>N} |u_n|. \quad (321)$$

Repeated use of (319) supplies arbitrarily high algebraic tail bounds. For moderate precision this method is attractive: all coefficients are exact, evaluation is stable on $[-1, 1]$, and the approximation respects evenness automatically.

18.5 Least-squares optimality of the partial sums

The Legendre truncation is not only a convenient polynomial approximation; it is the exact orthogonal projection in $L^2[-1, 1]$. Define

$$S_N(x) = \sum_{n=0}^N u_n P_{2n}(x). \quad (322)$$

Theorem 18.4 (Pythagorean error identity). *For every real polynomial p of degree at most $2N + 1$,*

$$\int_{-1}^1 (\text{up}(x) - p(x))^2 dx = \int_{-1}^1 (\text{up}(x) - S_N(x))^2 dx + \int_{-1}^1 (p(x) - S_N(x))^2 dx. \quad (323)$$

Consequently S_N is the unique least-squares approximant among all such polynomials.

Proof. Every polynomial of degree at most $2N + 1$ is a linear combination of $P_0, \dots, P_{2N+1}$. By construction, $\text{up} - S_N$ is orthogonal to each even basis element through P_{2N} ; by evenness it is also orthogonal to every odd basis element. Hence

$$\int_{-1}^1 (\text{up} - S_N)(p - S_N) dx = 0.$$

Expanding $\text{up} - p = (\text{up} - S_N) - (p - S_N)$ gives (323). Equality with the minimum forces the final nonnegative integral to vanish. A polynomial vanishing almost everywhere vanishes identically, so $p = S_N$. $\square$

The one-degree slack is genuine: the odd coefficient of P_{2N+1} is zero. Thus a partial sum visibly of degree $2N$ is already optimal in the larger space of degree at most $2N + 1$.

19 Additional summation and spectral identities

19.1 The cosine series on the support

Periodize up with period 2. Since neighboring support intervals meet only at zeros, the periodization agrees with up on $[-1, 1]$. Its Fourier coefficients are $\widehat{\text{up}}(n/2)/2$, so rapid Fourier decay gives the uniformly convergent cosine series

$$\text{up}(x) = \frac{1}{2} + \sum_{n=1}^{\infty} \widehat{\text{up}}(n/2) \cos(\pi n x), \quad -1 \leq x \leq 1. \quad (324)$$

Using (55), every coefficient is an explicit infinite product. Many coefficients vanish because one of the sinc factors is evaluated at a nonzero integer.

19.2 Scaled partitions

Equation (60) may be differentiated term by term. For $j \geq 1$,

$$\sum_{k \in \mathbb{Z}} \text{up}^{(j)}\left(t + \frac{k}{m}\right) = 0. \quad (325)$$

The sums are locally finite. Combining this with (68) turns the identity into exact finite cancellations among signed translates of $\mathcal{F}$ at dyadic scales.

19.3 Mean and variance of the Fabius law

From the random-series representation,

$$\mathbb{E}X = \sum_{n \geq 0} \frac{1/2}{2^{n+1}} = \frac{1}{2}. \quad (326)$$

Since $\text{Var}(U_n) = 1/12$ and the coordinates are independent,

$$\text{Var}(X) = \frac{1}{12} \sum_{n \geq 0} 4^{-n-1} = \frac{1}{36}. \quad (327)$$

For $Y = 2X - 1$, this gives $\mathbb{E}Y = 0$ and $\text{Var}(Y) = 1/9$, agreeing with $c_1 = 1/9$. Higher even moments are exactly the c_n of (79).

19.4 An exact self-convolution law

Splitting the random series into its even and odd coordinates gives two independent scaled copies with different weights. At the transform level this is the factorization

$$\widehat{\text{up}}(z) = \widehat{\text{up}}(z/2) \text{sinc}(\pi z). \quad (328)$$

In physical space,

$$\text{up}(x) = 2 \int_{-1/2}^{1/2} \text{up}(2x - 2t) dt = \int_{2x-1}^{2x+1} \text{up}(y) dy, \quad (329)$$

with the normalization understood as convolution with the uniform density on $[-1/2, 1/2]$. Differentiating this integral refinement recovers (28). Conversely, integrating the differential equation and using compact support recovers the convolution law.

20 Computation in exact and numerical regimes

The identities above do not lead to a single universally best numerical method. They lead to a collection of algorithms adapted to different questions. This section makes the distinctions explicit, since confusing an exact arithmetic formula with an endpoint asymptotic is as unhelpful as using a dense dyadic table to evaluate one value at 2^{-10^6} .

20.1 Which representation should be used?

Representation	Principal strength	Principal limitation
Contraction fixed point	A direct constructive proof; after m iterations the uniform error is geometrically small.	Produces approximating functions rather than exact individual values.
Finite convolution / probability	Positive approximants, natural sampling, and a transparent explanation of smoothness.	High resolution requires many convolution levels.
Centered finite spline	An explicit finite formula with certified uniform error at most 2^{-p} .	Direct summation has exponentially many terms unless its recursion is compiled.
Fourier sinc product	Entire transform, rapid frequency decay, and efficient evaluation away from extreme endpoint cancellation.	Recovering a tiny value of $F(x)$ by oscillatory inversion may lose many digits.
Exact dyadic formula	A closed rational answer and an algebraic proof of representation invariance.	The literal Thue–Morse sum may contain exponentially many terms.
Bit-recursive Taylor evaluator	Exact rational arithmetic with at most one recursive call per set bit of the numerator.	Rational numerators and denominators themselves become very large.
Legendre series	Uniform polynomial approximation on the full closed support, with exact coefficients.	Not specially adapted to resolving the boundary layer at a tiny x .
Lambert saddle expansion	Stable evaluation of $\log F(x)$ at extraordinarily small positive x .	Asymptotic rather than exact; the periodic correction must be retained.

For the fixed-point construction, let T be the transform of (24). If f_0 is any admissible continuous function, then

$$\|T^m f_0 - F\|_\infty \leq 2^{-m} \|f_0 - F\|_\infty \leq 2^{-m}. \quad (330)$$

Thus the contraction proof itself is a certified numerical scheme. It is excellent for coarse graphs and for constructing a first approximation without any prior arithmetic tables. Exact dyadic computation, however, should use the rational recurrences.

20.2 Precomputing the reciprocal-power table

Put

$$V_n = F(2^{-n}). \quad (331)$$

Eliminating d_n from (88) and (102) gives an especially convenient recurrence involving only earlier V_k :

$$V_0 = 1, \quad V_{n+1} = \frac{1}{2^{n+1} - 1} \sum_{k=0}^n \frac{V_k}{2^{\binom{n+1}{2} - \binom{k}{2}} (n+2-k)!}. \quad (332)$$

Every operation is rational. A practical implementation stores the values in an array, reduces each rational after every addition, and reuses factorials and powers of two. This table is precisely the data required by the Taylor blocks in (144).

There are three useful normalization choices.

1. Store V_n directly. This makes the public API simple and is ideal when values are eventually returned as rational numbers.
2. Store $d_n = n!2^{\binom{n}{2}} V_n$. The recurrence has smaller explicit powers of two, and the parity theorem provides a strong integrity check.
3. Store the division-free natural numerators D_n of the half moments, defined by (90). This avoids fraction reduction inside the recurrence but requires restoring a known structured denominator at the end.

The third choice is usually best for bulk generation; the first is usually best for a small library interface.

20.3 Operation count of the bit recursion

Suppose $0 < a < 2^n$, and write the set-bit positions of a in decreasing order as

$$b_1 > b_2 > \cdots > b_h \geq 0, \quad a = \sum_{j=1}^h 2^{b_j}. \quad (333)$$

The evaluator of [theorem 9.11](#) makes exactly $h = \text{popcount}(a)$ nontrivial recursive calls. At the j th call, the Taylor order is $n - b_j$. Hence the number of rational multiply-add steps in straightforward Horner evaluation is

$$\sum_{j=1}^h (n - b_j) \leq nh \leq n^2. \quad (334)$$

This is an arithmetic-operation count, not a bit-complexity estimate. The bit lengths of the exact numerator and denominator grow quadratically in n already at $a = 1$, as the factor $2^{\binom{n}{2}}$ in (102) makes unavoidable.

By contrast, the literal formula (130) sums over every $0 \leq h < a$. When a is comparable with 2^n , it therefore has exponentially many outer terms. Its value is conceptual rather than algorithmic: it proves rationality, exposes Thue–Morse cancellation, and connects the answer to q -binomial identities. The Taylor recursion is the compiled form of that cancellation.

Unreduced inputs cause no mathematical difficulty. The exact formula and the executable evaluator satisfy

$$F\left(\frac{2a}{2^{n+1}}\right) = F\left(\frac{a}{2^n}\right). \quad (335)$$

Reducing the pair (a, n) first merely avoids unnecessary work. Signed inputs should be handled at the outermost layer: clamp to 0 for $a \leq 0$, clamp to 1 for $a \geq 2^n$, and call the unit-interval recursion only in the interior.

20.4 Stable endpoint evaluation

For a very small non-dyadic x , direct evaluation of $F(x)$ is the wrong numerical variable. One should compute $\log F(x)$. Set

$$S = -\log x, \quad L = \log 2. \quad (336)$$

The exact saddle coordinate is the positive solution of

$$L\lambda - \log \lambda = S. \quad (337)$$

Although a library implementation may use the W_{-1} branch directly, Newton iteration is simple and stable:

$$\lambda_{j+1} = \lambda_j - \frac{L\lambda_j - \log \lambda_j - S}{L - \lambda_j^{-1}}. \quad (338)$$

For large S , the starting value

$$\lambda_0 = \frac{S + \log S - \log L}{L} \quad (339)$$

already has small relative error. Once λ is known, evaluate (262) and, when needed, add $A_1(\lambda)/\lambda$ from (286).

The periodic function Ψ should be evaluated by its Fourier series (251). The Gamma factor in (251) decays exponentially with the frequency, while the zeta factor has only moderate growth on the relevant vertical line. Consequently a handful of modes usually suffices for precision far beyond that of ordinary floating-point arithmetic. Since Ψ is real,

$$\Psi(u) = \widehat{\Psi}(0) + 2\Re \sum_{k=1}^K \widehat{\Psi}(k) e^{2\pi i k u} + \text{exponentially small Fourier tail}. \quad (340)$$

Reducing u modulo 1 before evaluating the exponentials avoids loss of phase accuracy.

The saddle expansion and the exact dyadic evaluator overlap over a substantial range. Their agreement is a powerful independent test. For $x = 2^{-n}$, an implementation can compare the exact rational logarithm (computed at high precision) against (308). The difference after the first correction should be $O(n^{-2})$, with the phase evaluated at the exact λ_n , not at n .

20.5 A verification suite suggested by the mathematics

An exact implementation should test identities, not merely decimal snapshots. Particularly effective checks are

$$F(a/2^n) + F((2^n - a)/2^n) = 1, \quad (341)$$

$$F(2a/2^{n+1}) = F(a/2^n), \quad (342)$$

$$v_2(F(2^{-n})) = -\binom{n}{2} - 1 - v_2(n!), \quad (343)$$

$$F(5/16) = \frac{305857}{2073600}, \quad (344)$$

$$R_n \equiv 1 \pmod{2}. \quad (345)$$

The Taylor-block identity can be tested for every a on a modest dyadic grid; the q -binomial expression can be evaluated at several unrelated translations q to test its proved translation independence. For floating-point code, the partition of unity (60), Fourier refinement (56), and agreement between the Legendre and Fourier reconstructions provide complementary global tests.

20.6 Effective uniform continuity and sequential computability

Computability has a precise recursion-theoretic meaning here, not merely the informal meaning that a numerical routine can be written. We use the classical Grzegorzczuk formulation [10], expressed through fast dyadic names.

Definition 20.1 (Fast signed-dyadic name). A signed-natural numerator is a pair $c = (c^+, c^-) \in \mathbb{N}^2$. At precision p it represents

$$\langle c \rangle_p = \frac{c^+ - c^-}{2^p}. \quad (346)$$

A real sequence $(x_i)_{i \geq 0}$ is *computable* if one recursive algorithm, uniform in both i and p , produces pairs $a(i, p)$ such that

$$|x_i - \langle a(i, p) \rangle_p| \leq 2^{-p}. \quad (347)$$

A function $f: \mathbb{R} \rightarrow \mathbb{R}$ is *sequentially computable* if it sends every computable real sequence to a computable real sequence.

The two-natural representation is deliberately redundant. It is extensionally the usual signed dyadic representation, but its arithmetic can be certified using primitive recursion on natural numbers without hiding an integer or rational coding convention.

Definition 20.2 (Effective uniform continuity). A function $f: \mathbb{R} \rightarrow \mathbb{R}$ is *effectively uniformly continuous* if there is a recursive $d: \mathbb{N} \rightarrow \mathbb{N}$ such that, for every positive n ,

$$d(n) > 0, \quad |x - y| < \frac{1}{d(n)} \implies |f(x) - f(y)| < \frac{1}{n}. \quad (348)$$

It is a computable real function in this sense if it is both sequentially computable and effectively uniformly continuous.

Restricting (348) to $n > 0$ and requiring $d(n) > 0$ are substantive, not cosmetic. At $n = 0$ the reciprocal accuracy request is meaningless; in a totalized division formalism, allowing $d(n) = 0$ would make the implication vacuous.

Theorem 20.3 (Computability of the bounded Fabius function). *The constant-tail extension of the bounded Fabius distribution function is sequentially computable and effectively uniformly continuous. A primitive-recursive modulus is*

$$d(n) = 2n \quad (n > 0). \quad (349)$$

Consequently $F: \mathbb{R} \rightarrow [0, 1]$ is a computable real function in the preceding sense.

Proof. The effective-continuity clause follows immediately from the global 2-Lipschitz estimate: if $|x - y| < 1/(2n)$, then

$$|F(x) - F(y)| \leq 2|x - y| < \frac{1}{n}.$$

Multiplication by the constant 2 is primitive recursive, so (349) is an effective modulus.

It remains to give one recursive transformer of fast names. Suppose an output of precision p is requested, and put $s = p + 3$. Read the input numerator $c = (c^+, c^-)$ at precision s , and clamp it to the unit interval by the natural-number operation

$$a = \begin{cases} 0, & c^+ \leq c^-, \\ 2^s, & 2^s + c^- \leq c^+, \\ c^+ - c^-, & \text{otherwise.} \end{cases} \quad (350)$$

Because F has constant tails, this clamp does not change its value. Evaluate the centered spline at the matched grid point $a/2^s$. Formula (46) simplifies to the exact rational number

$$S_s\left(\frac{a}{2^s}\right) = \frac{\sum_{r=0}^{a-1} \varepsilon_r (2a - 2r - 1)^s}{2^{\binom{s+1}{2}} s!}. \quad (351)$$

On this clamped grid the numerator is nonnegative. It can nevertheless be computed without signed arithmetic: keep separate positive and negative accumulators according to the Thue–Morse bit.

Every part of this evaluator is primitive recursive. More explicitly, compute the Thue–Morse bit from

$$b(0) = 0, \quad b(m) = (b(\lfloor m/2 \rfloor) + (m \bmod 2)) \bmod 2 \quad (m > 0), \quad (352)$$

fold the finite range $0 \leq r < a$ using addition, multiplication, subtraction, and powers on natural numbers, and compute the denominator from

$$\delta_0 = 1, \quad \delta_{s+1} = \delta_s 2^{s+1} (s+1), \quad \delta_s = 2^{\binom{s+1}{2}} s!. \quad (353)$$

Finally round the rational in (351) to the nearest multiple of 2^{-p} , breaking ties upward. If N and D are its nonnegative numerator and positive denominator, the returned numerator is

$$Q = \left\lfloor \frac{2(2^p N) + D}{2D} \right\rfloor. \quad (354)$$

Bounded range folds, natural division, and the displayed recursion are all closed under primitive recursion; thus $(c, p) \mapsto (Q, 0)$ is one certified recursive algorithm.

It remains to verify its precision. The centered-spline error (52) at level $s = p + 3$ costs one unit of 2^{-s} . Nearest-grid rounding costs

$$\frac{1}{2} 2^{-p} = 4 \cdot 2^{-(p+3)}.$$

Hence the evaluator itself has error at most

$$5 \cdot 2^{-(p+3)}. \quad (355)$$

When the input is the s th approximation to an arbitrary real x , propagation through the 2-Lipschitz function costs another

$$2 \cdot 2^{-(p+3)}.$$

The total is therefore

$$(2 + 5)2^{-(p+3)} = 7 \cdot 2^{-(p+3)} \leq 8 \cdot 2^{-(p+3)} = 2^{-p}.$$

Applying the same algorithm uniformly to the two indices (i, p) produces a fast name for $F(x_i)$ from any fast name for (x_i) . This proves sequential computability and completes the theorem. $\square$

Remark 20.4 (Scope of the formal computability theorem). The signed global extension $\mathcal{F}$ is also globally 2-Lipschitz, so the same modulus $d(n) = 2n$ proves its effective uniform continuity. The formal development packages the primitive-recursive spline evaluator and the combined sequential-computability theorem for the bounded constant-tail function F . It does not claim a packaged global sequential computability theorem for $\mathcal{F}$.

21 Corrections, conventions, and proof audit

A formal development is most useful when it records not only what is true but also exactly where an attractive informal shortcut ceases to be valid. The following points are the ones most likely to create a silent error in a human treatment.

21.1 Bounded versus signed-global Fabius functions

The bounded function F takes values in $[0, 1]$ and is constant to the right of 1. The signed extension $\mathcal{F}$ is not bounded in that way and instead obeys

$$\mathcal{F}'(x) = 2\mathcal{F}(2x) \quad (x \in \mathbb{R}).$$

They agree for every $x \leq 1$, hence in particular on $[0, 1]$, but they are globally distinct. Thus a statement such as “ $\mathcal{F}'(x) = 2\mathcal{F}(2x)$ for all real x ” is correct only when F denotes the signed extension. The explicit Thue–Morse series (11) makes the distinction unavoidable and proves that the global sum is locally finite rather than merely conditionally convergent.

Likewise, the support statement for up concerns the topological support: $\text{supp up} = [-1, 1]$, even though $\text{up}(\pm 1) = 0$. Endpoint formulas must distinguish closed support from the open set on which the function is positive.

21.2 The Reshetnikov normalization

Equation (124) contains the factor $2^{+\binom{n-1}{2}}$. The negative exponent printed in the abstract of one version of the arithmetic paper is incompatible with its own equation (27) and with the integrality theorem. The positive exponent is the one proved by the exact recurrence. The resulting first values are

$$R_1, R_2, \dots, R_7 = 1, 5, 15, 1001, 5985, 2853675, 26261235, \quad (356)$$

all odd as predicted.

The strongest common-denominator formula in the arithmetic paper depends on a separate divisibility conjecture. The formal development keeps the unconditional multiplier bound and the conditional exact denominator statement logically separate. A computational match for the first many levels is evidence for the conjecture, not a proof.

21.3 The local asymptotic derivation

Write the logarithmic endpoint profile in a dyadic variable as

$$g(t) = \log F(2^{-t}). \quad (357)$$

A proposed elementary derivation substitutes a main term $G(t)$ into the exact difference-differential relation. Two steps in that derivation cannot be made for an unspecified remainder $R = g - G$:

1. a coefficient comparison drops the actual difference $R(t) - R(t-1)$;
2. a later estimate uses a bound on $R'(t)$ that has not been assumed or proved.

Even the explicit main term has a residual of order

$$\left(\frac{\log t}{t} \right)^2, \quad (358)$$

with a nonzero leading coefficient; it is not $O(t^{-2})$. Therefore the printed iteration does not establish the claimed bounded periodic remainder. The rigorous route is the exact negative-Laplace decomposition of [section 13](#), followed by Bromwich inversion and a quantitative saddle analysis.

There is also a small but conceptually useful conversion issue. Substituting $x = 2^{-t}$ in the literal elementary small- x expression does not give the separately displayed dyadic expression term for term. Their difference is

$$\frac{(\log \log 2)^2}{2(\log 2)^2 t}, \quad (359)$$

which is $O(t^{-1})$. It is harmless for a coarser theorem but must be retained when the display itself claims accuracy through the inverse-logarithmic term.

21.4 Why the periodic term cannot be averaged away

Replacing $\Psi(\lambda)$ by its mean changes the logarithm by a bounded quantity, so it preserves only very coarse statements such as (304). It does not preserve a sharp multiplicative asymptotic. Indeed, for every nonzero integer k ,

$$\widehat{\Psi}(k) = -\frac{\Gamma(-\chi_k)\zeta(1-\chi_k)}{\log 2} \neq 0.$$

The nonvanishing follows from the absence of zeros of Γ , the fact that $\zeta(s) \neq 0$ on $\Re s = 1$ away from its pole, and $\chi_k \neq 0$. Hence Ψ is nonconstant. Choosing two fractional saddle phases where it takes different values produces two sequences along which the ratio to an uncorrected main term has different limits. This is a structural obstruction, not a matter of choosing a better constant.

21.5 A visually good quotient of exponentials has the wrong boundary scale

A proposed elementary approximation to the displaced bump has the form

$$Q(x) = \frac{1 + \exp(-|x|^{-\alpha})}{1 + \exp((1-2|x|)/(x^2 - |x|))}, \quad \alpha = \frac{3}{4}\sqrt{2\pi}. \quad (360)$$

It can look strikingly accurate on a plot. Put $y = x + 1$, the distance from the left support endpoint. For $0 < y \leq 1/4$ one has the rigorous bound

$$Q(-1+y) \leq 2 \exp\left(-\frac{1}{2y}\right). \quad (361)$$

But [theorem 17.2](#) shows

$$\exp(-c/y) = o(F(y)) \quad (y \downarrow 0) \quad (362)$$

for every $c > 0$. Therefore

$$Q(-1+y) = o(F(y)), \quad (363)$$

so the quotient is not asymptotically equivalent at the endpoint. This is a useful warning: a graph on a linear scale is almost blind to the distinction between $\exp[-\Theta((\log(1/y))^2)]$ and $\exp[-\Theta(1/y)]$ once both values are tiny.

21.6 The literal Thue–Morse prefix normalization does not converge

The construction of [section 12](#) has a shorter and more attractive reading, in which the normalized grid $\mathcal{V}_k$ of (223) is plotted directly against the abscissae $x_{k,j} = j/2^k$ and the resulting broken lines are asserted to converge, uniformly on $[0, 1]$, to the Fabius function. That assertion is false, and it fails in the strongest possible way: not merely without a rate, and not merely nonuniformly, but pointwise at the right endpoint.

Theorem 21.1 (Endpoint obstruction). *For every $k \geq 0$,*

$$\mathcal{V}_k(2^k) = -\frac{1}{2^{\binom{k}{2}}} < 0, \quad |\mathcal{V}_k(2^k) - 1| = 1 + \frac{1}{2^{\binom{k}{2}}} > 1. \quad (364)$$

In particular the sequence $k \mapsto \mathcal{V}_k(2^k)$ of values at $x = 1$ does not converge to $F(1) = 1$; every term is negative, so any limit is at most 0.

Proof. The index 2^k is the right boundary index of [theorem 12.7](#) with $r = k$, so $\sigma_k(2^k) = -1$ by (215). Divide by $2^{\binom{k}{2}}$ to obtain the first identity, and note $-2^{-\binom{k}{2}} < 0 < 1$ for the second. A sequence of nonpositive numbers cannot converge to 1. $\square$

The obstruction is not an artifact of reading the displayed floor function literally, which produces a step function rather than a curve. Let Π_k denote the polygon obtained by joining the consecutive points $(x_{k,j}, \mathcal{V}_k(j))$ by line segments, which is the object the construction intends. By construction Π_k interpolates the grid: $\Pi_k(x_{k,j}) = \mathcal{V}_k(j)$ for every j , and in particular

$$\Pi_k(1) = \mathcal{V}_k(2^k) = -\frac{1}{2^{\binom{k}{2}}}. \quad (365)$$

So $\Pi_k(1) \not\rightarrow 1$ either, whether the polygon is formed over $\mathbb{Q}$ or over $\mathbb{R}$. Passing from the step reading to the polygonal reading changes nothing at the right endpoint, because the two agree there.

The failure is localized precisely. It is not a failure of the iterated prefix sums, whose values (213)–(215) are exactly right; it is a failure of the pair consisting of the prefix order and the abscissa convention. Shifting the order from k to $k + 1$ and reading the index as a cell center rather than a left endpoint yields genuine pointwise convergence, by [theorem 12.16](#). In that corrected reading the right endpoint is still exceptional, but for a reason that is now visible and harmless: the limit there is $\text{up}(1) = 0$, and the exact value $-2^{-\binom{n}{2}}$ of (215) converges to it. The corrected statement is a separate theorem about a different scheme, not a reinterpretation that rescues the literal one.

21.7 A proposed maximum proxy has no maximum

A second construction attached to the same prefix sums proposes to approximate F near the origin by the largest term of an explicit family. Write

$$M_n(x) = \frac{(2^n x)^n}{n! 2^{\binom{n}{2}}} \quad (n \geq 1), \quad (366)$$

and consider the claim that for small $x > 0$ the quantity $\max_{n \geq 1} M_n(x)$ exists and serves as a proxy for $F(x)$. The maximum does not exist for any $x > 0$.

Theorem 21.2 (Unboundedness of the proxy family). *Fix $x > 0$. Then*

$$\frac{M_{n+1}(x)}{M_n(x)} = \frac{x 2^{n+1}}{n+1} \rightarrow \infty \quad (n \rightarrow \infty), \quad (367)$$

the family $\{M_n(x) : n \geq 1\}$ is unbounded above, and there is no index $n \geq 1$ with $M_m(x) \leq M_n(x)$ for all $m \geq 1$.

Proof. Since $\binom{n+1}{2} - \binom{n}{2} = n$,

$$\frac{M_{n+1}(x)}{M_n(x)} = \frac{(2^{n+1} x)^{n+1}}{(2^n x)^n} \cdot \frac{1}{(n+1) 2^n} = \frac{2^{2n+1} x}{(n+1) 2^n} = \frac{x 2^{n+1}}{n+1},$$

which tends to infinity because $2^{n+1}/(n+1) \rightarrow \infty$ and $x > 0$ is fixed. Choose N with $x 2^{n+1}/(n+1) \geq 2$ for all $n \geq N$. Then $M_{N+m}(x) \geq 2^m M_N(x)$ for every $m \geq 0$, and $M_N(x) > 0$, so the family is unbounded above. Unboundedness contradicts the existence of a largest term. $\square$

Since $0 \leq F \leq 1$, an unbounded family cannot serve as a pointwise proxy for F , and the displayed maximum cannot be evaluated at any positive argument. The monotone structure is in fact the reverse of the one a maximum requires. The ratio (367) is nondecreasing in n , because $2^{n+2}/(n+2) \geq 2^{n+1}/(n+1)$ for every $n \geq 0$, so it crosses the value 1 at most once. Hence for $0 < x < 1/2$ the terms first decrease to a single interior minimum and then increase without bound, and for $x \geq 1/2$ they increase from the start. The distinguished index is therefore a minimum of the family, not a maximum, and the analytically controlled way to select a scale is the saddle-point argument of section 14, where the contour radius is $r = 2^\lambda$ with λ given by theorem 12.20.

21.8 A stationary substitution that leaves a linear term

Even after the index in (366) is treated as a real variable, so that a stationary point exists, a separate algebraic defect remains. The Stirling proxy attached to (366) is

$$\mathcal{H}(x, n) = \frac{L}{2}n^2 + n \log x - n \log n + n, \quad L = \log 2, \quad (368)$$

and the claim under discussion is that substituting the stationary equation into (368) produces $-\frac{L}{2}n^2$ up to an error $\mathcal{O}(\log n)$ inherited from Stirling's formula.

The Stirling input itself is correct and we record it in the sharp form actually used.

Lemma 21.3 (Stirling with logarithmic error). *As $n \rightarrow \infty$,*

$$\log(n!) = n \log n - n + \mathcal{O}(\log n). \quad (369)$$

More precisely, for every $n \geq 1$,

$$0 \leq \log(n!) - \left(n \log n - n + \frac{1}{2} \log n + \frac{1}{2} \log(2\pi) \right) \leq \frac{1}{12n}.$$

Proof. The two-sided bound is Robbins' form of Stirling's formula, obtained from the Stirling sequence $s_n = n! e^n n^{-n} (2n)^{-1/2}$: the increments $\log s_n - \log s_{n+1}$ are positive and bounded by $\frac{1}{12n} - \frac{1}{12(n+1)}$, so telescoping and passing to the limit $\log s_n \rightarrow \frac{1}{2} \log \pi$ gives $0 \leq \log s_n - \frac{1}{2} \log \pi \leq \frac{1}{12n}$, which is the display after unwinding the definition of s_n . Since $\frac{1}{2} \log n + \frac{1}{2} \log(2\pi) + \frac{1}{12n} = \mathcal{O}(\log n)$, the display implies (369). $\square$

Theorem 21.4 (The residual linear term). *Let $x > 0$ and $n > 0$ be real with $n \log 2 + \log x = \log n$, equivalently $n2^{-n} = x$. Then*

$$\mathcal{H}(x, n) = -\frac{L}{2}n^2 + n. \quad (370)$$

The residual term n is not $\mathcal{O}(\log n)$ as $n \rightarrow \infty$; consequently the substitution does not yield $-\frac{L}{2}n^2 + \mathcal{O}(\log n)$.

Proof. Differentiating (368) in n gives $\partial_n \mathcal{H}(x, n) = nL + \log x - \log n$, so the stationary equation is $nL + \log x = \log n$. For $x, n > 0$ it is equivalent to $n2^{-n} = x$, since both sides of the latter are positive and taking logarithms is injective there: $\log(n2^{-n}) = \log n - nL$. Substituting $\log x - \log n = -nL$ into (368),

$$\mathcal{H}(x, n) = \frac{L}{2}n^2 + n(\log x - \log n) + n = \frac{L}{2}n^2 - Ln^2 + n = -\frac{L}{2}n^2 + n.$$

Suppose $n = \mathcal{O}(\log n)$ as $n \rightarrow \infty$. Composing with $\log n = o(n)$ would give $n = o(n)$, which is impossible for a quantity that is eventually nonzero. Hence the term n in (370) cannot be absorbed into a logarithmic remainder. $\square$

The defect sits inside the main term rather than in the error term, and the amount by which it does is computable. The stationary equation of [theorem 21.4](#) is $n2^{-n} = x$, which is exactly (260); so the stationary index is the Lambert phase, $n = \lambda(x)$. Discarding the term n in (370) therefore replaces the coefficient of λ in the answer by 0, whereas (262) has the coefficient $1 + \frac{L}{2}$.

Remark 21.5. The proxy (368) loses a second linear term even before the substitution. Since $\binom{n}{2} = \frac{1}{2}n^2 - \frac{1}{2}n$, the exact logarithm of (366) is

$$\log M_n(x) = \frac{L}{2}n^2 + \frac{L}{2}n + n \log x - \log(n!),$$

so [theorem 21.3](#) gives $\log M_n(x) = \mathcal{H}(x, n) + \frac{L}{2}n - \frac{1}{2} \log n - \frac{1}{2} \log(2\pi) + \mathcal{O}(n^{-1})$; the proxy (368) omits the term $\frac{L}{2}n$. Reading the index as a real variable and substituting $\log x - \log n = -nL$ at $n = \lambda(x)$ turns this into

$$-\frac{L}{2}\lambda^2 + \left(1 + \frac{L}{2}\right)\lambda - \frac{1}{2} \log \lambda - \frac{1}{2} \log(2\pi) + \mathcal{O}\left(\frac{1}{\lambda}\right),$$

which reproduces the quadratic, linear, and logarithmic coefficients of the corrected main term (262). What it does not reproduce is the constant C_F and, more importantly, the nonconstant periodic function $\Psi(\lambda)$. Both omitted linear terms, n and $\frac{L}{2}n$, must be restored to reach even that much; and restoring them still leaves the periodic obstruction described earlier in this section.

21.9 The uniform local slope bound fails at the second level

The last claim attached to the literal grid is a quantitative one: that the polygon Π_k of [section 21.6](#) has, on each cell of width $h = 2^{-k}$, a slope within $4h$ of the derivative of F there, and hence a uniform error controlled by the same quantity. The bound fails at the first index at which it can be tested.

Theorem 21.6 (A counterexample at $k = 2$). *Let $k = 2, j = 1$, so that $h = 2^{-2}$ and $x_{2,1} = 1/4$. The forward slope of Π_2 on $[1/4, 1/2]$ equals -2 , whereas $F'(1/4) = 1$. The discrepancy is 3, while $4h = 1$.*

Proof. The relevant values are collected in the following table, computed from (206) and $\binom{2}{2} = 1$.

j	0	1	2	3
ε_j	1	-1	-1	1
$\sigma_1(j)$	1	0	-1	0
$\sigma_2(j)$	1	1	0	0
$\mathcal{V}_2(j)$	$\frac{1}{2}$	$\frac{1}{2}$	0	0

Hence

$$\frac{\mathcal{V}_2(2) - \mathcal{V}_2(1)}{h} = 2^2 \left(0 - \frac{1}{2}\right) = -2. \quad (371)$$

On the other side, (4) and the symmetry value $F(1/2) = 1/2$ give $F'(1/4) = 2F(1/2) = 1$. Therefore $|-2 - 1| = 3$, while $4h = 4 \cdot 2^{-2} = 1$. $\square$

The sign is the informative part. The bounded Fabius function is strictly increasing on $(0, 1)$, so any approximation whose slope is negative in the interior is not approximating F at all on that cell. This is the same phenomenon as [theorem 21.1](#), seen at the level of first differences rather than at the level of values: the literal grid samples a compactly supported bump on a centered window, and a bump has decreasing stretches. The corrected reading of [section 12.5](#) makes that identification explicit, and (213) shows that the zero runs responsible for the flat stretches sit exactly at the top of each dyadic block.

21.10 What the formalization proves, and what it deliberately does not claim

The public aggregate contains exact proofs of the two source papers' named results, together with the additional dyadic, q -binomial, Legendre, and sharp-asymptotic theorems described here. It does not promote a source conjecture to a theorem, does not infer derivative bounds for an unspecified asymptotic remainder, and does not identify two global functions merely because they coincide on $[0, 1]$. These negative design decisions are part of the mathematics: they preserve the hypotheses needed by every later theorem.

22 Conclusion

The Fabius function is unusual not because it has one exotic property, but because several mathematical structures coincide in one object.

- A contraction on symmetric distribution functions constructs it and proves uniqueness.
- An infinite sum of independent uniform variables explains positivity, symmetry, and smoothing.
- Rvachev's fold is a compactly supported C^∞ density satisfying an exact functional-differential equation.
- Its Fourier transform is an entire sinc product with superalgebraic real-axis decay.
- Thue–Morse splitting creates a signed global solution and exact formulas for every derivative.
- Rational moment recurrences evaluate every dyadic argument exactly.
- A Taylor recursion turns the closed Thue–Morse formula into an efficient bit-sensitive algorithm.
- Binary-prefix reduction gives a finite telescope with an explicit residual, an absolutely convergent global series, and the corrected parity-power presentation.
- Half-base Gaussian finite differences explain the translated complex dyadic formulas, the point-wise global q -series, and the Toeplitz discrete limit.
- A negative-Laplace product and a lower-Lambert saddle expose the true endpoint scale, its Gamma–zeta periodic fluctuation, and its complete asymptotic expansion.
- A uniformly convergent Legendre series converts the same exact dyadic data into globally optimal polynomial approximants on the support.

The dyadic and asymptotic theories are not separate curiosities. The exact values $F(2^{-n})$ determine the moments and the Legendre coefficients; their logarithmic growth is governed by the same saddle geometry that controls arbitrary small arguments. The periodic term is the analytic shadow of dyadic self-similarity, while the Thue–Morse signs encode the same binary splitting algebraically. Seen together, these descriptions turn a famous “pathological example” into a remarkably coherent piece of analysis.

A Exact data

The following tables collect enough exact values to reproduce the examples in the text and to seed independent implementations.

A.1 Moment and half-moment sequences

n	c_n	d_n
0	1	1
1	$\frac{1}{9}$	$\frac{1}{2}$
2	$\frac{675}{583}$	$\frac{2}{5}$
3	$\frac{59535}{132809}$	$\frac{1}{18}$
4	$\frac{32531625}{46840699}$	$\frac{6}{143}$
5	$\frac{24405225075}{4068990560161}$	$\frac{1350}{19}$
6	$\frac{4133856862760625}{583}$	$\frac{270}{1153}$
7	—	$\frac{23814}{17010}$

The c_n column satisfies (79); the d_n column satisfies (88). Substitution in (102) gives the next table.

A.2 Reciprocal powers and normalized odd integers

n	$F(2^{-n})$	$v_2(F(2^{-n}))$	R_n
0	1	0	—
1	$\frac{1}{2}$	−1	1
2	$\frac{5}{72}$	−3	5
3	$\frac{1}{288}$	−5	15
4	$\frac{143}{2073600}$	−10	1001
5	$\frac{19}{33177600}$	−14	5985
6	$\frac{1153}{561842749440}$	−20	2853675
7	$\frac{583}{179789679820800}$	−26	26261235

The valuation column is computed from (122); it is not obtained by factoring the displayed decimal-sized denominators. This is exactly the kind of redundancy that makes the table useful as a test vector.

Example A.1 (A worked consistency check at $F(1/16)$). The half-moment recurrence (88) at $n = 4$ reads $5(2^4 - 1)d_4 = \sum_{k=0}^3 \binom{5}{k} d_k$. With the values $d_0 = 1$, $d_1 = \frac{1}{2}$, $d_2 = \frac{5}{18}$, $d_3 = \frac{1}{6}$ from (89),

$$\sum_{k=0}^3 \binom{5}{k} d_k = 1 + 5 \cdot \frac{1}{2} + 10 \cdot \frac{5}{18} + 10 \cdot \frac{1}{6} = \frac{18 + 45 + 50 + 30}{18} = \frac{143}{18},$$

so $75 d_4 = \frac{143}{18}$ and $d_4 = \frac{143}{1350}$, in agreement with (89). Now apply (102) with $\binom{4}{2} = 6$:

$$F\left(\frac{1}{16}\right) = \frac{d_4}{4! 2^{\binom{4}{2}}} = \frac{143/1350}{24 \cdot 64} = \frac{143}{2073600},$$

which is the entry for $n = 4$ in (109). The valuation formula (122) predicts $v_2(F(2^{-4})) = -6 - 1 - v_2(4!) = -6 - 1 - 3 = -10$, and indeed $2073600 = 2^{10} \cdot 2025$ with 2025 odd and 143 odd. The two computations are independent, which is what makes this a test rather than a restatement.

B Proof dependency and Lean module guide

The Lean development is intentionally split into small files, while this article groups results by mathematical theme. The tables below cover every module of the development, not merely its principal entry points. File names are relative to

Analysis/FabiusFunction/Lean/FabiusFunction/,

the sole exception being the root aggregate `FabiusFunction.lean`, which sits one directory higher and is the module a user actually imports.

At the time of writing the development consists of 175 Lean files: 174 modules in the directory named above, together with that root aggregate. They comprise roughly 62,400 lines, 2967 top-level theorem or lemma declarations — of which 537 are private, leaving 2430 in the public interface — and 536 top-level `def` or `abbrev` declarations.

Dependency order runs broadly downward through the clusters. The exact-arithmetic module `Arithmetic.lean` imports nothing from the project at all, and `Basic.lean` imports only it; between them they sit below essentially everything else. The moment, dyadic, Fourier, and Legendre layers depend on the foundations and on each other but not on any asymptotic module. Within the endpoint asymptotics the order is genuinely less linear: the dyadic-cumulant branch and the vertical-Bromwich branch are developed in parallel from a shared negative-Laplace root and first meet in `FabiusDyadicSharpAsymptotic.lean`, then again in `FabiusSharpAsymptotic.lean`, which imports both.

A handful of modules are pure infrastructure with no counterpart anywhere in the article: the generic Gaussian-contraction and Gaussian-tail estimates, the formal exponential and logarithm coefficient algebra of the `SaddleExpansion/SaddleLog` files, the abstract Toeplitz row lemma, and the generic weighted-path-sum solver are all stated for arbitrary inputs and are used only to keep the named Fabius theorems short.

B.1 Foundations: definitions, existence, and the original characterization

These modules underlie everything else and correspond to [Section 1](#), [Section 2](#), and [Section 3](#).

Module	Role in the proof
<code>Arithmetic.lean</code>	Keeps all exact quantities in $\mathbb{Q}$ and $\mathbb{N}$: binary weight, Thue–Morse signs, c_n and d_n , division-free numerators, reciprocal values, closed dyadic formulas, and executable rational evaluators. Imports nothing from the rest of the project.
<code>Basic.lean</code>	Separates the three functions that share the letter F : the bounded CDF-valued <code>BoundedFabius</code> subtype with its characterization <code>IsFabius</code> , the signed extension on all of $\mathbb{R}$, and the compactly supported up. Defines the Fourier products and moment integrals used later.
<code>Existence.lean</code>	Constructs F as the fixed point of a strict contraction on symmetric $[0, 1]$ -valued continuous functions, extracts the integral fixed-point equation, and proves uniqueness from agreement on the dense set of dyadic rationals.
<code>Differential.lean</code>	Derives the Rvachev differential equation, gluing the two one-sided derivatives at the fold point $x = 0$, and bootstraps the equation to smoothness of every finite order.

Module	Role in the proof
<code>OriginalCharacterization.lean</code>	Formalizes Theorem 1 of the compact-support paper starting from an unspecified constant k , and derives the exact support, nonnegativity, translate identity, normalization, and the forced value $k = 2$.
<code>OriginalUniqueness.lean</code>	Proves the uniqueness half of that theorem by the Fourier refinement argument: every solution has transform value one at the origin and satisfies the same finite and infinite dyadic sinc-product formulas.
<code>OriginalPaperSupplement.lean</code>	Packages Theorems 4, 6, and 7 of the compact-support paper, the finite Taylor expansion at dyadic points, the nowhere-analyticity corollary, and the exact even and odd integer values of the signed extension.

B.2 Regularity, shape, and the global extension

The sharp shape statements collected here are used throughout [Section 2](#) and supply the exact derivative identities of [Section 6](#).

Module	Role in the proof
<code>Regularity.lean</code>	Records the global 2-Lipschitz estimate for both functions, proves that the constant is attained at the midpoint, and isolates the linear majorants $F(x) \leq 2x$ and $\text{up}(x) \leq 2(1 - x)$ used as quantitative input elsewhere.
<code>Monotonicity.lean</code>	Collects the order-theoretic consequences of the derivative formula: strict monotonicity of F on $[0, 1]$ and the resulting bijection of $[0, 1]$ onto itself, the exact open support $(-1, 1)$ of up , and its strict unimodality.
<code>Convexity.lean</code>	Turns the monotonicity of up into convexity of F on $(-\infty, 1/2]$ and concavity on $[1/2, \infty)$, strictly inside the unit interval, so that the midpoint is exactly the inflection point.
<code>GlobalExtension.lean</code>	Proves that the Thue–Morse translate series defining the signed extension is locally finite, then derives the global differential equation and the closed formulas for every iterated derivative of both $\mathcal{F}$ and up .
<code>GlobalBounds.lean</code>	Observes that exactly one translate survives on each block $[2b, 2b + 2]$, hence $ \mathcal{F} \leq 1$ everywhere, and converts the derivative formula into the sharp attained bound $ \mathcal{F}^{(k)} \leq 2^{\binom{k+1}{2}}$.
<code>BoundedDerivatives.lean</code>	Transfers those sharp derivative bounds from the signed extension to the bounded CDF by a germ argument below one, a local-constancy argument above one, and continuity at the remaining point $x = 1$.
<code>NowhereAnalytic.lean</code>	Determines the exact real-analytic locus: F is analytic at x if and only if $x \notin [0, 1]$. The right endpoint needs a separate argument, since the reflection identity is not an identity of germs there.
<code>FabiusFlatness.lean</code>	The qualitative flatness statement at the origin: every derivative vanishes there, so Taylor's theorem gives $\mathcal{F}(x) = o(x^n)$ for every n , with no constant.
<code>EffectiveFlatness.lean</code>	An elementary effective form $F(x) \leq 2^{\binom{n+1}{2}} x^n$ on $2^n x \leq 1$, proved by iterating the mean value theorem, kept because it avoids importing any integral machinery.
<code>SharpFlatness.lean</code>	Recovers the factorial that the mean value theorem discards by integrating instead, giving $F(x) \leq 2^{\binom{n+1}{2}} x^n / n!$; the remaining overshoot at $x = 2^{-n}$ is exactly $1/d_n$.

B.3 Probability, finite splines, and computability

This cluster realizes the probabilistic model of [Section 4](#) and supplies the evaluators and error bounds of [Section 20](#) and [Section 20.6](#).

Module	Role in the proof
<code>ProbabilityRepresentation.lean</code>	Builds the infinite product probability space, proves independence and the uniform law of each coordinate, identifies the CDF of the random weighted series with F globally on $\mathbb{R}$, and obtains μ as the corresponding density.
<code>FabiusComplexMGF.lean</code>	Identifies the entire generating function with Mathlib’s complex moment-generating function of that law, so that all complex derivatives are tilted moments and vertical-line norms are dominated by real tilted moments.
<code>EarlyApproximants.lean</code>	Formalizes the generating polynomials, their product and recursion laws, exact degree and coefficient normalization, the associated atomic measures and finite Bernoulli convolutions, and the corrected half-endpoint step representatives.
<code>EarlyMeasureBridge.lean</code>	Proves that the atomic measure encoded by the generating polynomial really is the finite Bernoulli convolution appearing in the source.
<code>WeakConvergence.lean</code>	Proves weak-* convergence of the finite convolution measures to μ , with explicit test-function formulations over an arbitrary $\mathbb{R}$ -like field.
<code>StepMeasureBridge.lean</code>	Identifies half-endpoint cells with ordinary interval indicators almost everywhere, computes their masses exactly, and sandwiches each histogram integral between shrunken and enlarged interval masses.
<code>StepApproximationLimit.lean</code>	Supplies coefficient unimodality, monotonicity of the step functions, and the interval average squeeze that upgrades convergence of interval integrals to pointwise convergence against a continuous limit.
<code>FabiusUniformSpline.lean</code>	Identifies the centered Thue–Morse power sums with distribution functions of finite weighted sums of independent uniform coordinates, giving support bounds, monotonicity, block-translation identities, and pointwise convergence.
<code>FabiusComputability.lean</code>	Formalizes both clauses of computability in the sense of computable analysis: effective uniform continuity from the global derivative equation, and preservation of computable sequences by rounded dyadic spline evaluation.
<code>FabiusComputableSpline.lean</code>	Certifies a primitive-recursive natural-number evaluator: clamp a signed-natural numerator, evaluate a finite centered spline, round half-up, and inherit the explicit error bound $5 \cdot 2^{-(p+3)}$.

B.4 Fourier transform, Poisson summation, and the Laplace bridge

These are the analytic identities of [Section 5](#), together with the spectral consequences of [Section 19](#) and the transform normalization that the asymptotic analysis later requires.

Module	Role in the proof
<code>FourierProduct.lean</code>	Proves that the removable sinc factors form a genuine convergent infinite product, that the transform satisfies the finite refinement identity, and that the limit identifies the two functions.
<code>FourierAnalytic.lean</code>	Shows that the transform of μ is entire, establishes Fourier inversion, and relates its values at imaginary arguments to the half-moment generating function.
<code>PoissonSummation.lean</code>	Formalizes Poisson summation and the partition-of-unity identities, correcting two typographical errors in the source: a missing dependence on t in the exponent and a spurious factor $1/a$ after multiplication.

Module	Role in the proof
<code>LaplaceTransform.lean</code>	Proves that the negative generating function divided by the spectral variable is the one-sided Laplace transform of F , in both complex and real form, with the integrability needed to avoid detouring through complex integrals.

B.5 Moments, generating functions, and exact arithmetic

Everything here stays in $\mathbb{Q}$ or $\mathbb{N}$, so no denominator or 2-adic valuation is ever taken of a real number. The cluster carries [Section 7](#) and [Section 8](#).

Module	Role in the proof
<code>MomentPowerSeries.lean</code>	Proves in exact rational arithmetic that the half moments are the even-binomial transform of the moments, by packaging the defining recurrence as a formal power series identity.
<code>NormalizedMoments.lean</code>	Shows that the executable rational recurrence for d_n agrees with the division-free recurrence for its natural numerator, independently of any analytic construction.
<code>NormalizedEvenMoments.lean</code>	The same normalization statement for the even moments c_n , identifying the division-free natural recurrence with the numerator of the rational one.
<code>AnalyticMoments.lean</code>	Supplies the integral interpretations of both exact sequences, their inverse-power specializations, and the generating-function dilation identity.
<code>BernoulliRecurrences.lean</code>	Connects the executable moment recurrences to Mathlib's formal generating series for the Bernoulli numbers.
<code>LaplaceMoments.lean</code>	Differentiates the negative Laplace transform under its compact integral, producing the tilted moment family with $M'_k = -M_{k+1}$ and $M_k(0) = d_k$, the exact moment input for the endpoint comparison.
<code>Parity.lean</code>	Proves the binary-coefficient count behind the oddness of every integral moment numerator, using Lucas's theorem at the prime 2, together with the truncated odd-coefficient count needed two-adically.
<code>TwoAdic.lean</code>	Establishes the exact numerator and denominator parity of the half moments and converts it into the dyadic valuation $v_2(F(2^{-n}))$, using Legendre's formula to trade the factorial valuation for binary weight.
<code>HalfMomentDenominator.lean</code>	Proves that the reduced denominator of $2d_n$ is odd for every n , inductively at even indices and through $d_{2m+1} = (2m+1)c_m/2$ at odd ones.
<code>DenominatorBound.lean</code>	Puts the moments in the closed dyadic formula over a common denominator, shows the scaled value is an integer after the factorial and power-of-two cancellations, and derives the lcm denominator bound.
<code>ExactInversePower.lean</code>	Arithmetic proofs of the explicit inverse-power identities relating $F(2^{-n})$ to the half moments and to the normalized odd integers R_n .
<code>FabiusRecurrenceSequence.lean</code>	Records the factorially normalized half-moment sequence $a_n = d_n/n!$, its elementary recurrence, its exact relation to $F(2^{-n})$, and the associated entire series and dyadic product.
<code>FabiusInverseDyadicClosedForm.lean</code>	Solves the triangular recurrence for $F(2^{-n})$ nonrecursively as a weighted sum over ordered compositions of n , via a generic path-weight layer with last-edge and edge-count decompositions; the empty composition covers $n = 0$.

B.6 Dyadic closed forms and finite q -binomial identities

This is the exact evaluation machinery of [Section 9](#), whose algorithmic side is discussed in [Section 20](#).

Module	Role in the proof
<code>ScaleTranslation.lean</code>	Proves that the Thue–Morse series changes sign under translation by a power of two over the corresponding initial block, and combines this with the derivative formula to obtain the translated Taylor-remainder identity.
<code>TaylorReduction.lean</code>	Turns that cancellation into the finite recursive reduction formula, treating nonnegative indices by Taylor expansion and negative ones directly by the sign-translation law.
<code>DyadicCorrectness.lean</code>	Discharges the structural half of the evaluator proof: strong induction upgrades the single-step dyadic recurrence to agreement of the terminating bit recursion with the closed formula at every unit dyadic grid point.
<code>DyadicClosedForm.lean</code>	Proves the affine Thue–Morse cancellation and the Taylor-block identity, identifies the inverse-power lookup table, and concludes that the fast evaluator agrees with the closed formula on $[0, 1]$.
<code>DyadicAnalytic.lean</code>	Closes the loop analytically: endpoint compatibility of the finite Taylor recurrence determines the inverse-power values, so the rational evaluator really computes the analytic function.
<code>GlobalDyadic.lean</code>	Extends exact evaluation beyond the unit interval by reducing modulo a length-two block, folding the residue, and applying the Thue–Morse sign; local finiteness identifies the one contributing translate.
<code>HalfQBinomial.lean</code>	Develops the finite q -Pochhammer symbol and the Gaussian binomial coefficient at $q = 1/2$ over $\mathbb{Q}$: vanishing criteria, Mersenne evaluation, symmetry, both Pascal recurrences, and the finite q -binomial theorem.
<code>ThueMorseExponential.lean</code>	Packages the centered Thue–Morse power sums into a formal exponential generating series, shows Prouhet cancellation forces divisibility by X^k , and computes the effect of translating the inner power by a constant.
<code>FabiusQBinomialFormula.lean</code>	Proves the source-literal centered q -binomial formula, first at inverse powers and then for arbitrary numerators, together with the stronger statement that a common rational translation leaves the numerator unchanged.
<code>FabiusQBinomialTaylor.lean</code>	Identifies that finite q -binomial polynomial with the Taylor reduction sum by viewing the translation as an indeterminate over $\mathbb{Q}$; constancy then permits evaluation in any characteristic-zero field.
<code>FabiusQBinomialScalarFormula.lean</code>	Restores the normalizing prefactor at scalar level, exposing the centered identity for arbitrary real and complex translations.
<code>FabiusRawQBinomialFormula.lean</code>	Proves the raw-coordinate variant with inner power $(r + q)^{n+k}$ by reflecting each dyadic Thue–Morse block, the combined sign being cancelled exactly by the normalization $(-2)^{n^2}$.
<code>FabiusRawQBinomialScalar.lean</code>	Lifts the raw identity from $\mathbb{Q}$ to any $\mathbb{Q}$ -algebra by the same constant-polynomial argument, with real and complex wrappers and fully expanded finite-sum statements.
<code>FabiusDyadicQBinomialScalar.lean</code>	Does the same for the arbitrary-numerator dyadic formula, so that a single statement covers $\mathbb{R}$, $\mathbb{C}$, and every other characteristic-zero scalar field.

B.7 Global binary reduction and the discrete limit

These modules prove the telescope of [Section 10](#) and the convergence statement of [Section 11](#).

Module	Role in the proof
<code>FabiusBinaryReductionSeries.lean</code>	Iterates the finite Taylor reduction along the binary expansion, keeping the outer index at $m = 0$ so that the even and odd unit blocks are distinguished, and upgrades the signed residual identity to absolute summability.
<code>FabiusParityPowerSeries.lean</code>	Formalizes the corrected scale-zero parity-power series, valid for every nonnegative real input; the series indexed from $m = 1$ remains valid only on $0 \leq x < 1$.
<code>FabiusGlobalQBinomialSeries.lean</code>	Combines the constant-polynomial identity with the binary telescope to give the absolutely convergent pointwise q -binomial expansion, over $\mathbb{R}$ and $\mathbb{C}$ alike, for every real or complex translation.
<code>FabiusComplexShiftSpline.lean</code>	Identifies the complex-shifted finite spline with a fixed polynomial branch, expands it around the centered shift $q = 1/2$, and proves an exponentially small translation bound, so every fixed shift has the same limit.
<code>FabiusDiscreteLimitComplexShift.lean</code>	Isolates the complex-analytic ingredient: translating a normalized finite Thue–Morse power sum has an exact finite Taylor expansion in the lower-degree branches.
<code>FabiusDiscreteLimitToeplitz.lean</code>	Proves that each row of finite q -binomial weights has mass one, gives an exact formula and a uniform bound for its total variation, and supplies a general finite-row Toeplitz convergence theorem.
<code>FabiusDiscreteLimitIntegration.lean</code>	Performs the outer q -binomial reindexing, applies Toeplitz convergence to the complex-shift spline limit, and extends the result from the nonnegative axis to all of $\mathbb{R}$ by exact empty-prefix vanishing.

B.8 The Fourier–Legendre expansion

The whole cluster corresponds to [Section 18](#).

Module	Role in the proof
<code>LegendrePolynomial.lean</code>	Supplies the ordinary Rodrigues-normalized Legendre polynomials on $[-1, 1]$, which Mathlib does not carry, with endpoint, parity, norm, and differential-equation facts and the affine rule relating them to the shifted basis.
<code>LegendreSeriesConvergence.lean</code>	Provides the analytic convergence layer from the self-adjoint Sturm–Liouville operator $f \mapsto -((1 - x^2)f')'$, whose weight makes both boundary terms vanish, and iterates the Green identity to move eigenvalue powers onto the expanded function.
<code>FabiusLegendreCoefficients.lean</code>	Evaluates the coefficient u_n of the even function up in closed form in terms of the inverse-power values $F(2^{-k})$.
<code>FabiusLegendreSeries.lean</code>	Combines coefficients and convergence: absolute and uniform convergence on $[-1, 1]$, including both endpoints, plus the full orthogonality relation and absolute summability of the coefficient sequence.
<code>FabiusTranslatedLegendreSeries.lean</code>	Translates the even expansion to $[0, 2]$, where $\mathcal{F}(x) = \text{up}(x - 1)$, and expands the shifted Legendre polynomials into monomials to give the exact double finite inner sum.
<code>FabiusLegendreLeastSquares.lean</code>	Proves a generic Pythagorean identity for the unweighted $L^2[-1, 1]$ error and deduces that S_N is the unique least-squares minimizer through degree $2N + 1$, one degree better than its visible class.

B.9 The negative-Laplace product and its periodic correction

This is the exact decomposition of [Section 13](#) together with the Mellin evaluation of the constant appearing in [Section 14](#).

Module	Role in the proof
<code>NegativeLaplace.lean</code>	Constructs the logarithm of the dyadic product as an absolutely convergent series and splits its dilation equation into a quadratic solution, an exact one-periodic correction, and a forward tail bounded by $4e^{-s}$, using only elementary bounds.
<code>PeriodicCorrection.lean</code>	Proves continuity of that exact correction by treating the two infinite sums locally on compact positive intervals with geometric majorants, establishing the regularity interface used downstream.
<code>PeriodicRegularity.lean</code>	Differentiates the forward tail termwise through order four with superexponential majorants, proving that Ψ is C^4 and that its first two derivatives are one-periodic and globally bounded.
<code>PeriodicSmooth.lean</code>	Upgrades that to C^∞ by encoding the $(k+1)$ st derivative of each summand as a recursively generated polynomial in e^{-s2^n} , then propagates smoothness through the saddle-jet recurrence.
<code>PeriodicMean.lean</code>	Integrates the negative-Laplace logarithm and its forward tail over a dyadic interval, evaluates the mean of the exact correction, and pins down the unit-interval integral of the centered normalization Ψ .
<code>PeriodicFourier.lean</code>	Computes every Fourier coefficient of Ψ through the regularized Bose–Mellin kernel, proves the first coefficient nonzero — hence Ψ nonconstant — and gives the summable Gamma–zeta Fourier series.
<code>StieltjesConstant.lean</code>	Fixes the sign convention for the first Stieltjes constant and records the Laurent expansion of ζ at its pole.
<code>MellinBose.lean</code>	Proves the real Mellin identity for $\log(1 - e^{-x})$ with value $-\Gamma(a)\zeta(a+1)$ by expanding the logarithm and justifying termwise integration.
<code>MellinFinitePart.lean</code>	Regularizes $\Gamma(s)\zeta(1+s)$ at the double pole $s=0$ and identifies the finite part with the Euler–Stieltjes combination.
<code>BoseFinitePartIntegral.lean</code>	Realizes that finite part as a convergent split integral — singular logarithm removed near zero, kernel divided by x near infinity — and supplies the two kernels reused by the mean and Fourier computations.
<code>GammaSecondOrder.lean</code>	Supplies the second-order Taylor expansion of Γ at 1 with an explicit remainder, obtained from analyticity and the reflection formula.
<code>FabiusSharpConstant.lean</code>	Combines the mean of the periodic correction with the Gaussian saddle prefactor and records the resulting constant in both compact Gamma–zeta and expanded Euler–Stieltjes form.

B.10 Tilted moments, endpoint cumulants, and the sharp dyadic formula

This branch turns [Section 13](#) into the unconditional dyadic statement quoted in [Section 14](#).

Module	Role in the proof
<code>NegativeLaplaceDerivatives.lean</code>	Expresses the first four logarithmic derivatives in terms of normalized tilted moments $R_k = M_k/M_0$ satisfying $R'_k = -R_{k+1} + R_k R_1$, so that they become the usual cumulant polynomials.
<code>NegativeLaplaceDerivativeBounds.lean</code>	Differentiates the dilation equation, bounds the elementary dilation kernel with polynomial weights, and iterates to prove $q^{(j)}(n) = \mathcal{O}(\log n/n^j)$ for $1 \leq j \leq 4$.
<code>NegativeLaplaceAllOrderJets.lean</code>	Packages every ordinary derivative into one recursive sequence agreeing with <code>iteratedDeriv</code> , splitting each dyadically rescaled derivative into a linear drift, a smooth periodic jet, and a forward-tail remainder.
<code>NegativeLaplaceScaledTailFlat.lean</code>	Bounds every positive-order forward-tail summand by a polynomial times an exponential on $[1, \infty)$ and shows the scaled dyadic sum is o of every real power of the radius.

Module	Role in the proof
LaplaceMoments.lean	(See Section B.5 .) Its tilted moment family is the entry point of this branch.
LaplaceMomentBounds.lean	Makes the endpoint comparison unconditional through $R_k(s)^2 \leq s((4/s)^k k!)^2$ for $s \geq 2$, proved by Hölder log-convexity at the intermediate tilt $3s/4$ with no periodic regularity input at all.
LaplaceCumulantAsymptotics.lean	Inverts the first four cumulant polynomials and packages the bookkeeping that turns the logarithmic derivative bounds into an $\mathcal{O}(1/n)$ endpoint error.
LaplacePeriodicSecondOrder.lean	Differentiates the quadratic-plus-periodic decomposition, controls the forward-tail derivative, and shows the linear logarithmic terms cancel in $n(q'' + (q')^2)/2$, leaving the explicit periodic-derivative expansion.
EndpointLaplaceComparison.lean	Compares $(1-x)^n$ with $e^{-nx}(1-nx^2/2)$ pointwise, bridges to survival-function moments by Fubini, and transfers d_n to the negative Laplace transform at scale n with an explicit logarithmic correction.
DyadicSharpDecomposition.lean	Performs the exact algebraic assembly before any estimate: elementary main term, exact periodic correction, exponentially small product tail, endpoint-versus-Laplace error, and sharp Stirling remainder, with no hidden cancellation.
DyadicSharpConditional.lean	Quantifies every contribution except the two isolated moment estimates, reducing the complete dyadic formula to exactly those.
FabiusDyadicSharpCumulant.lean	Combines the exact Euler–Stieltjes constant with the unconditional endpoint comparison, reaching the final $\mathcal{O}(1/n)$ precision while keeping the second-order cumulant exact.
FabiusDyadicSharpAsymptotic.lean	Recognizes that cumulant as the first-order displacement of the periodic correction from $\log_2 n$ to the exact Lambert phase, and performs the Taylor transfer with a uniform quadratic remainder.

B.11 Vertical estimates, Bromwich inversion, and the central saddle

The contour-free analytic input to [Section 14](#). There is no contour-shifting argument anywhere: every hypothesis of Fourier inversion is discharged explicitly.

Module	Role in the proof
NegativeLaplaceVertical.lean	Proves the canonical complex dyadic product and extracts finitely many factors to obtain the minor-arc bound $\ P(-rw)\ /P(-r) \leq C(r, N)\ w\ /\ w\ ^N$; already $N = 2$ gives integrability of the vertical kernel.
NegativeLaplaceMinorArc.lean	Isolates the elementary criterion bounding the finite coth product by $\exp(4Ne^{-b})$, separating that estimate from any later choice of the number of extracted factors.
NegativeLaplaceVerticalLog.lean	Constructs a branch-safe logarithm on each vertical line by integrating the logarithmic derivative rather than selecting a branch of log, and proves that exponentiating it recovers the transform ratio.
NegativeLaplaceVerticalSmooth.lean	Promotes that normalized logarithm to C^∞ , since the underlying transform curve is smooth and nonvanishing and the fundamental theorem of calculus transfers the regularity.
NegativeLaplaceVerticalTaylor.lean	Gives the branch-safe cubic Taylor expansion with explicit integral remainder, together with continuity and C^4 smoothness of the normalized vertical moments.
NegativeLaplaceVerticalFourthBound.lean	Supplies the uniform off-axis fourth derivative bound used by the first-order central estimate.

Module	Role in the proof
NegativeLaplaceVerticalAllOrderBound.lean	Continues the one-factor derivative holomorphically and applies Cauchy’s estimate on disks of radius $1/2$ to give the scale-independent increment bound $m! 5/(1/2)^m$, then iterates to $\mathcal{O}(b)$ at $r = 2^b$ for every order.
NegativeLaplaceVerticalOrdinaryJets.lean	Defines the holomorphic logarithmic derivative on the open right half-plane and matches it on the real axis with the ordinary derivatives of the negative-Laplace logarithm.
BromwichSaddle.lean	Derives the real Bromwich formula directly from Mathlib’s Fourier inversion theorem and performs the exact change of variables putting a transform $P(z)/z$ into dimensionless saddle coordinates.
FabiusBromwichInput.lean	Discharges the hypotheses of that framework for F : exponential tilting gives an integrable continuous input whose transform is the negative generating function over the vertical coordinate.
QuantitativeSaddle.lean	The measure-theoretic end of the argument. Bounds on a central set and its complement give a relative estimate $1 + \mathcal{O}(1/b)$, with the vanishing of the signed odd correction made explicit.
FabiusSaddleReduction.lean	Separates the exact Bromwich formula into a real prefactor and a normalized kernel integral, so that no positivity or logarithm manipulation survives into the central-arc estimates.
FabiusSaddleCentral.lean	Bounds the central L^1 error against a standard Gaussian plus the retained odd linear-plus-cubic correction, at a general radius and then at the standardized one.
FabiusSaddleCentralLambert.lean	Instantiates that estimate on the dyadic Lambert orbit, extracting the concrete linear, cubic, and curvature coefficients and proving the normalized kernel mass is $1 + \mathcal{O}(1/\lambda)$.

B.12 The lower-Lambert phase and the corrected sharp asymptotic

The exact saddle coordinate and the source-facing statements of [Section 14](#).

Module	Role in the proof
LowerLambertW.lean	Supplies the minimal real W_{-1} infrastructure: a totalized branch with specification and uniqueness on $(-e^{-1}, 0)$ and the standard first two terms of its expansion.
FabiusLambertPhase.lean	Defines the coordinate λ solving $\lambda 2^{-\lambda} = 2^{-t}$, proves its exact fixed-point equation, and records its first two asymptotic terms; keeping it exact is what avoids losing a logarithmic factor.
FabiusLambertSaddle.lean	Packages λ with the positive Laplace radius $r = 2^\lambda$, turning the defining equation into the identity $rx = \lambda$ without committing to an implicitly defined saddle point.
FabiusLambertRates.lean	Compares the reciprocals of the Lambert and logarithmic coordinates by $\mathcal{O}$ and converts the rate into the literal $1/(-\log x)$ scale.
FabiusLambertHigherExpansion.lean	Retains the next phase term, since substitution into a quadratic action magnifies phase errors by one power of t ; without it the final error would be $\log t/t$ rather than $1/t$.
FabiusLambertDerivativeBounds.lean	Bounds the first three logarithmic derivative residuals along the real orbit at a Lambert radius and expresses the second and third derivatives in periodic form.
FabiusLambertMinorArc.lean	Chooses the factor count $m(t) = \lceil \lambda(t)/4 \rceil$, so that $2m(t) = \lambda(t)/2 + \mathcal{O}(1)$ and the finite coth product is uniformly bounded by e^4 .
FabiusLambertTailFlat.lean	Records once and for all that the forward Laplace tail is smaller than every inverse power of λ , so it can be discarded at every finite order.

Module	Role in the proof
FabiusSmallArgumentScale.lean	Packages the exact inverse relationship between $t \mapsto 2^{-t}$ and $x \mapsto -\log_2 x$ and transfers $\mathcal{O}$ estimates between the two forms.
FabiusSharpExactReduction.lean	Isolates the exact identity behind the transfer: on the lower-Lambert branch, $\log F$ minus the corrected main term is exactly the logarithm of the real saddle ratio plus the exponentially small forward tail.
FabiusSharpLambertMain.lean	Isolates the purely algebraic part of the corrected small-argument asymptotic, leaving the normalized Bromwich kernel estimate as the sole analytic input.
FabiusSharpLambertTransfer.lean	Converts a relative kernel error of order $1/\lambda$ into the corrected sharp logarithmic asymptotic, absorbing the forward tail unconditionally via $\lambda \leq 2^\lambda$.
FabiusSharpAsymptoticTransfer.lean	Packages the last source-facing reduction, producing the corrected formula with its literal $\mathcal{O}(1/(-\log x))$ error as $x \downarrow 0$.
FabiusExplicitSharpTransfer.lean	Combines the compact Lambert transfer with the expansion into logarithms and Euler–Stieltjes constants, discharging every phase and formula-conversion error.
FabiusFirstSaddleCorrection.lean	Expands the normalized kernel one order further and records the resulting periodic coefficient of $1/\lambda$, contracted against the Gaussian moments $(1, 3, 15)$, independently of the analytic remainder estimate.
FabiusSharpAsymptotic.lean	The headline module: it closes the quantitative saddle argument, shows the elementary printed expression is missing a genuine nonconstant periodic term, and imports the quotient-approximation obstruction.

B.13 The all-orders expansion

The machinery behind [Section 15](#). The first eight modules are generic: they know nothing about the Fabius function and would serve any saddle-point problem with bounded periodic coefficients.

Module	Role in the proof
SaddleExpansionAlgebra.lean	Turns a formal exponent $E(\varepsilon) = \sum_{j \geq 1} E_j \varepsilon^j$ into the correction coefficients of $\exp E$, in any commutative $\mathbb{Q}$ -algebra, and defines the bounded-coefficient Poincaré expansion API.
SaddleExpansionFiniteRemainder.lean	Makes the coefficient identity a finite polynomial one: the discrepancy is an exact multiple of X^L , with canonical quotient defined by iterating div_X so that parameter dependence stays algebraic.
SaddleExpansionFlat.lean	Bridges from estimates at every algebraic order to the parameter-dependent expansion API.
SaddleExpansionSmallArgument.lean	Transports a full expansion from the dyadic logarithmic scale to $x \downarrow 0$.
SaddleLogExpansionAlgebra.lean	Builds the coefficient sequence of the formal logarithm of a unit-constant series from the recurrence $B'A = A'$.
SaddleLogExpansionPowerSeries.lean	Identifies that recurrence with Mathlib’s <code>PowerSeries.logOf</code> , connecting the finite recursion to the standard substitution calculus.
SaddleLogAsymptoticTransfer.lean	Turns a full real expansion with constant coefficient one into the expansion of its logarithm, combining the formal divisibility statement with the analytic Taylor remainder and a stability estimate.
SaddleAllOrders.lean	Supplies the two-source exponential remainder bound, the Gaussian-weighted reference, and the criteria under which a normalized integral matches its even expansion.
GaussianPolynomialContraction.lean	Identifies normalized Gaussian integration with a linear functional on complex polynomials and records the integrability and coefficient estimates reused throughout.

Module	Role in the proof
GaussianPolynomialTail.lean	Bounds the complementary-interval integral of a Gaussian times a finite polynomial coefficient by coefficient, making any fixed reference smaller than any requested inverse power.
GaussianPolynomialTailAllOrders.lean	Upgrades that estimate to the order-dependent radius, so a phase-dependent polynomial family contributes a tail of order $b^{-(N+1)}$ whenever its coefficient weight stays bounded.
GaussianPolynomialWholeIntegral.lean	Controls the full L^1 norm of the same references by the same factorial coefficient weight.
FabiusSaddleCoefficientRecurrence.lean	Records the concrete periodic-jet algebra: $r^{n+1}q^{(n+1)}(r) = s_n t + p_n(t)$ with $s_n = (-1)^{n+1}n!$, together with the parity $E_m(t, -v) = (-1)^m E_m(t, v)$ that kills all odd half-powers.
FabiusSaddlePolynomialCoefficients.lean	Packages that exponent coefficient as an element of $\mathbb{C}[X]$ and propagates coefficientwise conjugation, so every even exponential coefficient is fixed.
FabiusSaddleExpansionCoefficients.lean	Exponentiates and contracts against Gaussian moments, proves every mass coefficient real, lifts the recurrence to continuous maps for continuity and periodic boundedness, and applies the logarithmic recurrence.
FabiusSaddleExponentialOrders.lean	Splits the exact centered exponent jet of each order into its bounded part, its linear-in- t slope part, and the flat forward-product correction.
FabiusSaddleCentralAllOrders.lean	Identifies the finite polynomial reference on the dyadic Lambert orbit, contracts its odd terms, and decomposes the exact centered exponent into a periodic truncation plus explicit remainders.
FabiusSaddleCentralRadiusAsymptotics.lean	Shows the logarithmic central radius grows more slowly than $\sqrt{\lambda}$ for every fixed degree, controlling all fixed-order Taylor polynomials uniformly on the expanding interval.
FabiusSaddleReferenceWeight.lean	Proves the Gaussian tail weight of the finite reference polynomial uniformly bounded, in the periodic phase and in every substitution parameter of norm at most one.
FabiusSaddleReferenceTail.lean	Supplies the explicit Gaussian moment tail estimates on the complement of $[-A, A]$ used by that weight argument.
FabiusSaddleTail.lean	Integrates the polynomial kernel bound $C(r, 2m)/(1 + v^2/b)^m$ by splitting into an intermediate Gaussian region and a far geometric region, giving an $\mathcal{O}(1/b)$ tail at the standardized radius.
FabiusSaddleTailAllOrders.lean	Repeats that at the order-dependent radius $\sqrt{32(N+1)\log b}$, so the complementary tail is below every requested inverse power.
FabiusSaddleMassAllOrders.lean	Combines vertical Taylor expansion, finite exponential substitution, Gaussian contraction, and the minor-arc estimates into the full Poincaré expansion of the normalized kernel mass.
FabiusFullAsymptoticExpansion.lean	Transfers that expansion through the exact real saddle identity and the logarithm, adds the flat forward tail, and moves the result to $x \downarrow 0$, keeping every coefficient evaluated at the exact Lambert phase.
FabiusLambertAllOrderAlgebra.lean	Defines the displacement coefficient polynomials $a_n(\ell)$ by their exact recursive algebra and connects the first nontrivial truncation to the quantitative expansion already proved.
FabiusLambertFormalLog.lean	Identifies those coefficients with the logarithm of the formal unit series $1 + uA(u)$, supplying the cancellations needed to make the formal series analytic.
FabiusLambertAllOrderRemainder.lean	Upgrades the formal recurrence to a rigorous expansion of every finite order, bounding the varying-coefficient Taylor remainder and using stability of $u - \log u / \log 2$ on the large positive branch.
FabiusLambertAllOrderSmallArgument.lean	Transports the all-order phase expansion to $x \downarrow 0$ and writes it purely in terms of $-\log x$ and $\log(-\log x)$.

B.14 Elementary log–log forms and endpoint consequences

These modules carry [Section 16](#) and [Section 17](#), and several of the negative results of [Section 21](#).

Module	Role in the proof
<code>FabiusLogScale.lean</code>	Introduces the profiles $F(2^{-t})$ and $-\log F(2^{-t})$ and proves their exact differential-delay identity, carrying the explicit hypothesis $t \geq 1$ that the source omits and without which the statement is false.
<code>FabiusDyadicLogBounds.lean</code>	Combines the exact identity $d_n = n! 2^{\binom{n}{2}} F(2^{-n})$ with elementary integral bounds to prove $ \log F(2^{-n}) + (\log 2/2)n^2 \leq 3n \log(n+1)$, with no periodic ansatz.
<code>FabiusLogSquaredAsymptotic.lean</code>	Promotes that natural-index estimate to the real logarithmic scale by squeezing t between consecutive integers, giving the rigorous leading asymptotic $\log 2/2$.
<code>FabiusLogMainDefect.lean</code>	Substitutes the draft’s proposed main term into the exact delay equation and determines its residual: a nonzero multiple of $(\log t/t)^2$, which is exactly the term the source later discards.
<code>FabiusWikipediaMain.lean</code>	Defines the compact lower-Lambert form of the published expression, including the centered periodic correction, and proves it equals the Lambert-coordinate saddle main term.
<code>FabiusWikipediaExpansion.lean</code>	Expands that compact form into the elementary printed logarithms, retaining the periodic correction at its exact phase; the fourth phase term is what keeps the error at $1/t$.
<code>FabiusWikipediaObstruction.lean</code>	Shows the periodic correction is nonconstant and its phase visits every far integer translate, so deleting the term makes an $\mathcal{O}(1/(-\log x))$ error impossible.
<code>FabiusDecayComparison.lean</code>	Proves the second qualitative comparison: although F is flatter than every power, it decays strictly more slowly than $\exp(-c/x)$ for every $c > 0$.
<code>FabiusQuotientExponentialMismatch.lean</code>	Formalizes the endpoint obstruction for the visually accurate quotient of exponentials, whose ordinary $\exp(-c/y)$ decay is strictly faster than the true bump.
<code>StirlingAsymptotics.lean</code>	Records the precise $\mathcal{O}(\log n)$ form of Stirling’s formula invoked without proof in the draft.

B.15 Iterated Thue–Morse prefixes and draft refutations

The machine-checked half of [Section 21](#): what the local drafts get right, stated correctly, and what they get wrong, refuted.

Module	Role in the proof
<code>ThueMorsePrefix.lean</code>	Defines the iterated inclusive prefix sums, proves the convolution kernel formula, the Prouhet vanishing of low-degree power sums, the affine power-sum identity, and the exact zero runs at dyadic indices.
<code>ThueMorseGenerating.lean</code>	Builds the signed Thue–Morse block polynomial and its finite product form, then shows the draft’s literal grid normalization has endpoint value bounded away from one, refuting its convergence claim.
<code>ThueMorseApproximation.lean</code>	Identifies the corrected iterated prefixes with the polynomial and histogram approximants, exactly at dyadic cell centers and asymptotically along moving cells, including the exceptional right endpoint and the rescaling that recovers F .
<code>ThueMorseBinomialLog.lean</code>	Proves the binomial-parity formula for the Thue–Morse sequence, interpreting the finite sum in $\mathbb{Z}$ and the outer division in $\mathbb{Q}$, and showing the logarithm’s argument is exactly $2^{\text{wt}_2(n)+1}$.

Module	Role in the proof
<code>DraftCounterexamples.lean</code>	Records three explicit obstructions while preserving the draft’s definitions literally: the nonexistent maximum, the uncanceled linear term in the printed proxy, and a local error of 3 where the draft claims $4h = 1$.

B.16 Public aggregates

The root module `FabiusFunction.lean` is what a user imports. It pulls in five paper aggregates — `Paper05442.lean`, `Paper06487.lean`, `PaperStatements.lean`, `PaperFabiusAsymptotic.lean`, and `PaperKFoldThueMorse.lean` — the last two of which cover the two local drafts audited in [Section 21](#), together with the regularity, flatness, Legendre, negative-Laplace, and computability layers. Only `PaperStatements.lean` is reached indirectly, through `Paper06487.lean`.

Module	Role in the proof
<code>FabiusFunction.lean</code>	The root aggregate, one directory above all the others. Its module docstring is the authoritative summary of the public surface: the two formalized papers, the two audited drafts, exact arithmetic, the Legendre expansion, the sharp asymptotic, the regularity layer, and Grzegorzczak computability.
<code>Paper05442.lean</code>	Public import for the compact-support paper, naming a proved Lean counterpart for every theorem, lemma, corollary, and prose proposition, and flagging where the development proves strictly more than the source-facing domain.
<code>Paper06487.lean</code>	Public import for the arithmetic paper, exposing the analytic, integrality, divisibility, and two-adic statements, both orders of summation in the dyadic formula, and the denominator consequences of the conjecture.
<code>PaperStatements.lean</code>	The numbered index for the arithmetic paper: every proposition, theorem, lemma, question, definition, and conjecture, with the two deliberate corrections — the bounded versus signed codomain split, and the missing order hypothesis in Lemma 1.
<code>Paper06487Supplement.lean</code>	Exposes the prose-level and proof-internal assertions of the same paper that are not numbered theorem environments, several with weaker hypotheses or wider boundary cases than the prose needs.
<code>PaperFabiusAsymptotic.lean</code>	Claim-level audit of the local asymptotics draft, which contains no theorem environments at all: it tracks numbered equations and prose claims, proves the valid ones, disproves the $\mathcal{O}(t^{-2})$ residual claim, and points to the correct saddle analysis.
<code>PaperKFoldThueMorse.lean</code>	Claim-level audit of the local K -fold Thue–Morse draft: the identities, zero runs, and corrected approximation theorem are proved, while the printed normalization, the maximum, and the proxy’s missing linear term are refuted.

C Notation crosswalk

Different sources use F , φ , θ , and $\uparrow$ for overlapping objects. The following crosswalk records the convention used here.

This article	Definition	Frequent source notation
$F(x)$	Bounded CDF on all of $\mathbb{R}$, constant 0 to the left and 1 to the right.	Fabius function on $[0, 1]$; sometimes the same letter is reused for the global extension.
$\text{up}(x)$	Even compactly supported density $F(1 - x)$ on $[-1, 1]$.	$\text{up}(x)$ or $\varphi(x)$.
$\mathcal{F}(x)$	Signed Thue–Morse extension, zero on $(-\infty, 0]$ and satisfying $\mathcal{F}' = 2\mathcal{F} \circ (2 \cdot)$.	$F(x)$ or $\theta(x)$ in papers that work globally.
c_n	Even moment $\int x^{2n} \text{up}(x) dx$.	c_n in the arithmetic paper.
d_n	Half moment $n \int_0^1 x^{n-1} \text{up}(x) dx$, with $d_0 = 1$.	d_n in the arithmetic paper.
Ψ	Centered or explicitly mean-separated one-periodic correction in the negative-Laplace finite part.	Often absent from elementary accounts.
λ	Exact positive solution of $\lambda 2^{-\lambda} = x$.	Lower-Lambert saddle coordinate $-W_{-1}(-x \log 2)/\log 2$.

When comparing formulas, one should first identify which global convention is in force and only then compare signs, supports, and endpoint values.

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